



UI POST UTME PAST QUESTIONS & ANSWERS 2019-till date

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ADEAGBO MERCY	LAW
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ABISOLA MARIAM	THEATER ART
NWAORGWU CHIAMAKA	LAW
OYEDIRAN TOHEEBAT	PHARMACY
NNUBIA ESTHER	NURSING
OGUNBIYI PRAISE	LAW
KAMORUDEEN ABUBAKR	CIVIL ENGINEERING
ADEAGBO GOODNESS	ELECT ENGINEERING
SHARON ONYENE	LAW
AKANJI JEHU	ELECT ENGINEERING

And So Many More...

SUBJECT

COMBINATION: ENGLISH, CHEMISTRY, BIOLOGY, PHYSICS

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2023

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OLORUNTOYIN (LAW)

FRANK MIRACLE (LAW)

OGUNBIYI PRAISE (LAW)

SHANNOYE WARIS (PETROLEUM

ENGR) OMOLAYO PETIREMI (ELECTRICAL ENGR)

PLEASANT (LAW)

AFOLABI SAMUEL (PETROLEUM ENGR)

KAMORUDEEN ABUBAKIR (CIVIL
ENGR)ONWUAGHA CHIDERA (CIVIL ENGR)

BIOLOGY 2021

1. The only living cells in the xylem are they I Collenchyma cells II parenchyma cells III tracheids IV

companion cell V none of the above

A II only B III, IV C II and IV D I-IV E IV only

2. I xerophthalmia II cataract III Night blindness IV astigmatism

Which of the following eye defects are nutritional problem

A II and IV

B I and III

C II-IV

D III only

3. Which of the following does form in the nucleolus

A proteins B DNA C RNA D ribosomes

4. Dryopteris is a A moss B fungi C virus D ferns

5. Which of the following is a discontinuous variation I height II weight III skin colour IV ability to taste PTC

A I-IV

B III and IV

C I and II

D IV only

6. Which of the following are bacterial infection I whooping cough II syphilis III Gonorrhea IV ringworm

A II-IV

B III and IV

C I and III

D I and III

7. Normally any character shown by an individual is due to the effect of I hormones II

chromosomes III mutation IV

genes V environment

A I-V B I and IV C I IV and V D IV and V E I III and C

8. When a colour blind man marries a carrier what is the probability of their offspring being color blind

A. 25% B 50% C 75% D 100%

9. Mumps is a _____ infection

A bacteria B viral C protozoan D fungi

10. Which is not found in plant cells

I nucleolus II cell wall III chloroplast IV cell membrane A all except I B II and III C I only D I and IV only E

IV only

11. Which of the following is an abiotic factor I CO_2 II bacteria III Temperature IV light A none except II B I

III and IV C III only D III and IV E I and II

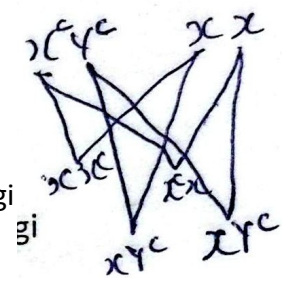
12. What is the probability of a colour blind mother and a normal father giving birth to a normal male child

A. 25% B 50% C 75% D None

13. I) fat soluble vitamins are precursors to coenzyme II water soluble vitamins are precursors to coenzyme.

enzymes III) cataract can be caused by deficiency of vitamin A

(Inability of blood to clot can be a deficiency of vitamin K



Which are correct

A II III IV

BIIIV

C IV only

D II and IV

E I and IV

14 I sunken stomata II Tiny leaves III Thick cuticle IV Big broad leaves

Which isn't a adaption mode for desert plants

A lonly BIICIIIII DIV ONLY X

15 which is not concerned with excretion I kidney II prostate gland III ureter IV urethra A II only B III, IV C I, II, III D I, II, III, IV

ot C III only D II and IV E IV only

16 which of these is a disease of the respiratory tract A bronchitis B malaria C cholera D none

17 Herbivores are A producer B decomposer C autotrophs D consumer

18 Interaction of group of population is called
A Ecosystem B Community C biosphere D niche

19 Turbidity of water is measured with
A thermometer B secchi disk C wind
vane

20. The end product of light stage of photosynthesis produce A oxygen and water

B glucose and starch C NADPH and ATP

D protein

21 which of the following is a producer I
spirogyra II paramecium III mucor IV
amoeba

A I only

B II and III only

Cland IV only

D none

22 the part of the ear involved in balance is A cochlea B eustachian tube C semi circular canals D

pinna

23 which of the following is peculiar to birds I feather II scales III hair IV lungs

x All-IV B I and IV only C II Only D I and III only
E I only

24 Meiosis takes place in the I sperm li ovary III testes IV ovum A I and II B II and III C IV and I only D IV

and III only

25 at times hyenas feed on the remains of animals killed by others animals and other hyenas

themselves kill animal for food there hyenas
may be best described as

A scavengers and herbivores

B.scavengers nd parasite

C scavengers and predators

D scavengers and parasimus

26 several incidence of Spina bifida can be reduced by preventing the administration of which vitamin

I vitamin A II vitamin B 3 III vitamin B 7 IV
vitamin B9 A I to IV B II and III C IV D I and II only
E II only

27 which of these can be prevented by BCG vaccine I tuberculosis II poliomyelitis III malaria IV HHV/AiDS

A II only B I to IV C IV only D II and IV E I only

28 A disease in haemostasis is I Liver stones II haemophilia III night blindness IV dysmenorrhea A II

only B I and IV C IV and III only D IV E I II and IV

29. Grasses are pollinated by

A wind B insect C man D animals

30. Lysosomes are called the suicide bag of a cell because it is responsible for packaging and distribution. I Ribosomes produce nucleolus II nucleolus produces ribosomes III adenine pairs with

guanine

Which is correct

A III only B II only C III and IV D II and IV

31. Which of the following is an invertebrate? A millipede B fish C toad D lizard

32. The structures in plant which can be compared to arteries and veins in mammals are

A xylem and cortex

B. xylem and phloem

C phloem and parenchyma

D cortex and chlorophyll

33. The mesophyll cells of plants consist of I photosynthesis cells II cells with lignin III parenchyma cells

IV cortex. A I and III

B. II and III

C II III and IV

D III and III

34. Which of the following is not associated with respiration

I lungs II oesophagus III trachea IV alveoli

A II only B I III and IV C III only D II and III E none

35. I carbon cycle

II nitrogen cycle

III water cycle

IV phosphorus cycle

V sulphur cycle which is required by leguminous plant

A I II and IV B II and IV C I II III IV D all

36. Which of the following organelles are double membrane? I cell membrane II nucleus III chloroplast IV mitochondria

A II B II and IV C III III D II III IV

37. Enzymes are I protein II carbohydrates III lipids IV minerals

A I only B I and IV only C I III and IV D I and III

38. Amoeba moves by I pseudopodia II cilia III axopodia IV flagella

A III B II C II IV D I III IV

39. Which of the following has DNA? I nucleus II mitochondria III chloroplast IV vacuole

A I only B II and IV C I and II D III III

40. Which of the following are primates? I lemurs II apes III gorilla IV monkey

A II and IV B I and IV C II III and IV D all

41. Pepsin acts on I carbohydrates III starch III fats IV sugar

A I and IV B II only C IV only D III only

42. How many nuclei are found in pollen grain tube during germination? A 4 B 2 C 3 D 1

43. Animal cells include the following except I cell wall II chromosomes III nucleus IV cell membrane A II

only B III C II and III D I

44. The name for the collective female reproductive organ in flowers is I gynoecium II Androecium III

ovary IV ovum

A I only B I and IV only C IV only D I II and IV only

45: I oviparous II viviparous III ovoviviparous

Which of the following is based on egg laying alone

A I only B I and IV only C IV only D I II IV only

46 Which of the following is mainly involved in absorption of water

A large intestine B small intestine C oesophagus
D stomach

47 PCR means

A polymer carbon reaction
B polymerase Chain reaction
C polymerase Carbon reaction
D None

48. An example of osmosis is

I translation of water in the phloem
II Conduction of water via the xylem
III absorption of water in the ~~plp~~ ^{phloem}
IV absorption of water from the root

A I I-IV B II and III C IV only D II only E II and IV only

49. the sum total of all the roles an organism plays in a habitat is referred to as A Ecosystem

B. Ecology C niche D habitat

50 which of the following possess flagellum A euglena spp B hydra spp C amoeba spp

51 which is the correct food chain

A diatom fish crustacean man

B. diatom crustacean fish man

C fish crustacean diatoms man

D fish man crustacean diatoms

52 which of the following exhibit external fertilization A rat B lizard C echinoderms D mammals

53 The reproductive organ of reptiles is hidden in the A scales B prostate gland C hemipenis D corpus

callosum

54: Crocodile is a/an A amphibian B mammals C aves D none of the above

55 Emulsification of fats and oils is caused by A lipase B ptyalin C bile D maltase

56 cellular respiration is divided into 3 stages called

57 which of the following is involved in photosynthesis

I glucose is synthesized

II Energy is absorbed from the sun

III Oxygen gas is given off

IV carbon dioxide is evolved

A II and IV B III and IV C I II and III D All

58 The 2 main compounds in the plasma membrane are I protein II carbohydrates III phospholipids

A I and II B II and III C I and III D all

59 which of the following diseases can be treated by bacteriophages I malaria II athlete's foot III staphylococcus lung infection IV HIV/AIDS

A III only B II and IV C I II and III D II and III

E none of the options

60 Open Circulation is found in which of the following I fish II insect III toad IV lizard

A I and II B II only C II and IV D I only

61 I stomata II lenticel III cuticle IV mitochondria. which of the following is found in breathing root

A II only B II III and IV C II and III only D all of the options E None of the options

Answers

1. Xylem parenchyma is the only living tissue
note Companion cell is
the living tissue Phloem (A)

2. B

3. D .Nucleus is ribosomes factory

4. D. A wood fern (D) 2019 questions repeated

5. D

B. Dringworm are caused by fungus

7. D Jamb past question repeated

8. B

9. B

10. E this is the most correct answers in some
plant cells the cell

membrane is fused with the cell wall

II. Non living factors (A)

12. D all the Male children will be colour blind
while the female will

be carriers

13. (D) Two of the fat soluble are precursors
to coenzyme (Vit. A AND K

14. D

15. A

16. A

17. D

18. B

19. B

20. c

21. Green plants are producer (A)

22. C. Cochlear is involved in hearing

23. Feather (E)

24. B Meiosis takes place in the testes and ovary
producing sperm and

ova

25. C UI practice questions repeated

26. C

27. E

28. A inability Of blood to clot

29. All grasses are pollinated by wind (C)

30. (B) Nucleolus produces Ribosomes and
not vice versa

31. B

32. B ;

33. A

34. A

35. c

36. B

37. PROTEIN (Go for protein and mineral
cos minerals are cofactors **Ans B**)

for Enzymes)

38. D FALSE FOOT JAMB

39. D

40. D

41. NO PROTEIN IN THE OPTION JUST GO
FOR SUGAR (C)

42.c

43.D

44.A

45.A-

46.A

47 B

48.E

49.c

50.A

51.B 52.C

53.c

54.D

55.c

56. Glycolysis, krebs cycle and electron
transport chain

57.c

58.c

59.A

60.B

61.A

BIOLOGY

1. Which of the following functions in the mammalian eye

- I ossicles
- li pinna
- III auditory meatus
- IV cornea
- V sclera
- VI iris

- AI
- B I to III
- C IV V and VI
- DI-VI

2 which of the following is a part of the neurone I glia li meninges III axon IV synapse

- AI-IV
- B II III IV
- C III and IV
- D II and III

3Which has the least efficient energy link transfer in the chain?

- I Sun to producer
- III.Primary consumer to secondary consumer
- IV.Secondary consumer to decomposer
- V.Decomposer to scavenger
- II.Producer to primary consumer

- A.II only
- B III
- CV
- DI

E IV
4 the total sum of the biotic and abiotic component In the environment is

- II.Biome
- I.Habitation
- III.Ecosystem
- IV.Ecological niche

- A land IV
- B II and III
- C iii
- DII

E IV
5 which of the following is a plant waste

- I oil
- li cytokinin
- III tannin
- IV resin

- A III
- B III and IV
- CI-IV
- D III and IV

E IV
6an advantage of food webover food chain is

- I organism have alternative means of survival
- li it can't be destroyed easily
- lii there's competition for food and space
- IV it gives a complete overview of how the nutrient is cycled in the ecosystem

A I to iV B III CI II IV D I and IV EI III and IV.

7.Which of the following functions in the eye

- I ossicles li pinna III auditory meatus iv cornea
V sclera Vi Iris A I and li B I to III C I to Vi D IV/ v
VI E none of the
- above
- 8 which of the following is/are eye defects
I long sight.ii short sight III glaucoma iv colour
blindness V river blindness A I - IV B IIIIII CIII III IV
DIIIIV
9. Which of these is true of a type of climax
community I it's affected by season and climatic
changes li it's the first stage of succession III it is
stable IV it has a high biodiversity
- A.I-IV
C.I&III
D.III&IV
B.II,III&IV
E.I,III&IV
- 10 differences between bryophytes and
pteridophytes is I absence of flowers li water is
needed for fertilization III
vascular bundles
- A.I only
D.III only
C I-iii
B.II only
E.I&III
- 11 In which of this part is smooth muscle
found?
- II.Stomach
III.intestines
I.Blood vessels
IV.uterus
V.urinary bladder
- A.I-V
D.I,IV&V
B.I,IV&V
C.II,III&IV
E.I&IV
- 12 which of this is a type of muscle
- I.Smooth
II.Cardiac
III.Tendon
IV.Trapezius.. A.I-IV
C.I&II
B.I,II&III
D.I,II&IV
- 13 Angiosperms and Gymnosperms belong to
which group?
- II.Schizophyte
I.Pteridophyte
III.Spermatophyte
IV.Thallophytes
A.II only
B.IV only
C.III only
D.I only
- 14 Which of the following reduces soils
fertility I. Leaching II. Erosion III.compartment
- '15.which of these is/are absent in prokaryotes?
- III.plasmid
I.mitochondrion
II.cytoplasm
IV.nucleus

A.I only

B.II&III

C.all of I-IV

D.III only

16 16.The basis for growth and asexual reproduction is

A.meiosis

B.cytokinesis

D.Cytogenesis

C.mitosis

E.Cell elongation. 17 The process which ensures that the chromosome number for each species of organism remain constant from

generation to generation is called A.fission

B.fusion

C.meiosis

D.Mitosis

E.oogenesis

18Which of the following species exhibit an aerial root

I.Ficus

II.Centresema

IV.lantana

III.Antigonon

V.Dahlia

A.I-V

D.III-V

E.II only

B.I-III

C.I only

19..Which of these plants has an emerginate apex?

II.Pistia

III.Ixora

IV.Musa

I.Bauhinia

V.Terminalia

A.1-III

C.III &IV

B.I only

D.I&II

20.Organism with two nucleus is/are

I.Euglena

IV.paramecium

II.volvox

III.amoeba

V.fungi

A.I-V

C.II IV&V

B.IV only

D.III only

E.IV&V

21.The pigment of the lymph responsible for the transport of oxygen

I.lymph corpuscles

III.lumthane

II.heamoglobin

IV.lymphatic

A.All except II

B.III only

D.I only

C.I&IV

E.None of the above

22..Which of the following is/are made up of dead cells

I.meristematic

II.xylem tissue

III.cambium

IV.mesophyll

B.I&IV

A.III only

C.II only

D.II&III

E.I-IV

23 Which of the following is/are not true of mucor

II.Bears spores in sporangia

I.its grows saprophytically

III.consists of hyphae

IV.reproduce by fragmentation

A.All except IV.

B.I&III

E.none of the above

24 Which of the following features of euglena is found only in animal

I.paramylum granule

III.pellicle

II.flagellum

IV.pyrenoid

A.I&IV

C.II&III

B.II only

D.III only

25.Which of the following do/does not found in the nucleolus

I.Protein

II.DNA

III.RNA

IV.Ribosome. A.I&IV

D.IV only

C.All except III

B.II&III

E.II&IV

26.All are carnivores except

III.seals

II.porcupines

I.Rhinoceros

IV.Civets

A.all except IV

D.I,II&IV

B.I&IV

C.II&III

E.none of the combinations

27Which of the following instruments has to do with the aquatic habitat?

II.Salinometer

I.Thermometer

III.pH scale

IV.Secchi disc

V.Anemometer

A.I&III only

C.I&V

B.I IV

D.II&IV

E.II III&IV

28. Less than ----of bacteria are harmful

A.5%

B.10%

D.1%

C.0.1%

E.none of the above

29.I.it's unicellular II.it's prokaryotic III.it's plant like.The

following are characteristics of

A.Anaebena

C.Euglena

B.Candida

D.Herpes

30.Which of the following does sperm not travel through.

C.vas deferens

A.ureter

B.urethra

D.epididymis

Answers

1.(C)

2.(A)

3.The most efficient energy is transferred from sun to producer

while least is transferred from secondary consumer to

decomposers.The sequence is as follow

Sun→Producer→Primary

Consumer→Secondary Consumer

→Decomposers (E)

4C

5.Cytokinin is a plant hormone (A)

6.(C)

7.(D) All are tissues in the eye

8.C River blindness is a disease not a eye defect

9.(E)

10. Bryophytes lack vascular bundles (D)

11. (A)it's found

12d

13c

16c

17d

18c

19b

20e

21e

22c

23e

24b

25d

e0M

e 6 Z

p 8

e L C

p 9

BIOLOGY 2023

1 Which of these organisms belong to the second largest phylum in the animal kingdom

I squids II insects III earthworm IV flatworms
A II and III B I and III C IV D I E none

2. bacteriophage is a ---?

A land IV B II C I D IV E II

3 which of the following is not a biotic factor
I oxygen II bacteria III temperature IV light

A III B I III IV C I and II D III and IV

4 caryopsis is an example of I dry dehiscent II fleshy III schizocarpic IV dry indehiscent
A III only B I only C IV only D II and IV only E II only

5 which of the following is a dry dehiscent fruit
I Samara II achene III nut IV legume

A I and IV B II C IV D III III E I

6 the following are monocots and dicots except
I oranges II pines III bananas IV oaks
A II and IV B I and III C II only D I and IV only

7 which of the following is a biome .I aquatic habitat II swamps III mangrove swamp forest IV savannah

A II and III B I C I and IV D II-IV

8 presbyopia is associated with I aging II Nutrition III Neoplasm IV heredity
A I II IV B II only C I and IV D

III only E I only

9 which is not a difference between eubacteria and archaea domain

I they are prokaryotic. II they are flagellated
III they possess plasmids A I -III B II C II and III D none

10 which of the following is not a characteristic of phylum mollusca

I they have no body cavity
II their body is divided into head visceral mass muscular foot and mantle

III they are in the organ level of organization

IV they are bilaterally symmetrical

A II III IV B I and III C III D II and IV

11. which of the following is used in lumbering I fir II cycads III ginkgos IV pinus V gnetus

A I to V

B II III and V

C II and V

D I III and V

E I and II

12 which is not a major classification in plants

I thallophyta

II chlorophyta

III bryophyta

IV spermatophyta

A I to III B II to IV C I only D II only

13 which of the following species exhibit an aerial root

I ficus

II centrosoma

III Antigonon

IV Antana

V dahlia

AltoV	E Iv 17 When the anther matures before the stigma
B Ito III	,it is said to be what
Clonly	A.Protandry
D III to V	B.Protogyny
E II only	C Dichogamy
14.Which of these plants has an emerginate apex?	D.cleistogamy
II.Ixora	18.The agency in charge of aid of children
III.Musa	A Fund, New York, United States
I.Bauhinia Pistia	B und, Geneva, Switzerland
IV.Terminalia	
(a)I-III (b)I only (c)III &IV(d)I&II	D.United Nations Children's Fund,New York
15 When a heterozygous tall plant(Tt) and a homozygous short plant(tt),what will be the ratio of tall to short	C.United Nations Children's Fund, Geneva,
B.1:1	B.United Nations Children Emergency
D.4:0	A.United Nations Children Emergency
C.3:1	Switzerland United State
A.2:1	19.Which of these is the general classification of living organism
E.2:2	I plants animal and fungi
16.Which of these are prokaryotes	II plant, animals, bacteria
III.Fungi and virus	III.plants, animals, archaea
IV.Bacteria and archaea	IV.Bacteria, fungi, archaea
I.Virus and bacteria	V.prokaryotes, archaea eukaryotes
II.Fungi and bacteria	C.I only
A I-IV	A.I-III
B.1&11	B.III-V
CI	D.II only
D II	E.V only
	20.Which of these is not part of the whorl of a flower
	A.Androecium

B.Gymnocieum

C.Antheridium

D.Corolla

21 The agency in charge of protecting species on the Red List

A.International organization for

B.International organization for

Conservation of Nature and Natural

conserving of Nature. resources

C international organization for conservation of Nature

22 Which is the order of preference of waste area

I.Waste reduction

II.Waste prevention

V.Disposing in lands far from residential

III.Re-use

IV.Recycling

A.V>11>1>1V>111

B.11>1>V>N>111

C.11>111>N>V

D.11>111>1>N>V

E.V>N>111>1>11

23 which of these are pioneers in a Xenarch succession

I Algae

li lichen

III mosses

Iv fungi

V phytoplanktons

AltoV

Bii and III

CIto IV

D II only. E I to III only

24 in a climax community what are observed

I nom vascular plants are the dominant plants

li species diversification

III the organisms are more adapted to the environment

AI-III

B II

c II and iii

D ii

E I to III

25 all are cryptograms except

lferns

li bryophytes

lii algae

IV spermatophytes

A I to IV

B II and III

CIIIII

D IV

E none of the above

26 in a community bacteria and fungi are referred to as I producers li consumers III scavengers IV tertiary

consumer V decomposer

A II and V

B III and V

Cvonly

D II

ElandV

27 which of the following structures is found in reptiles and birds I scales li beaks III legs IV claws

AI and iV

BI-IV

C1

Dland III

28 the theory of natural selection was developed by I lamrcak li Darwin III mallace IV Mendel V hooke A I and li B ii and III Clto IV DI to III E IV and V

29 which of the following is major factor of variation among organisms

I inbreeding

li sexual reproduction

III back crossing

IV gene

V test crossing

AI-V

B II and IV

C III and V

D IV

EIIIV

30 the plumule is best described as

I structure near the micropyle

li a miniature shoot

III part of the endosperm

IV structure opposite the hillum

V part of the cotyledon

AltoV

Bii and IIII

C II only

D II IV and V

E II and V . 31 the cell membrane covering the vacuole is

A cell wall

B cell membrane

C tonoplast .d ribosomes

32 The animal granules in plant is

I starch grain

li mitochondria

III ribosomes

IV glycogen

AIbII C III D IV

33 which of the following is the function of sclerenchyma I storage of water li storage of food III transport IV strengthening and rigidity

A I and li only

Biii

C IV

D II only

34 the following are simple tissue except

I sclerenchyma

li xylem

III parenchyma

IV physiochyma

V fiber.AIIIV

B IV

C V only

D IV and V

35 which of these is an effect of deforestation

I soil erosion

II Biodiversity

III pollution

IV flooding

V global warming

Alto III

B I to V

C I iii and V. D I iii IV and V

36 which of the following is a vascular tissue I

xylem II parenchyma III sclerenchyma IV

Collenchyma

A I only B I - IV C II III IV D II I-IV

37 which of the following has the highest energy level A producer B primary consumer C secondary consumer D

decomposer

38 which of the following stores food in it's root

I casava II Irish potato III yam IV cocoa A I and II

B I and III C II and III D II and IV

39. which of the following processes is responsible for keeping a constant number of chromosomes I fusion II

fission III meiosis

IV mitosis

A I and II

B II and III C I-V D none. 40. which of the following devices is used to measure the nutrient content of a biscuit I bomb calorimeter II food

calorimeter III food meter IV hydrometer

A I B I C I D IV

41. which of the following does not belong to the main group of the plant kingdom I

bryophytes II chlorophyta III

pteridophytes IV spermatophytes and V

thallophyta

Alto V

B I to IV. C II only

D III only

42 the class of plants that doesn't have membrane bound organelles is A fungi and bacteria B archaea and Bacteria

C protist and archaea D fungi and protist

43 the plasma membrane and cell membrane do not allow water to pass through them because of I diffusion II

osmosis III selectively permeable membrane IV permeable membrane

A I-IV

B I and II

C III only

D III and IV only

44 which of the following organisms exist in the nucleus A paramecium B euglena C chlamydomonas D Amoeba

45 The international organisation that deals with natural resources

D. Economic community for conservation of nature

A. International Union for conservation of with the conservation of natural resources is

C. Economic community for conservation

has its three organs above

the ovary

D.none of the above

46.Arrange the following in sequence of succession in a climax community

I.trees

III.lichen

II. herbaceous plants and grasses

IV. bare rocks

V. shrubs

A.IV, III, II, V, I

B.IV, III, II,I,V

C.. IV,V, III, II,I

D.IV, II, III,V,I

47Arrange the following from the least to the greatest

I.vulnerable

IV. extinction

II. endangered

III. critically endangered

V. least concerned

A A.V, I, II, II,IV

C.III,IV, V,I, II

B.IV, V,III,II,I

D.V,I, II, IV, III

48 Millipede and earthworms are

A. Decomposers

C. detritivores

B.heterotrophs

D.consumers

49 The living part of phloem tissue is

I. Sieve plate

II. Sieve element

III. Cytoplasm

IV. Companion cell

A.I and II

B.III only

C.IV only

D.I only

ANSWERS

1. The largest phylum is arthropoda followed by molluscs (D)

2. This a virus that attacks bacterias (B)

3. Abiotic are non living factors e.g climates,temperature e.t.c(A)

4. Caryopsis is a single-seeded indehiscent fruit .Other examples of

dry indehiscent fruit are achene,nut&samara(C)

5. Dry dehiscent are Legume, follicles, capsules e.t.c (C)

6. Angiosperm are divided into monocots and dicots .Oaks is an Angiosperm while pines are neither dicots nor monocots ,they are conifers not flowering plants.(C)

7. There are five major types of biomes;aquatic, grassland, forest, deserts, and tundra(B)

8.(E)

9.(D)They are both prokaryotic,flagellated and possess plasmid

10.Mollusc have small

coelom(COELOMATE).The main

body cavity is a Hemocoel through which blood circulates(D)	28.B
	29.(D)
	11 All gymnosperms are used in lumbering (A)30 .C
12(D)	
	31.C
13.(C)	
	32.D
14.(B)	
	33.A
	15.(B)
	34.C
16. Prokaryotes are divided into eubacteria and archaea (E)35.B	
17. Anther before stigma is Protandry while stigma before anther is protogyny(A)	
18. Full meaning of UNICEF(united nations children's fund)	38.B
	39.D
and the headquarters(new york USA)	40 B
19. (E)	41.C
20.(C)	42. PROKARYOTES B
21.(B)	43.C
22.prevent-Reduce-Reuse-Recycle-Recover-Dispose (I)	46.C
	44.A
23. Xerarch succession occurs in a very dry 45.A environments such	48.A
a rock deserts and sand dunes.Sequence of	47.A
stages in xerarch	
succession are as	49.C
follow:Lichen→Moss→herb→shrub→climax	
community (D)	
24 D	
25. Cryptogams are non-seeded plants e.g thallophytes, bryophytes,pteridophytes .seeded plants are called	
Phanerogams e.g spermatophytes (D)	

26.(C)

27.(C)

BIOLOGY 2024^{Alto IV}

1 A phenotype is due to only one genotype this statement

A True

C partially true

D partially false

is

A mitosis

B.meiosis

C Nutrition

D reproduction

3 A major function of the xylem is

A. Reduction of water loss

C.Conducting food materials

4 which of the following external features li scrotum shows that Rat is a mammal

I possession of lungs

li possession of well developed kidneys

III possession of sweat glands

IV mill and sweat production

V possession of furs and whiskers

B.IV and V

CIIIIII

D lii and V

E v only

B false 5 which of the following carries blood to the heart

I superior vena cava

2 The basis of growth and asexual reproduction li inferior vena cava

IV Renal vein

V phrenic arteries

A111

B I-IV

cl-V

DliIIII

B strengthening the stem 6 which of the following is responsible for the production

and storage of sperm

III Epididymis

D none

of the

options

I testes

IV sperm duct

AIIIIII

B II and III

C I to IV

D I and III

7 which of the following does not function in

the kidney

A. Capsule

B. loop of Henle

Cureter

D.PC T proximal convoluted tubule

8 which of the following is a biome

A. Aquatic habitats

B estuarine

C mangrove swamp

D none of the above

9 Which of the following essential protein present in the plasma is responsible for blood clotting

I Fibrinogen

II prothrombin

III fibrin

IV serum

V thrombin

VI-V

VI-IV

VI-III

VI-III

10. Which of the following is not an adaptation of desert plants

A sunken stomata

B extended tap root

C shallow root with broad leaves

D spines

11. Which of the following plants has a palmate leaf?

A. Banana

B. Mango

C. Papaya

D.Coconut

12 . THE BOTANICAL NAME OF guinea grass is

A Zea Mays

B Carica papaya

C Panicum maximum

D Pennisetum purpureum

13 which of the following leaves is imparipinnate

A ixora

B Tecoma stans

C grass

D Terminalia

14 Which of the following is the botanical name of cassia plants

A filliformis

B Nepenthes

C dodder . D None of the options

15 which of the leaves is Completely Bipinnate

A cassava

B Periwinkles

C delonix

D hibiscus

16 Which of the following plants has leaves that are typical of the order Cycadales?

A. Cycas revoluta

B. Mangifera indica

C. Zea mays

D. Tectona grandis

17 which of the plants exhibit climbing roots

A boerhaavia

B pumila

C raphia

D white mangrove

18 which of this is a compound palmate leaf

A ixora

B delonix

C musanga cecropoides

D Allamanda

19. Which of the following is a brightly pinnate leaf

A Neem

B delonix

C pride of Barbados

D none

20. The sensitive plant is a type of which compound leaf . A simple palmate leaf . B Bipinnate leaf

C pinnate leaf . D imparipinnate leaf

21 what is the botanical name of spear grass

A Vigna sp

B. Zea Mays . C imperata cylindrical

D solanum tuberosum

22 what is the botanical name of bahama grass

A cyndom dacton

B aspilla africana

C ipomoea botata

D none of the options

23 The botanical name of pigeon pea. Is

A Glycin max

B Euphorbia aorta . C sida acuta

D none of the options

24 which of the following exhibit decusate arrangements

A sensitive plant

B independence plant

C silk cotton

D all of the above

25 The shape of yam leaf is

A oblong

B cordate

C Oval

D ovate

These are all the past questions on botanical name and leaf arrangements

ANSWER

1. B it is false a lot of genetic constitution make up the phenotype included the effect of the environment

2. B mitosis

3 B apart from the conduction of water through its vessels

4 E only furs and whiskers are external and they are indicative that rat is a mammal!

5 D renal vein will and phrenic vein will empty into the inferior vena cava not to the heart directly

6 A sperm duct is not necessarily included

7 C ureter function in the neuro pelvic junction and not in the kidney

8 C mangrove swamp

9 C prothrombin and fibrinogen are two protein present in the plasma alongside with enzymes fibrin,serum

22A

23 D none of the options

, thrombin are not in the plasma

10.C the root should be deep not shallow and also leaves should be reduced to spines to conserve water

11 C papaya leafs are palmate leafs

12 C panicum maximum

13 B the leaves of tecoma stans are imparipinnate!

Imparipinnate leafs are also called trifoliate leafs

14 A cassytha filliformis ,this is a trap put cassytha in the option will reveal the

15:C the leaves of delonix regia are bipinnate!Having oppositely arronaged leaflets

16 A cycas revoluta remember this is a cycad under the class gymnosperms

17 B ficus pumila exhibit climbing roots

18 C musanga cecropoids is a palmate compound leaf

We have both simple palmate leafs (cassava and pawpaw)

And compounf palmate leafs including musanga silk cotton and umbrella tree

These are the questions that fuck people up yearly

Congratulations on getting this PQ

?

19 A Neem remember Neem is a stimulant

20 B the sensitive plant which is mimosa is a bipinnate leaf

21 C imperata cylindrical

Pigeon pea is *cajanus cajan*

24 B independence plant Also known as
chromolaena odorata

25 B it is cordate or heart shaped

BIOLOGY 2019

1 The sign + is used to indicate an association where an organism gains, while 0 is used where an organism is unaffected. An association indicated as +0 is known as:

- ☐ (a) Predation
- ☐ (b) Commensalism
- ☐ (c) Parasitism
- ☐ (d) Competition

2. The haustorium is a feature found in:

- ☐ (i) Dodder
- ☐ (ii) Cassytha
- ☐ (iii) Mistletoe
- ☐ (iv) Scaldweed
- ☐ (A) I & II only
- ☐ (B) I & III only
- ☐ (C) II and III only
- ☐ (D) All of the above

3. All the following are common to both plant and animal cells except:

- ☐ (a) Nucleus
- ☐ (b) Mitochondrion
- ☐ (c) Chloroplast
- ☐ (d) Cell membrane

4. All the following are groups of plants except:

- ☐ (i) Spermatophyta
- ☐ (ii) Pteridophyta
- ☐ (iii) Coelentrata
- ☐ (iv) Shizophyta
- ☐ (a) I
- ☐ (b) III
- ☐ (c) I and IV
- ☐ (d) None of I-IV

5. The incorrect match for the blood genotype of a proposed couple:

- ☐ (i) AS and AA
- ☐ (ii) AA and AA
- ☐ (iii) AS and AS
- ☐ (iv) SS and SS
- ☐ (a) I and II
- ☐ (b) II only
- ☐ (c) IV only
- ☐ (d) III and IV

6. Which is the most organized protist?

- ☐ (a) Euglena
- ☐ (b) Paramecium
- ☐ (c) Amoeba
- ☐ (d) Plasmodium

7. The function of xylem is to:

- ☐ (a) Conduct food
- ☐ (b) Transport waste product
- ☐ (c) Strengthening stem
- ☐ (d) Transport water

8. Which of these is a disease of the

respiratory tracts?

- ☐ (a) Bronchitis
- ☐ (b) Malaria
- ☐ (c) Cholera

9 Which of the following animals is photosynthetic?

- ☐ (a) Euglena
- ☐ (b) Plasmodium
- ☐ (c) Amoeba
- ☐ (d) Paramecium

10. Typhus is caused by:

- ☐ (i) Rickettsia
- ☐ (ii) Body louse
- ☐ (iii) Rat flea
- ☐ (iv) Aedes mosquito
- ☐ (A) I only
- ☐ (B) IV only
- ☐ (C) II and III
- ☐ (D) I, II and III

11. Wastages in the blood during aerobic respiration are removed by:

- ☐ (a) Kidney
- ☐ (b) Liver
- ☐ (c) Alveoli in the lungs

12. Which of these bacteria is a nitrogen-fixing bacteria?

- ☐ (a) Rhizobium
- ☐ (b) Nitrosomonas
- ☐ (c) Nitrobacter

13. A gene which is homozygous is expressed as:

- ☐ (a) Dominant
- ☐ (b) Recessive
- ☐ (c) Masked
- ☐ (d) Not expressed

14. Which of the following natural resources cannot be conserved?

- ☐ (a) Air
- ☐ (b) Soil
- ☐ (c) Wildlife
- ☐ (d) Forest

15. Meiosis takes place in the:

- ☐ (i) Sperm
- ☐ (ii) Ovary
- ☐ (iii) Testes
- ☐ (iv) Ovum
- ☐ (a) I and II
- ☐ (b) II and III
- ☐ (c) IV and I
- ☐ (d) IV and III

16. Which is not a form of asexual reproduction?

- ☐ (a) Regeneration
- ☐ (b) Fragmentation
- ☐ (c) Fertilization
- ☐ (d) Binary fission

17. Which does not produce glucose exclusively?

- o (a) Cellulose
- o (b) Starch
- o (c) Maltose
- o (d) Sucrose

18. All cell activities are controlled by:

- o (a) Mitochondrion
- o (b) Cytoplasm
- o (c) Nucleus
- o (d) Ribosome

19. Which of these is responsible for the green color of plants?

- o (a) Centrioles
- o (b) Chloroplast
- o (c) Cytoplasm
- o (d) Chlorophyll

20. Urea is formed during:

- o (a) Krebs cycle
- o (b) Citric acid cycle
- o (c) Nitrogen cycle
- o (d) Ornithine cycle

21. Cells in which system have the ability to reproduce:

- o (i) Circulatory System
- o (ii) Transport System
- o (iii) Excretory System
- o (iv) Reproductive System
- o (v) Nervous System
- o (A) IV only
- o (B) II, III and IV

o (C) I, II, III and IV

o (D) All of the above

22. The cell is enclosed by:

- a) Cell wall
- o (b) Cell membrane
- o (c) Nuclear membrane

23. The enzyme in the stomach converts:

- o (a) Carbohydrates to starch
- o (b) Starch to sugar
- o (d) Fat to glycerol
- o (d) Protein to peptones

24. A plant is green because:

- o (a) It absorbs all white light and reflects green light
- o (b) It reflects all white light and absorbs green
- o (c) It reflects blue and red
- o (d) It reflects white and red

25. Which of the following plants has a stem that runs horizontally along the ground?

- o (a) Runners
- o (b) Rhizomes
- o (c) Corms
- o (d) Tubers

26. The enzyme that curdles milk is:

- o (a) Pepsin
- O (b) Ptyalin

☐ (c) Rennin

☐ (d) Maltase

27. Compound eyes are found in:

☐ (a) Ant

☐ (b) Snail

☐ (c) Rat

☐ (d) Hydra

28. The spectrometer is used to measure:

☐ (a) Sunlight intensity

☐ (b) Light spectrum

arrangement

☐ (c) Turgidity

☐ (d) Tidal strength

29. An example of an acellular organism is:

☐ (i) Chlamydomonas

☐ (ii) Plasmodium

☐ (iii) Diatoms

☐ (iv) Paramecium

☐ (A) III only

☐ (B) II and III only

☐ (C) I, III and IV only

☐ (D) All of the above

30. Which of the following is not a biomass of an ecosystem?

☐ (a) Food chain

☐ (b) Producers

☐ (c) Consumers

☐ (d) Saprophyte

☐ (e) Microorganisms

31. State of matter, molecular size, difference in concentration, and temperature are factors influencing diffusion. Which of the following is correct?

☐ (A) II only

☐ (B) II and IV only

☐ (C) All of the above

☐ (D) None of the above

32. The innermost membrane of the meninges is the:

☐ (A) Pia

☐ (B) Dura

☐ (C) Arachnoid

☐ (D) Vagus

33. The causative organism of the following can be observed in a culture medium:

☐ (i) Influenza

☐ (ii) Herpes

☐ (iii) Rabies

☐ (iv) Ebola

☐ (v) Mumps

☐ (A) II only

☐ (B) II and V only

☐ (C) All of the above

☐ (D) None of the above

34. An example of a purine is:

- o (A) Cytosine
- B) Guanine
- o (C) Uracil
- o (D) Thymine

35. The animal that will likely possess a four-chambered heart is:

- o (A) Bats
- O (B) Ducks
- o (C) Swans
- o (D) Sea gulls

36. A plant vacuole contains:

- o (A) Dissolved chemicals, salts, sugars, pigment, and crystals
- o (B) Waste products of metabolism, DNA, RNA, and crystals
- o (C) Pigments, crystals, DNA, and waste products of metabolism
- o (D) RNA, DNA, sugars, and salts

37. Sexual reproduction in Spirogyra is called:

- o (A) Conjugation
- o (B) Isogamy
- o (C) Hologamy
- o (D) Vegetative reproduction

38. In lower plants like mosses, the structure that performs the functions of roots of higher plants is called:

- o (A) Root hair
- o (B) Rootlets
- o (C) Foliage
- o (D) Rhizoids

39. The odd member in the group:

- o (A) Cholera
- o (B) Typhoid
- o (C) Malaria
- o (D) Diphtheria

40. The following gases are released during decay:

- o (i) Oxygen
- o (ii) Hydrogen Sulphide
- o (iii) Ammonia
- o (iv) Carbon dioxide
- o (A) I and II only
- o (B) II and III only
- o (C) II, III, and IV
- o (D) All of the above

ANSWERS

- 1B
- 2d
- 3c
- 4b
- 5d

6b	35a
	7d
	36a
8a	37a
9a	38d
10a	39c
11c	40c
12a	
13b	
14a	
15b	
16c	
17d	
18c	
19d	
20d	
21a	
22b	
23d	
24a	
25a	
26c	
27a	
28b	
29 none	
30a	
31c	
32a	
33D	
34b	

BIOLOGY 2021

1 Which of the following is used for lumbering? I. Fir

II. Ginkgos

III. Pines

IV. Cycads

V. Gnetum

(a) I & III

(b) I, II & IV

(c) 1, II, III & IV

(d) All of the above

2. Which of the following has flagella?

(a) Euglena

(b) Paramecium

(C) Spirogyra

(D) Seaweed

3. Pepsin acts on: (a) Carbohydrates

(b) Starch

(c) Fats

(d) Proteins

4. Grasses are pollinated by: I. Wind

II. Insect

III. Man

IV. Animals

(a) I only

(b) 1 & II

(c) 1, II & III

(d) All of the above

5. Which is peculiar to birds? (a)

Feathers

(b) Scales

(c) Hair

(d) Lungs

6. Which of the following is found in breathing roots? I. Stomnata

II. Lenticel

III. Cuticle

IV. Mitochondria

(a) I only

(b) II only

(c) II & III

(d) All of the above

7. Which of the following is not an adaptation of desert plants? I.

Sunken stomata

II. Tiny leaves

III. Thick cuticle

IV. Big and broad leaves

(a) 1

(b) II

(c) I, II and III only

(d) IV

8. Which of the following is not concerned with excretion?

(a) Kidney

(b) Prostate gland

(c) Ureter

(d) Urethra

9 Which of these is a disease of the respiratory tract? (a) Bronchitis

(b) Malaria

(c) Cholera

10. Which of the following is a producer?

I. Spyrogyra

II. Paramecium

III. Mucor

IV. Amoeba

(a) I

(b) II & III

(c) I & IV

(d) None of the above

11. Possession of pinna is a characteristic feature of: (a)

Mammals

(b) Reptiles

(c) Fishes

(d) Amphibians

(e) All of the above

12. A type of biological interaction between two species where one species is harmed or inhibited by the other, while the other species is not affected and does not gain from the act is known as: I. Parasitism

II. Commensalism

III. Ammensalism

IV. Symbiosis

(a) I, II, IV

(b) 1, 1I

(c) II only

(d) III only

(e) None of the options

13. What is the due order of observing the following scientific methods? I.

Measurement

II. Observation

III. Identification

IV. Inference

(a) III, II, I, IV

(b) I, II, III, IV

(c) II, IV, I, III

(d) IV, III, II, I

14. Auxins are produced in the

A Petiole of leaves

B Apical regions of roots and shoots

C Epidermis of roots and shoots

D Parenchyma of roots and shoots

15 The pituitary gland is called the master gland because

A it is located in the brain

B its secretions are more numerous than those of any glands

C its secretions control other glands

D it's regulated muscular movement

16 which of the following organs is responsible for controlling the body temperature regulation and water balance in mammals

- A kidney b hypothalamus c parathyroid
- d adrenal

17 the bicuspid valve is located between the .
A left auricle and left ventricle b aorta and the left ventricle c superior vena cava and the inferior vena cava d right auricle and left ventricle

18 Deamination is the process during which

- A excess amino acid is broken down into ammonia and keto acid
- B fats and oils are converted into an emulsion for absorption
- C excess amino acid is converted into proteins
- D excess glucose is converted and stored as glycogen in the liver

19 if the phloem of a healthy plant is killed by heat treatment, the

- A upward movement of salts will cease
- B food manufacture in the leaves will accumulate
- C whole plant will die immediately
- D leaves of the plant will become yellow

20 The main waste products formed in plants are

- A alkaloids ,tannins,resins
- B water , alkaloid, carbon dioxide
- C oxygen, CO₂, and alkaloids
- D water ,CO₂ and oxygen

21 a secchi disc is used in the determination of

- A tides b waves c current d turbidity

22 osmic acid boiled with a solution of food substance gave a black precipitate.

This indicates the presence of

- A proteins b fat and oils c amino acid d starch

23 Which of the following is not an evidence of evolution

- A fossil b anatomy c embryology d behaviour

24 species can be defined as the group of organism that

- A resemble each other and live in the same habitat
- B are of a common origin and are always found together
- C resemble each other and can interbreed freely
- D resemble each other and occupy the same niche

25 in aerobic respiration, oxidative phosphorylation takes place in

- A cytoplasm b mitochondria c lysosome d ribosome

Answer

1a

2a

3d

4a

5a

6b

7d

8b

9a

10a

11a

12d

13a

14b

15c

16b

17a

18a

19d

20d

21d

22b

23d

24c

25b

BIOLOGY 2022

1 The membrane covering the vacuole

is: (a) Cell wall

(b) Cell membrane

(c) Tonoplast

(d) Ribosome

2. The animal granules in plants are: I.

Starch grain

II. Mitochondria

III. Ribosome

IV. Glycogen

(a) I only

(b) II only

(c) III only

(d) IV only

3. Which of the following is or is not a

cryptogram? I. Schizophyta

II. Spermatophyta

III. Thallophyta

IV. Bryophyta

V. Pteridophyta

(a) IV

(b) II only

(c) IV and V

(d) III, IV, and V

4. Which of the following are involved

in Xerarch succession? I. Lichen

II. Mosses

III. Algae

IV. Fungi

V. Phytoplankton

(a) I and II

(b) I only

(c) I-V

(d) IV only

5. The international organization that

deals with children's welfare is: (a)

United Nations Children's

Emergency Fund (UNICEF); New

York in America

(b) United Nations Children's

Emergency Fund (UNICEF); Geneva

in Switzerland

(c) United Nations Children's

Emergency Fund (UNICEF); Addis

Ababa in Ethiopia

(d) United Nations Children's

Emergency Fund (UNICEF); New

Delhi in India

6. Which of the following is the function

of Sclerenchyma? I. Storage of

water

II. Storage of food

III. Transport

IV. Straightening and rigidity

(a) I and II

(b) III only

(c) IV only

(d) II only

7. Which of the following are simple tissues except? I. Sclerenchyma

II. Xylem

III. Parenchyma

IV. Physiochyma

V. Fiber

(a) II and IV

(b) IV only

(c) V only

(d) IV and V

8. Which of these is an effect of deforestation? I. Soil erosion

II. Biodiversity

III. Soil pollution

IV. Flooding

V. Global warming

(a) I-III

(b) I-V

(c) I, III and IV

(d) I, III, IV, and V

9. Which of the following is a vascular tissue? I. Xylem

II. Parenchyma

III. Sclerenchyma

IV. Collenchyma

(a) I only

(b) I-IV

(c) II, III and IV

(d) II-IV

10. Which of the following has the highest energy level? (a) Producer

(b) Primary consumer (c) Secondary consumer

(d) Decomposer

11. Which of the following stores food in its root? I. Cassava

II. Irish potato

III. Yam

IV. Cocoa yam

(a) I and II

(b) I and III

(c) II and III

(d) II and IV

12.

13. Which of the following devices is used for measuring the nutrient

content of a biscuit? I. Bomb calorimeter

II. Food calorimeter

III. Food meter

IV. Hydrometer

(a) I only

(b) II only

(c) III only

(d) IV only

14. Which of the following does not belong to the main group of the plant

kingdom? I. Bryophyte

II. Chlorophyte

III. Pteridophyte

IV. Spermatophytes

V. Thallophyte

(a) I-V

(b) I-IV

(c) II only

(d) III only

15. The class of plants that does not have membrane-bound organelles

is: (a) Fungi and Bacteria

(b) Archaea and Bacteria

(c) Protist and Archaea

(d) Fungi and Protist

16. The plasma membrane and cell membrane do not allow water to pass through them because of:

I. Diffusion

II. Osmosis

III. Selectively permeable membrane

IV. Permeable membrane

(a) I-IV

(b) I and II

(c) III only

(d) III and IV

17. In a cross between a tall variety of maize (Tt) with a short variety (tt), the ratio of tall plants in the offspring

will be: (a) 2:1

(b) 1:2

(c) 1:1

(d) 1:3

18. The temperature of a mesophilic organism is: (a) 20 - 45 degrees Celsius

(b) 45 - 70 degrees Celsius

(c) 78 - 90 degrees Celsius

(d) 90 - 105 degrees Celsius

19. A plumule can best be described as:

I. Part of a cotyledon

II. Part of the endosperm

III. A miniature shoot

IV. Structure near the micropyle

(a) I only

(b) II only

(c) III only

(d) III and IV

20. Which has an emarginate apex

a) Bauhinia

(b) Ixora

(c) Terminalia

(d) Pistia

21. Which of the following possess a palmate leaf? (a) Bauhinia

(b) Ixora

(c) Terminalia

(d) Pistia

22. The international organization that deals with the conservation of natural resources is: (a) International Union for Conservation of Natural Resources

(b) International Union for Conservation of Nature

(c) Economic Community for Conservation of Natural Resources

(d) Economic Community for Conservation of Nature

23. Arrange the following in sequence of succession in a climax community: I. Trees

II. Herbaceous plants and grasses

III. Lichen

IV. Bare rocks

V. Shrubs

(a) IV, III, II, V, I

(b) IV, III, II, I, V

(c) IV, V, III, II, I

(d) IV, II, III, V, I

24. Arrange the following in order of preference: I. Disposal of waste in an environmentally friendly manner

II. Waste prevention

III. Waste reduction

IV. Waste recycling

V. Incineration

(a) III, IV, V, II, I

(b) II, III, IV, I, V

(c) II, III, IV, V, I

(d) III, II, IV, I, V

25. Arrange the following from the least to the greatest: I. Vulnerable

II. Endangered

III. Critically endangered

IV. Extinction

V. Least concerned

(a) V, I, II, III, IV

(b) IV, V, III, II, I

(c) III, IV, V, I, II

(d) V, I, II, IV, III

26. Millipedes and earthworms are: (a)

Decomposers

(b) Heterotrophs

(c) Detritivores

(d) Consumers

27. Which of these plants have an emerginate apex? (a) Bauhinia

(b) Ixora

(c) Terminalia

(d) Pistia

28. Which of these plants exhibit aerial roots?

I. Dahlia

II. Ficus

III. Centrosema

IV. Antigonon

V. Lantana

(a) I only	15b
(b) II only	16c
(c) III only 17b	
	(d)
	IV
	and
	V
	18a
29. The living part of phloem tissue is:	19c
I Sieve plate	
	20a
II. Sieve element	
	III.
	Cyt
	opl
	as
	m
	21a
	IV.
	Co
	mp
	ani
	on
	cell
	22
	b
(a) I and II 23a	
	(b)
	III
	onl
	y
	24
	b
(c) IV only	25a
(d) I only	26c
	27a
	AN
	SW
	ER
	28
	b
1c	29c
2d	
3b	

4a

5a

6c

7a

o

8b

9a

10a

11 B

12bonus

13a

14c

BIOLOGY 2024

1. Density-dependent factors that affect the population of an area include?

- I. Corona virus
- II. Food shortage
- III. Global warming
- IV. Predation
- V. Overcrowding

- A. I, II, III and IV
- B. I, II, IV and V
- C. I, III, IV and V
- D. II, III, IV and V

2.The types of plants that exhibit naked seeds are__

- A. Grasses
- B. Conifers
- C. Legumes
- D. Palms

3. Which of these is an eye defect?

- I long sightedness
- II short sightedness
- III Glaucoma
- IV Cataract

- V River blindness

- A.1, II, III and IV
- B. I, II, IV and V
- C. I, II, III, IV and V
- D. II, III, IV and V

4. When a yam tendril grows on another plant what type of tropism does it exhibit

- A. Phototropism
- B. Thigmotropism
- C. Thermotropism
- D. Halotropism
- E. None of the above

5. What is part of the living component of the cell

- I. Organelles
- II. Protoplasm
- III. Cytoplasm
- IV. Plasma membrane
- V. Chromatin

- A. I, II, III and IV
- B. I, II, IV and V
- C. I, II, III, IV and V
- D. II, III, IV and V

6. Which of the following have emerginate apex

- A. Bauhinia
- B.Ixora
- C.Allamenda

7. Difference in the characteristics observed in individual plant of the same species is called

- I. Traits
- II. Phenotype
- III. Variation
- IV. Mutation
- A. I and II
- B. I and IV
- C. III and IV
- D. III only

8. Factors that affect organisms in terrestrial habitat

- I. Temperature
- II. Rainfall
- III. Relative humidity
- IV. Ph
- V Turbidity

- A. I-v
- B. I II and III
- C. III III IV
- D. II III IV
- E. None of the above

9. Which of the following is not an evidence of evolution

- I. Fossil record
- II. Comparative anatomy
- III. Comparative embryology

IV Human behavior

- A. I and IV
- B. III and IV
- C. I and III
- D. None of the above

10. Organisms compete for

- I. Space
- II. Mate
- III. Light
- IV. Food
- A. A. I, II, III and IV
- B. I, II, IV and V
- C. I, II, III, IV and V
- D. II, III, IV and V

11. Consequences of over crowding

- I. Inbreeding
- II. Epidemic
- III. Competition
- IV. Cannibalism
- V Crime

- A. I, II, III and IV
- B. I, II, IV and V
- C. I, II, III, IV and V
- D. II, III, IV and V

12..... Is/are not adaptation of xerophyte plant

I.sunken stomata

II.Tiny leaves

III.thick cuticles

IV.Big and broad leaves

A.I only

B.ii only

C.i,II,and IV only

D.IV only

13. Which of the following is CO₂ evolved to the atmosphere

A. Photosynthesis

B. Respiration

C. Decomposition

D. Sequestration

14. Which of the following is CO₂ absorbed from the atmosphere

A. Photosynthesis

B. Respiration

C. Decomposition

D. None of the above

15.What's the interaction between biotic and abiotic factors

A. Community

B. Ecosystem

C. Biome

D. Population

16. The theory of natural selection was developed by

A. Lamarck and Darwin

B. Darwin and Wallace

C. Darwin and Mendel

D. Mendel and Lamarck

17. Which of the following ensures chromosome is constant from generation to generation

I Mitosis

II Meiosis

III Fusion

IV Fission

A.I

B.II

C II and III

D.I and II

18. The plant found in areas where there is shortage of water is?

A.Halophyte

B. Mesophyte

C. Hydrophyte

D. Xerophyte

19. Which is a biotic factor

A.bacteria

C.Rainfall

B.temperature 24 which of the following indicates an association in which at least one organism benefits

D.oxygen I predation ii commensalism iii parasitism V competition ammensalism Iv

20. Onion bulb stores food where?

A ii,i ,V B I,ii,IV c ii,iii,IV d li,IV,V

A. Root

B.Stem

C. Hyphae 25 Differences in characteristics

D. Fleshy leaf A traits b variation c mutation d phenotype

I) blood 1b

21.Which of the following are tissues

Answers

II)

ly

m

ph

2b

III)

m

us

cle

3c

IV)

bo

ne

12a s

4b

V)

ski

n

5c

A.

I,

II,

III

an

d

IV

6a

B.I,

II,

IV
an
d V
7d
C.I,
III,
IV,
an
d V
8c
D.
II,
III,
IV
an
d V
9d
10c

11c

22 Which of the following undergoes Fehling test

A starch B protein c glucose d sucrose

13b

14a

23 Which of the following exhibit aerial roots

15b

A fiscus B Centrosema c Lantanna d Dahlia

16b

17c

18d

19a

20d

21c

22c

23a

24 B

25d

UI BIOLOGY 2024

1. Density-dependent factors that affect the population of an area include?

- I. Corona virus
 - II. Food shortage
 - III. Global warming
 - IV. Predation
 - V. Overcrowding
- A. I, II, III and IV
- B. I, II, IV and V
- C. I, III, IV and V
- D. II, III, IV and V

Answer;

B -I,II, IV and V

Density-dependent factors are influences on population growth that change as the population density changes. These factors typically have a greater impact as the population size increases. Let's analyze each option:

1. *Corona virus*: This is a density-dependent factor because the spread of infectious diseases like COVID-19 is more likely in densely populated areas where individuals are in close contact with each other.

II. *Food shortage*: This is also a density-dependent factor. As the population grows, the demand for food increases, which can lead to shortages if the food supply does not keep pace with the population growth.

III. *Global warming*: This is generally considered a density-independent factor because it affects populations regardless of their density. Global warming is a large-scale environmental change that impacts all populations, not just those that are densely populated.

IV. *Predation*: This is a density-dependent factor. Predators are more likely to find and prey on individuals in a population as the population density increases.

V. *Overcrowding*: This is a density-dependent factor. Overcrowding can lead to increased competition for resources, stress, and the spread of diseases, all of which can negatively impact population growth.

Summary: The density-dependent factors that affect the population of an area include:

- I. Corona virus
- II. Food shortage

- IV. Predation
- V. Overcrowding

2. The types of plants that exhibit naked seeds are

- A. Grasses
- B. Conifers
- C. Legumes
- D. Palms

Answer;

B. Conifers

Explanation;

Conifers are gymnosperms, a group of plants that produce seeds that are not enclosed within a fruit. Instead, their seeds are "naked" and typically found on the scales of cones. Examples of conifers include pines, spruces, firs, and cedars.

The other options (grasses, legumes, and palms) are angiosperms, which produce seeds that are enclosed within a fruit. Therefore, they do not exhibit naked seeds.

3. Which of these is an eye defect?

I long sightedness

II short sightedness

III Glaucoma

IV Cataract

V River blindness

A. I, II, III and IV

B. I, II, IV and V

C. I, II, III, IV and V

D. II, III, IV and V

Answer;

C-I, II, III, IV and V

All the options listed are eye defects or conditions that affect vision. Here's a breakdown of each:

1. *Long-sightedness (Hypermetropia)*: This is a refractive error where distant objects are seen more clearly than nearby objects. It occurs when the eyeball is too short or the cornea has too little curvature, causing light to focus behind the retina instead of on it.
- II. *Short-sightedness (Myopia)*: This is another refractive error where nearby objects are seen more clearly than distant objects. It occurs when the eyeball is too long or the cornea is too curved, causing light to focus in front of the retina instead of on it.
- III. *Glaucoma*: This is a group of eye conditions that damage the optic nerve, often due to increased

pressure in the eye. It can lead to vision loss and blindness if not treated.

IV. ***Cataract***: This is a condition where the lens of the eye becomes cloudy, leading to blurred vision. It is often age-related but can also result from injury, disease, or congenital factors.

V. ***River blindness (Onchocerciasis)***: This is a parasitic infection caused by the worm ***Onchocerca volvulus***, transmitted through the bites of black flies. It can cause severe itching, skin changes, and vision loss, including blindness.

4. When a yam tendril grows on another plant what type of tropism does it exhibit

- A. Phototropism
- B. Thigmotropism
- C. Thermotropism
- D. Halotropism
- E. None of the above

Answer;

B. Thigmotropism

Explanation;

Thigmotropism is a type of tropism where a plant responds to touch or physical contact with a solid object. Yam tendrils are specialized structures that coil around other plants or

objects for support as they grow. This response to touch helps the plant climb and stabilize itself.

The other options are not correct in this context:

- ***A. Phototropism***: Growth in response to light.
- ***C. Thermotropism***: Growth in response to temperature changes.
- ***D. Halotropism***: Growth in response to salt concentrations.

5. What is part of the living component of the cell

- I. Organelles
 - II. Protoplasm
 - III. Cytoplasm
 - IV. Plasma membrane
 - V. Chromatin
- A. I, II, III and IV
 - B. I, II, IV and V
 - C. I, II, III, IV and V
 - D. II, III, IV and V

Answer;

C - I, II, III, IV and V

The living components of a cell are the structures and substances that are actively involved in the cell's metabolic processes and maintenance of life. Let's analyze each option:

1. ***Organelles***: Organelles are specialized structures within the cell that perform distinct processes necessary for cellular function. Examples include the nucleus, mitochondria, ribosomes, and endoplasmic reticulum. These are part of the living component of the cell.

II. ***Protoplasm***: Protoplasm refers to the living content of a cell that is surrounded by the plasma membrane. It includes the cytoplasm and the nucleus. Protoplasm is the living substance of the cell.

III. ***Cytoplasm***: The cytoplasm is the gel-like substance within the cell membrane that contains the organelles and other cellular components. It is part of the living component of the cell.

IV. ***Plasma membrane***: The plasma membrane (or cell membrane) is a biological membrane that separates the interior of the cell from the outside environment. It regulates the movement of substances in and out of the cell and is essential for maintaining the cell's integrity and function. It is part of the living component of the cell.

V. ***Chromatin***: Chromatin is the complex of DNA and proteins found within the nucleus of eukaryotic cells. It is involved in the regulation of gene expression and the packaging of DNA. Chromatin is part of the living component of the cell.

All are part of the living component of the cell

6. Which of the following have emarginate apex

A. Bauhinia

B. Ixora

C. Allamanda

Answer;

A. Bauhinia

Explanation;

The term "emarginate apex" refers to a leaf tip that has a slight notch or indentation at its apex. Among the options provided:

A. ***Bauhinia***: Bauhinia leaves are known for their distinctive shape, often bilobed (divided into two lobes) with a notch at the apex, which can be considered emarginate.

B. ***Ixora***: Ixora leaves typically have an acute or pointed apex and are not generally described as having an emarginate apex.

C. ***Allamanda***: Allamanda leaves usually have an acuminate or pointed apex and are not

typically described as having an emarginate apex.

7. Difference in the characteristics observed in individual plant of the same species is called

I.Traits

II. Phenotype

III. Variation

IV. Mutation

A.I and II

B.I and IV

C.III and IV

D.III only

Answer;

D-III only

Explanation;

Variation: Variation refers to the differences in traits or phenotypes among individuals within a species. This variation can be due to genetic differences, environmental factors, or a combination of both.

Other options are peculiar to each organism and not a measure of observable difference between organisms in a specie.

8. Factors that affect organisms in terrestrial habitat

I. Temperature

II. Rainfall

III. Relative humidity

IV.Ph

V. Turbidity

A.I-v

B.III and III

C.IIIIIIIV

D.II III IV

E. None of the above

Answer;

C. I, II, III & IV

Explanation;

The factors that affect organisms in a terrestrial habitat are primarily related to the physical and climatic conditions of the land environment.Let's analyze each option:

1. *Temperature*: Temperature is a critical factor in terrestrial habitats as it influences the metabolic rates of organisms, their behavior, and their ability to survive and reproduce.

II. *Rainfall*: Rainfall is essential for providing water, which is crucial for the survival of all terrestrial organisms. It affects plant growth,availability of drinking water, and the overall ecosystem dynamics.

V. Human behavior

III. *Relative humidity*: Relative humidity affects the water balance in organisms, particularly in plants A.I and IV and animals that rely on specific moisture levels for physiological processes.

B. III and IV

D. None of the above

C.I and III

IV. *pH*: pH levels can affect soil quality and the availability of Answer; nutrients, which in turn affects plant growth and the organisms that depend on those plants.

*D - None of the above, IV only is the correct explanation;

The evidence of evolution is supported by various scientific observations and studies. Let's analyze each option:

V. *Turbidity*: Turbidity refers to the cloudiness or haziness of a fluid caused by suspended particles. This factor is more relevant to aquatic habitats rather than terrestrial ones.

1. *Fossil record*: The fossil record provides a historical sequence of life on Earth, showing the changes in species over time. It is a key piece of evidence for evolution.

Summary: The factors that affect organisms in terrestrial habitats are:

- I. Temperature

- IV. pH

Turbidity (V) is not a relevant factor for terrestrial habitats.

- III. Relative humidity
II. *Comparative anatomy*
*: Comparative anatomy involves studying the similarities and differences in the structures of different species. Homologous structures (structures with similar origins but different functions) provide strong evidence for common ancestry and evolution.

9. Which of the following is not an evidence

I. Fossil record of evolution

II. Comparative anatomy

III. Comparative embryology

III. *Comparative embryology*: Comparative embryology examines the similarities and differences in the embryonic development of different species. Similarities in early developmental stages suggest common ancestry and support the theory of evolution.

IV. *Human behavior*: While human behavior can be influenced by evolutionary processes, it is not considered direct evidence of evolution. Behavior is complex and influenced by a combination of genetic, environmental, and cultural factors.

10. Organisms compete for

I. Space

II. Mate

III. Light

IV. Food

A. A, I, II, III and IV

B. I, II, IV and V

C. I, II, III, IV and V

D. II, III, IV and V

Answer;

C -I,II, III & IV

Explanation;

Organisms compete for various resources that are essential for their survival, reproduction, and overall fitness. Let's analyze each option:

1. *Space*: Organisms often compete for space, whether it is for territory, nesting sites, or areas with favorable conditions. Space can be a limiting resource, especially in densely populated environments.

II. *Mate*: Competition for mates is a common aspect of sexual reproduction. Organisms may compete to attract and secure mates to ensure the continuation of their genetic lineage.

III. *Light*: In plants, competition for light is crucial as it is necessary for photosynthesis. Plants may grow taller or spread their leaves to capture more light, especially in densely vegetated areas.

IV. *Food*: Food is a fundamental resource for all organisms. Competition for food can occur within and between species, influencing survival, growth, and reproductive success.

11. Consequences of over crowding I.

Inbreeding

II. Epidemic

III. Competition

IV. Cannibalism

V. Crime

makes it easier for pathogens to transmit.

A. I, II, III and IV

B. I, II, IV and V

C. I, II, III, IV and V

D. II, III, IV and V

Answer;

C -I, II, III, IV& V

Explanation;

Overcrowding can lead to various negative consequences due to increased population density and competition for limited resources. Let's analyze each option:

1. ***Inbreeding***: Overcrowding can limit the availability of mates, potentially leading to inbreeding, which can result in a higher incidence of genetic disorders and reduced genetic diversity.

II. ***Epidemic***: High population density facilitates the spread of infectious diseases, increasing the likelihood of epidemics. Close contact among individuals

III. ***Competition***: Overcrowding intensifies competition for essential resources such as food, water, space, and mates. This can lead to stress, reduced reproductive success, and increased mortality rates.

IV. ***Cannibalism***: In some species, overcrowding can lead to cannibalism as individuals compete for limited resources. This behavior can help regulate population size but has negative effects on the population.

V. ***Crime***: In human populations, overcrowding can lead to increased crime rates due to competition for resources, stress, and reduced living conditions. Social tensions and conflicts may rise in densely populated areas.

VI.

All the listed options are potential consequences of overcrowding.

12..... Is/are not adaptation of xerophyte plant

I. sunken stomata

II. Tiny leaves

III. thick cuticles

IV. Big and broad leaves

- A.I only
- B.ii only
- C.i,li,and IV only
- D.iv only

Answer;
D.IV only

Explanation;

Xerophytes are plants adapted to survive in environments with limited water availability,such as deserts. They have specific adaptations to minimize water loss and maximize water storage. Let's analyze each option:

1. ***Sunken stomata***: Sunken stomata are an adaptation of xerophytes.They help reduce water loss by decreasing the exposure of stomata to air currents and reducing the rate of transpiration.

II. ***Tiny leaves***: Tiny leaves are an adaptation of xerophytes. Smaller leaves reduce the surface area for transpiration, helping to conserve water.

III. ***Thick cuticles***: Thick cuticles are an adaptation of xerophytes. A thick, waxy cuticle on the leaf surface helps to reduce wwater loss through evaporation.

IV. ***Big and broad leaves***: Big and broad leaves are notan adaptation of xerophytes. Large leaves increase the surface area for transpiration,leading to greaterwater is disadvantageous in arid loSS which environments.

13. Which of the following is CO2 evolved to the atmosphere
- A. Photosynthesis
 - B. Respiration
 - C. Decomposition
 - D. Sequestration

Answer;
B. Respiration

Explanation;

- ***Respiration***: During cellular respiration,organisms (including plants, animals, and microorganisms) release CO2 as a byproduct of breaking down glucose to produce energy.
- ***Decomposition ©***: When organic matter decomposes, microorganisms break it down,releasing carbon compounds (not necessarily CO2).
- ***Photosynthesis***: This process absorbs CO2from the atmosphere and converts it into oxygen and glucose, so it does not release CO2.
- ***Sequestration***: This refers to the capture and storage of compounds/elements e.g

carbon, preventing it from entering the atmosphere.

14. Which of the following is CO₂ absorbed from the atmosphere

- A. Photosynthesis
- B. Respiration
- C. Decomposition
- D. None of the above

Answer;

***A. Photosynthesis*.**

Explanation:

- ***Photosynthesis***: During photosynthesis, plants, algae, and some bacteria absorb CO₂ from the atmosphere and convert it into glucose and oxygen using sunlight. This process removes CO₂ from the atmosphere.
- ***Respiration***: Releases CO₂ into the atmosphere as a byproduct of energy production.
- ***Decomposition***: Releases CO₂ into the atmosphere as organic matter breaks down.

Thus, the only process listed that absorbs CO₂ from the atmosphere is ***photosynthesis***.

15. What's the interaction between biotic and abiotic factors

- A. Community
- B. Ecosystem
- C. Biome

D. Population

The correct answer is;

***B. Ecosystem*.**

Explanation:

- ***Ecosystem***: An ecosystem refers to the interaction between ***biotic factors*** (living organisms, such as plants, animals, and microorganisms) and ***abiotic factors*** (non-living components, such as water, air, soil, and sunlight). These interactions create a functional unit where energy and nutrients are exchanged.
- ***Community***: A community refers only to the biotic factors (all the living organisms in a particular area) and does not include abiotic factors.
- ***Biome***: A biome is a large geographical area characterized by specific climate conditions and dominant vegetation, but it does not specifically describe the interaction between biotic and abiotic factors.
- ***Population***: A population refers to a group of individuals of the same species living in a specific area, and it does not involve abiotic factors.

Thus, the interaction between biotic and abiotic factors is best described by ***Ecosystem***.

16. The theory of natural selection was developed by

- A. Lamarck and Darwin
- B. Darwin and Wallace
- C. Darwin and Mendel

D. Mendel andLamarck

Answer:

B. Darwin and Wallace.

Explanation:

- The theory of natural selection was independently developed by *Charles Darwin*and *Alfred Russel Wallace* in the 19th century. Both scientists contributed to the understanding of how species evolve over time through the process of natural selection.
- *Lamarck*: Jean-Baptiste Lamarck proposed an earlier theory of evolution based on the inheritance of acquired characteristics, which is not the same as natural selection.
- *Mendel*: Gregor Mendel is known as the father of modern genetics for his work on inheritance in pea plants, but he did not contribute to the theory of natural selection.

17. Which of the following ensures chromosome is constant from generation to generation

- I Mitosis
- II Meiosis
- III Fusion
- IV Fission

- A.1
- B.II
- C II and III
- D.I and II

Answer:

C. II and III

Explanation:

- *II. Meiosis*: This process reduces the chromosome number by half, producing haploid gametes (sperm and egg). When these gametes fuse during fertilization, the diploid chromosome number is restored, ensuring the chromosome number remains constant from generation to generation.
- *III. Fusion (Fertilization)*: The fusion of haploid gametes during fertilization restores the diploid chromosome number, maintaining chromosomal consistency across generations

Mitosis (I) and fission (IV) do not play a role in maintaining the chromosome number across generations in the context of sexual reproduction. Mitosis is for growth and repair,while fission is a form of asexual reproduction.

18. The plant found in areas where there is shortage of water is?

- AHalophyte
- B.Mesophyte
- C. Hydrophyte
- D. Xerophyte

Answer:

D. Xerophyte

Xerophytes are plants adapted to survive in environments with a scarcity of water, such as deserts or arid regions. They have special adaptations like *thick cuticles, reduced leaf surface area, and deep root systems to minimize water loss and maximize water uptake.* Examples include cacti and succulents.

19. Which is a biotic factor

- A. bacteria
- B. temperature
- C. Rainfall
- D. oxygen

Answer:

A. Bacteria

Biotic factors are living components of an ecosystem that influence other organisms. *Bacteria are living organisms and thus considered a biotic factor.* The other options (temperature, rainfall, and oxygen) are abiotic factors, as they are non-living physical and chemical components of the environment.

20. Onion bulb stores food where?

- A. Root
- B. Stem
- C. Hyphae
- D. Fleshy leaf

Answer:

D. Fleshy leaf

Explanation;

In an onion bulb, food is stored in the fleshy leaves. These leaves are modified to store nutrients, which the plant uses for growth and survival. The bulb itself is a short stem surrounded by thick, fleshy leaves that store the food.

21. Which of the following are tissues

- I) blood
 - II) lymph
 - III) muscle
 - IV) bones
 - V) skin
- A. I, II, III and IV
 - B. I, II, IV and V
 - C. I, III, IV, and V
 - D. II, III, IV and V

The correct answer is:

C - I, III, IV and V

I) Blood, III) Muscle, IV) Bones, and V) Skin

Explanation:- ***Blood***: A connective tissue composed of plasma, red blood cells, white blood cells, and platelets.

- ***Muscle***: A type of tissue responsible for movement (e.g., skeletal, smooth, and cardiac muscle).

- ***Bones***: A type of connective tissue that provides structure and support.

- ***Skin***: An organ made up of multiple tissues, including epithelial tissue (outer layer) and connective tissue (dermis).

Lymph is not a tissue; it is a fluid that circulates in the lymphatic system, carrying immune cells and waste products. However, the lymphatic system contains lymphoid tissues (e.g., lymph nodes).

22. Which of the following grows under the soil
- A Bulb
 - B Runner
 - C Rhizome
 - D. None of the above

Answer:

C. Rhizome

Explanation;

Rhizome: A rhizome is a modified stem that grows horizontally underground. It stores food and can give rise to new shoots and roots. Examples include ginger and turmeric.

Bulb: A bulb is an underground storage organ, but it is not a stem. It

consists of fleshy leaves surrounding a short stem. Examples include onions and garlic.

Runner: A runner (or stolon) is a horizontal stem that grows above the ground and produces new plants at nodes. Examples include strawberries.

So, ***rhizomes*** grow under the soil, while bulbs are underground storage organs, and runners grow above the ground.

23. Which of the following are components of Neurons

- I dendrites
- II Axons
- III meninges
- IV Glial cells
- V synapse

A. I, II, III and IV

B. I, II and V

C. I, II, III, IV and V

D. II, III, IV and V

Answer;

B -I, II and V

1. Dendrites-These are tree-like structures that receive signals from other neurons.

11. Axons - These are long, thread-like parts of a neuron that transmit electrical impulses away from the cell body.

A.I only

V. Synapse - This is the junction between two neurons where communication occurs via neurotransmitters.

III. Meningitis - This is a disease, not a neuron component. It refers to the inflammation of the meninges (protective membranes around the brain and spinal cord).

IV. Glial cells - While important for supporting and protecting neurons, glial cells are not part of a single neuron itself. They play roles in nourishment, insulation, and immune defense in the nervous system.

24. The mutation theory of organic evolution was propounded by

1. Hugo de Vries

II. Jean Lamarck

III. Charles Darwin

- B.II only
- C.III only
- D.I, II and III

The correct answer is:

A -I only. Hugo de Vries

The *Mutation Theory of Organic Evolution* was proposed by *Hugo de Vries* in the early 20th century. He suggested that evolution occurs through sudden, large changes in genetic material (mutations), rather than gradual changes as proposed by Charles Darwin.

- *Jean-Baptiste Lamarck* (II) proposed the theory of inheritance of acquired characteristics, which is different from mutation theory.
- *Charles Darwin* (III) proposed the theory of natural selection, which focuses on gradual changes over time.

So, the correct answer is *I. Hugo de Vries*

25. Which of these are the characteristics of white mangrove

- A. Prop roots
- B. Butress roots
- C. Breathing roots
- D. Stilts roots

Answer:

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C. Breathing roots

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Explanation;

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White mangroves are characterized by *breathing roots* (pneumatophores). These specialized roots grow upward from the soil and help the plant take in oxygen in waterlogged or anaerobic conditions.

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The other options refer to different mangrove species:

- *Prop roots* and *stilt roots* are

characteristic of red mangrove.

- *Buttress roots* are typically found in large tropical trees, not mangroves.

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BIOLOGY 2023

1. The membrane covering the vacuole is_____ (a) Cell wall (b) Cell membrane (c) Tonoplast (d) Ribosome
2. The animal granules in plants is I. Starch grain; II. Mitochondria; III. Ribosome; IV. Glycogen. (a) I only (b) II only (c) III only (d) IV only
3. Which of the following is or is not a cryptogram? I. Schizophyta; II. Spermatophyta; III. Thallophyta; IV. Bryophyta; V. Pteridophyta (a) IV (b) II, I - III only (c) IV and V (d) III, IV and V
4. Which of the following are involved in Xerarch succession? I. Lichen; II. Mosses; III. Algae; IV. Fungi; V. Phytoplankton. (a) I and II (b) I only (c) I-IV (d) IV only
5. International organization that deals with children welfare is and is located at_____ (a) United Nations Children Emergency Funds; New York in America (b) United Nations Children Emergency Fund; Geneva in Switzerland (c) United Nations Children Emergency Funds; Addis Ababa in Ethiopia (d) United Nations Children Emergency Funds; New Delhi in India
6. Which of the following is the function of Sclerenchyma? I. Storage of Water; II. Storage of Food; III. Transport; IV. Straightening and Rigidity. (a) I and II (b) III only (c) IV only (d) II only
7. Which of the following are simple tissue except? I. Sclerenchyma; II. Xylem; III. Parenchyma; IV. Physicocyma; V. Fiber. (a) I and II and IV (b) IV only (c) V only (d) IV and V
8. Which of these is an effect of deforestation? I. Soil erosion; II. Biodiversity; III. Soil pollution; IV. Flooding; V. Global warming. (a) I-III (b) I- V (c) I, III and V (d) I, III, IV and V
9. Which of the following is a vascular tissue. I. Xylem; II. Parenchyma; III. Sclerenchyma; IV. Collenchyma. (a) I only (b) I -IV (c) II, III and IV (d) II-IV
10. Which of the following stores food in its root? I. Cassava; II. Irish Potato; III. Yam; IV. Cocoyam. (a) I and II (b) I and III (c) II and III (d) II and IV
11. Which of the following has the highest in its root? I. Cassava; II. Irish Potato; III. Yam; IV. Cocoyam
12. Which of the following processes are responsible for keeping a constant number of chromosomes? I. Fusion; II. Fission; III. Meiosis; IV. Mitosis; (a) I and II (b) II and III (c) I-V (d) None

Answers

1.C

2. D

3.B

4. A

5. A

6.C

7. A

8.D

9.A

10.B

11..

12.D

BIOLOGY 2019

1. (i) State of matter (ii) Molecular size (iii) Difference in concentration (iv) Temperature. Which of the following factors influences diffusion?
(a) II only (b) II & IV only (c) All of the above (d) None of the above
2. The innermost membrane of meninges is the____(a) Pia (b) Dura (c) Arachnoid (d) Vagus
3. The causative organism of the following can be observed in a culture medium; I. Influenza; II. Herpes; III. Rabies; IV. Ebola; V. Mumps; (a) II only (b) II and V only (c) All of the above (d) None of the above
4. The SPECTROMETER is used to measure____(a) Sunlight intensity (b) Light Spectrum arrangement (c) Turgidity (d) Tidal strength
5. An example of an ACCELULAR organism is, except I. Chlymadomona; II. Plasmodium; III. Diatoms; IV. Paramecium. (a) III only (b) II & III only (c) I, III & IV only (d) All of the above
6. A plant vacuole contains (a) Dissolved chemicals, salts, sugars, pigment and crystals (b) Waste products of metabolism, DNA, RNA and Crystals (c) Pigments, Crystals, DNA and waste products of metabolism (d) RNA, DNA, sugars and salts
7. _____ will likely possess a four chambered heart. (a) Bats (b) Ducks (c) Swans (d) Seagul
8. An example of a purine is (a) Cytosine (b) Guanine (c) Uracil (d) Thymine
9. Which of the following is not a biomass of ecosystem? (a) Foodchain (b) Producers (c) Consumers (d) Saprophyte (e) Microorganisms
10. I. Plasmodium; III. Diatoms; IV. Paramecium. (a) III only (b) II and III only (c) I, III and IV only (d) All of the above
11. The following are gases released during decay____ I. Oxygen; II. Hydrogen Sulphite; III. Ammonia; IV. Carbon dioxide. (a) I only (b) II and III only (c) II, III and IV (d) IV only
12. Which member of ODD? (a) Infective Hepatitis (b) Rabies (c) Poliomyelitis (d) Diphtheria
13. Sexual Reproduction in spirogyra is called____ (a) Fertilization (b) Symbiosis (c) Conjugation (d) Binary Fission
14. In lower plants like mosses, the structure which performs the functions of roots of higher plants is called____ (a) Root hairs (b) Rootlets (c) Hyphae (d) Rhizoids (e) Thalli
15. Plant is green because____ (a) It absorbs all white light and reflect

green light (b) It reflects all white light and absorb green (c) It reflects blue and red (d) It reflects white and red

16. Which of the following plants its stem runs horizontally along the ground? (a) Runners (b) Rhizomes (c) Corms (d) Tubers

17. Enzymes that curdles milk. (a) Pepsin (b) Ptyalin (c) Rennin (d) Maltase

18. Compound eye is found in (a) Ant (b) Snail (c) Rat (d) Hydra

19. The living part of phloem tissue is I. Sieve plate; II. Sieve element; III. Cytoplasm; IV. Companion Cell. (a) I and II only (b) III only (c) IV only (d) I only

20. Arrange the following from the least to the greatest. I. Vulnerable; II. Endangered; III. Critically endangered; IV. Extinction; V. Least concerned (a) V, I, II, III, IV (b) IV, V, III, II, I (c) III, IV, V, I, II (d) V, I, II, IV, III

21. Millipede and earth worms are (a) Decomposers (b) Heterotrophs (c) Detritivores (d) Consumers

22. Which of these plants has an emarginated apex? I. Bauhinia; II. Ixora; III. Terminalia; V. Pistia. (a) I only (b) II only (c) III only (d) IV only

23. Which of the following are Cycadalis? I. Larix; II. Pinus; III.

None of the options (d) E

24. Which of the following leaf? (a) Buckeye (b) All A-C

25. In a cross between a (Tt) with a short variety (tt) the ratio of Tall to Short the offspring will be (a) 1:2 (b) 1:1 (c) 1:3

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III

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20.A
21.C

Answers

- 1.C
- 2.A
- 3.D
- 4.B
- 5.D
- 6.A
- 7.A
- 8.B
- 9.A
- 10..
- 11.C
- 12.D
- 13.C
- 14.D
- 15.A
- 16.A
- 17.C
- 18.A
- 19.C

22.A
23.C
24.B
25.C

BIOLOGY 2021

1. The sign + is used to indicate an association where an organism gains, while 0 is used where an organism is unaffected. An association indicated as +0 is known as
(a) Predation (b) Commensalism
(c) Parasitism (d) Competition
2. The haustorium will be a feature found in
1. Dodder; II. Cassytha; III. Mistletoe IV. Scaldweed (a) I & II only (b) I & III only (c) II and III (d) All of the above
3. All the following are common to both plant and animal cell except
(a) Nucleus (b) Mitochondrion
(c) Chloroplast (d) Cell membrane
4. All the following are groups of plants except _____ I. Spermatophyta; II. Pteridophyta; III. Coelenterata; IV. Shizophyta. (a) I (b) III (c) I and IV (d) None of I-IV
5. The incorrect match for the blood genotype of a proposed couple is _____ I. AS and AA; II. AA and AA; III. AS and AS; IV. SS and SS. (a) I & II (b) II only (c) IV only (d) III and IV
6. Which of the most organized protists? (a) Euglena (b) Paramecium (c) Amoeba (d) Plasmodium
7. The function of xylem is to _____ (a) Conduct food (b) Transport waste product (c) Strengthening stem (d) Transport water and mineral salts
8. Which of these is a disease of the respiratory tracts? (a) Bronchitis (b) Malaria (c) Cholera
9. Which of the following animals is photosynthetic? (a) Euglena (b)

Plasmodium (c) Amoeba
(d) Paramecium

10. Typhus is caused by _____. I. Rickettsia; II. Body louse; III. Rat flea; IV. Aedes mosquito. (a) I only (b) IV only (c) II and III (d) I, II and III
11. Wastages in the blood during aerobic respiration is removed by _____. (a) Kidney (b) Liver (c) Alveoli in the lungs

12. Which of these bacteria is a nitrogen-fixing bacteria. (a) Rhizobium (b) Nitrosomonas (c) Nitrobacter

13. A gene which is homozygous is expressed (a) Dominant (b) Recessive (c) Masked

14. Which of the following natural resources cannot be conserved? (a) Air (b) Soil (c) Wildlife (d) Forest
15. Meiosis takes place in the _____. I. Sperm; II. Ovary; III. Testes; IV. Ovum. (a) I and II (b) II and III (c) IV and I (d) IV and III

16. Which is not a form of asexual reproduction? (a) Regeneration (b) Fragmentation (c) Fertilization (d) Binary fission
17. Which does not produce glucose exclusively? (a) Cellulose (b) Starch (c) Maltose (d) Sucrose

18. All cell activities are controlled by (a) Mitochondrion (b) Cytoplasm (c) Nucleus (d) Ribosome

19. Which of these is responsible for the green colour of plant? (a) Centioles (b) Chloroplast (c) Cytoplasm (d) Chlorophyll

20. Urea is formed during (a) Krebs cycle (b) Citric acid (c) Nitrogen cycle (d) Ornithine cycle

21. Cells in which system have the ability to reproduce. System; II. Transport 1 Circulation Excretory System; IV. System. (a) IV only (b) II, Reproductive and w{ } 1, II, III and IV (d) All of the above

22. The cell is enclosed by _____. (a) Cell wall (b) Cell membrane (c) Nuclear membrane

23. The enzyme in the stomach converts (a) Carbohydrates to starch (b) Starch to sugar (c) Fat to glycerol (d) Protein of peptones
24. Measurement; II. Observation; III Identification; IV. Interference. What is the due order of observing the above scientific methods? (a) III, II, 1, IV (b) I, II, III, IV (c) II, IV, III (d) IV, III, 1, 1

25. A type of biological interaction between two species where one species is harmed or inhibited by the other, while the other species is not affected and does not gain from the act is known as _____. I. Parasitism; II. Commensalism; III. Ammensalism IV. Symbiosis

Answers

1. B
2. D
3. C
4. B
5. D
6. B
7. D
8. A
9. A

10.A

11.C

12.A

13.A

14.A

15.B

16.C

17.D

18.C

19.B

20.D

21.A

22.B

23.D

24.A

25.III

BIOLOGY 2025

1. Which of the following instruments has to do with the aquatic habitat?
I. Thermometer II. Salinometer III. pH scale IV. Secchi disc V. Anemometer
Options: A. I & III only B. I-IV C. I & V D. II & IV E. II, III & IV
2. The hereditary material plants are carried by the I nucleus ii cytoplasm iii chromosome iv DNA v genes
3. Which of these plants has a spiral leaf arrangement? I. Gmelina II. Almond III. Trix IV. Terminalia A. I only B. I and IV C. II only D. All of the above
4. Examples of compound leaf arrangements are found in: I. Manihot II. Adansonia III. Allamanda IV. Thevetia A. I only B. I & II only C. III & IV only D. All of the above
5. Examples of imparipinnate leaves are Found in: I. Manihot II. Adansonia III. Newbouldia IV. Gliricidia A. I only B. II only C. III & IV only D. All of the above
6. Which of the following does not belong to the main group of plant kingdom I Bryophyte II pteridophyte iii chlorophyte IV spermatophyta V Thallophyte A) I-v B) III only C) V only D) I-iv
7. Which of the following is responsible for strength and rigidity of cell wall in a plant I. Chitin II. Tannin iii. Peptidoglycan iv. Lignin A) III B) iv only C) I and II D)?
8. What is the part of the living component of cell I organelles II protoplasm III cytoplasm IV Plasma membrane V chromatin A) I, II, III and V B) I, II, III and IV C)? D) I, II, III, IV and V
9. Which of the following is living I) organelles II). cytoplasm III). plasmids IV). nucleus

10. Which of the following organisms have two nuclei A. *Chlamydomonas* B. *Fungi* C. *Paramecium* D. *Amoeba*

form

22. Then they asked for plants that can survive in desert. A. *Hydrophyte* B. *Xerophytes* C. *Epiphyte* D. Something I can't remember
23. Which of this is a characteristics of rat A. Long intestine B. Fur and whiskers C. Amnion and placenta D. Tail
24. Which of this is a micronutrient I Magnesium II Manganese III Sulphur IV Phosphorus A. I only B. II only C. III only D. IV only
25. Which of the following heart I Superior vena blood to the brings III. caval. Hepatic vein

Pulmonary vein ly Repal vein

11. The three domains of classification are
 Fungi, virus, bacteria
 B. Fungi, archae, eukarya C. Archae, eukaryotes, prokaryotes
12. Which of the following possess nucleust. plants V bacteria Fungi II algae III protists IV
13. Which of the following has aerial roots ficus II ixora III ? IV
14. Which of the following is involved in breathing I intercostal muscles II Ribcage I Diaphragm IV Stomach
15. Which of the following is involved in circulation I plasma II Blood III Analyse IV Hormone V serum
16. Which of the following differentiate mammals I Diaphragm with lungs II small intestine and tail III Amniotic with placenta IV Whiskers with sweat
17. The kidneys a structure and functions involved in I pelvis II cortex IV Medulla V Ureter And Cortex
18. Which of the following is not part of natural water I Rain water II water Vapour III deionised water IV distilled water
19. During photosynthesis oxygen is gotten —
20. which of the following is not transferred by Vectors 1 Malaria II..... III..... IV COVID 1921. Which is a macronutrient I Manganese II Sulphur

26. Which of the following is a biome I. Aquatic plants li. Freshwater forest III. Desert Iv. Tropical rainforest
27. Angiosperms and gymnosperms are classified in which group? I Bryophytes li. Thallophytes lii. Pteridophytes Iv. Spermatophytes V. Plantae
28. Which of these is an eye defect? I. Long sightedness II. Short sightedness III. Glaucoma IV. River blindness
- Use these to answer questions 29 and 30I blinking of the eye II. Knee jerk III. Removal of hand from hot object IV. Salivation V. Swimming
29. Which of the above is/are reflex action
30. Which is/ are conditioned actions
31. The type of plants that exhibit naked seeds borne in cones are A. Grasses B. Conifers C. Palms D. Legumes
32. Theory of natural selection was given by A. Lamarck and wallace B. Darwin and wallace C. Darwin and Mendel D. Lamarck and mendel
33. Plants adapted to survive in salty marsh are A. Mesophytes B. Halophytes C. Xerophytes D) Hydrophytes
34. The totality of biotic and abiotic factors in a community is A. Habitat B. Biome C. Ecosystem D. Ecological niche E. Biological spectrum
35. Which of the following are components of neurons I. Synapse li. Dendrites Ili. Axons Iv. Meninges V. Glial
36. angiosperm and gymnosperms are group as a. thallophyte
37. which of the following are plants with naked seed A. legume B. grasses C. conifers D. 38. which are associated with hearing I. iris li. cornea lii. ossicle Iv. pinna
39. which are associated with eye vision A. cornea B. iris C. pinna D. ossicle
40. A mesophilic organism is found within range of temperature I. 60-65°C li 35-45°C III. 25-35°C iv. 80-100°C v. 20-45°C A. I, li and III only B. II, III, and V C. i-v D. Iv only E. I and IV only.
41. An animal that feeds on the remains of dead animals and kill other animals for food is dash is _____ I. Predator II. Prey Ili. Autotrophs iv. Scavengers V. Heterotrophs A. I, III, iv B. I, iv, v. C. Iv only D. I only E. lii, iv, v only.
42. Arrange the organism below as they would occur in a food chain I. Hawk II. Grass III. Snake IV. Grasshopper V. Lizard a. li, iv, v, lii, I b. ii, M, jv, v, | C. li, jv, v, II, I. D. li, iv, v, I, III
43. The instruments used to measure pressure, wind speed, turbidity, temperature are. I. Thermometer II barometer III. sechi disk iv. Anemometer V wind vane (arrange accordingly)
44. Which of the following characterizes the white mangrove a. Prop roots b. Buttress roots c. Breathing roots D. Root hairs e. Stilt roots
45. Which of the following group of plants is the most advanced a. Thallophytes b. Pteridophytes c. gymnosperms D. Bryophytes E. Cryptophytes
46. Who stipulated the theory of inheritance between organisms A. Jean Lamarck B. Charles Darwin c. Gregor Mendel D. Hugo de Vries
47. In which of the following does at least one organism benefits. I. Predation li. commensalism III. Ammensalism iv. parasitic A. I. li iv. B. I and li only C. All of the above D. I, li, III only.
48. If an organism gain is represented by +a if it doesn't it is represented by -a which of the following is +0 A. Parasitic B. Commensalism C. Ammensalism D. competition E.

49. Bulbs are modified (A) Roots (B) Shoots (C) Flowers (D) Ovary
50. Fruit are from fertilized A Ovules B Ovary C Flowers D Seeds

51. Which feature of Euglena makes it a plant A Eye Spot B chloroplasts C pellicle D flagellum
52. Which of these state can Mangrove Swamp be found (I) Lagos (II) Calabar (III) Delta (IV) Rivers (V) Ede A I_V B I- IV C None of I-V D I, II and III

53. Which of the following is a Climatic Factor (I) Relief (II) Altitude (III) Land and vegetation Cover (IV) Rainfall A None of I -

IV B I - IV C I, III and IV D II and IV

54. Which of the Following is a function of Xylem I Storage of Unused Sugar II Strengthening the Stem III Manufacturing food IV Conduction Of Food Forgotten how the Roman numerals was structured

55. Which of the Following is concerned with photosynthesis A Epidermis B Endodermis C Mesophyll D Cambium

56. Which of this forms the lateral hairs of Angiosperms A Pericycle B Endodermis C Cortex D Epidermis E Pith

57. The Ovules forms the after fertilization A Ovary B Fruit C Ovule D flowers

58. Which of the following is used to measure Turbidity A Rain gauge B Secchi disc C Thermometer D Anemometers

59. Which of the following is associated with Aquatic Habitat (I) Turbidity (II) Salinity (III) Dissolved gases (IV) Solubility (V) Pressure A I -IV B I- V C I, II, III and V D I and II

60. In addition to temperature which of the following majorly determines an habitat A Rainfall B Light C Pressure D Dissolves gases

61. I - Respiration II - photosynthesis III - combustion IV - decomposition V -

Sequestration Which of the following adds CO₂ to the atmosphere? A. I and IV B. II only C I III IV and V D

I III and IV 62. I- Respiration II - photosynthesis III. combustion IV - decomposition V -Sequestration Which of the following removes CO₂ from the atmosphere

A. I and II B. II and III C. II III and IV D. II and V

63. Which of the following is the most advanced A. Thallophytes B. Bryophytes C. Gymnosperms D. Pteridophyte

64. Which Biome in Nigeria is also known as tropical Grassland A Savanna B Savannah

65. Which of the following is also known as the amphibians of the plant kingdom A Thallophytes B Spermatophytes C Bryophytes D Ferns
66. Which of these organism has 2 distinct nucleus I Chlamydomonas II. Euglena III paramecium IV. Amoeba

67. Which of these is a feature of aquatic animal I bulbous stalk II gill III streamlined bodies IV scales V hairs A I, II, III, IV B II, III, IV C I, IV, V D II only

68. Which of these is a abiotic factor in terrestrial habitat I temperature II rainfall III relative humidity IV pH V turbidity A I only B I-IV C none except III D I -III

69. Which of the abiotic measuring instrument strictly speak of aquatic habitat I secchi disc II thermometer III salinometer IV pH meter V rain gauge A I, III, IV B I, III, IV, V C II only D none of the above

70. Soil fertility can be lost through I compaction II leaching III erosion IV loss of organic matter V terracing

71. The unit of biosphere is A ecology B ecosystem C atmosphere D community

72. Which of the following is true about predation I it help Organism to evolve II one organism gain why the other lose III the weaker organism is eliminated IV it enables transfer of energy V the prey is hunted into elimination

73. The recommended soil type for crop production A Sandy soil B clay soil C loamy soil D Stony soil
74. The following are consequences of overpopulation in an organism I interbreeding ii epidemic III competition IV cannibalism v crime
75. Only these factor form basis of competition I food ii space III water IV mates v light
76. Highland of montane forests and grasslands in Nigeria can be found in I obudu,cross river ii Jos plateau iii mambilla, adamawa IV ogun, Abeokuta v enugu,enugu

- 14.1,11,III
15. I,II,IV,V
16. 1, III & IV
17.1,II,III & IV
18. III & IV
19. WATER
20. IV ONLY
21.II ONLY
22.B
23.B
24.B
25. ALL
26. III&IV
27. IV ONLY
28. ALL
29.1,II,III
30. IV,V
31.B
32.B
33.B
34.C
35. 1,1I,III
36.C
37.C
38.III,IV
39. A&B
40.B
41.B
42.A
43. II,IV,III,I,
44.C
45.C
46.C
47.A
48.B
49.B
50.B
51.B
52.B
53.D
54.II
55.C
56.A
57.C

ANSWERS

- 1.B
2. V GENES
3. IV ONLY
4. B
5.C
6.B
7.B
8.A
9. I,II,IV
10.C
11.C
12.I,II,III,IV
13.I ONLY

58.B
59.C
60.A
61. I, III and IV
62. II and V.
63.D
64.B
65.C
66.III
67.B
68.B
69.A
70. ALL
71.B
72.I-IV
73.C
74.ALL
75. ALL
76.1-11I

PHYSICS 2019

1. In addition to a neutron, which of the following particles also exists in the nucleus? (a) An electron (b) An α -particle (c) γ -Particle (d) A proton
2. Which of the following emit(s) radiant heat energy? I. Forest fire; II. The human body; III. Ice Cubes placed on a table; IV. The walls of a room; (a)
3. A fixed mass of gas is compressed isothermally until its volume is appreciably reduced. Which of the following statements is/are correct about the gas? I. The average kinetic energy of its molecules is increased; II. The rate of molecular collisions is increased; III. The average inter-molecular spacing is reduced. (a) II only (b) I, II only (c) II, III only (d) I, II, III
4. Which of the following may produce a virtual image? I. Convex Mirror; II. Concave Mirror; III. Convex Lens IV. Concave Lens. (a) I, IV only (b) I, III only (c) I, II, IV only (d) I, II, III, IV
5. Light Waves of different frequencies will have different (a) Speeds (b) Amplitude (c) Colours (d) Intensities
6. The Horizontal Component of the Earth's Magnetic field is _____. I. Zero at the Magnetic equator; II. Zero at the magnetic poles; III. Maximum at the magnetic equator; IV. Maximum at the magnetic poles. Which of the above statements is/are correct? (a) I only (b) II only (c) II, III only (d) I, IV only
7. The following radiations are electrically neutral EXCEPT? (a) α -rays (b) γ -rays (c) X-ray (d) Neutrons
8. The centripetal acceleration of an object moving at constant speed around a circular path is _____. I. Directly proportional to the speed of the object; II. Directly proportional to the mass of the object; III. Directly proportional to the square of the speed of the object; IV. Inversely proportional to the radius of the circular path. Which of the following statements are correct? (a) I, II only (b) II, III, IV only (c) III, IV only (d) II, III only
9. The escape velocity of an object from the gravitational field of a planet _____. I. Increases with the mass of the object; II. Increases with the mass of the planet III. Is smaller on the moon than on the earth. Which of the above statements is/are correct? (a) I only (b) II only (c) II, III only (d) I, II, III
10. Two objects P & Q have the same weight W in air. When completely immersed in a liquid, the apparent weight of P is $\frac{3W}{4}$ while that of Q is $\frac{W}{2}$. It can be concluded that _____. I. Volume of P is twice the volume of Q; II. Volume of Q is twice that of P; III. Density of P is twice the density of Q; IV. Density of Q is twice that of P. Which of the above conclusions is/are valid? (a) I only (b) II only (c) I, IV only (d) II, III only
11. Which of the following statements is/are not correct? I. The pitch of a note emitted by a vibrating string

depends on its length; II. Two identical strings may emit notes of the same pitch and intensity but different quality; III. The Loudness of a note emitted by a vibrating string depends on the number of Overtones. (a) I only (b) II only (c) III only (d) II, III only

12. A beam of light is incident normally on an air/glass interface. Upon passing into the glass, which of the following properties of the light beam will change?

I. Frequency;

II. Speed; III. Direction; IV. Amplitude. (a) I, III only (b) II, IV only (c) I, II, IV only, (d) None of the above

13. 12V battery of internal resistance 0.12 is connected across a lamp of resistance R. If a current of 10A flows through the circuit, the value of R is (a) 0.92 (b) 1.00 (c) 1.10 (d) 1.20

14. An alternating current with RMS voltage of 50V as frequency of 60Hz. The AC represented by the equation. (a) $v = 50 \sin 60\pi t$ (b) $v = 50 \sin 120\pi t$ (c) $v = 70.7 \sin 60\pi t$ (d) $v = 70.7 \sin 120\pi t$

15. The readings on the Celsius and the Fahrenheit scale are numerically equal at temperature of _____ (a) -80° (b) -40° (c) 40° (d) 40°

16. In an electric kettle the heating element is usually located near the bottom of the kettle becomes _____. I. cold water is denser than hot water; II. Heat is transmitted to all parts of the water primarily; III. Heat is quickly conducted to all parts of the water; IV. Loss to the surroundings is

minimized. (a) III only (b) III, IV only (c) I, II, III only (d) I, II, IV only

17. A body initially at rest falls freely under gravity. If $e = 10 \text{ m/s}$ the distance covered by the body in the fourth second of fall is _____. (a) 35m (b) 40m (c) 45m (d) 80m

18. A spring balance measures _____. I. The mass of the hung on it; II. The Earth's Gravitational pull on the body hung on it; III. The weight of the body hung on it. Which of the following statements is (are) correct? (a) I only (b) III only (c) II & III only (d) I, II & III

19. A positive point charge B is placed near a positively charged metal sphere. As the charge B is moved closer to the sphere _____. I. Its electric potential energy increases; II. Its electric potential energy decreases; III. Work is done by the point charge; IV. Work is done on the point charge B. Which of the above statements is (are) correct? (a) I & IV only (b) II & III only (c) I & II only (d) II & IV only

20. A rainbow is viewed through a blue sheet of glass. The rainbow will appear _____. (a) White (b) Yellow (c) Blue (d) Black

21. Which of the following waves cannot be polarized? (a) Infra-red (b) Yellow light (c) X-rays (d) Sound

22. Two sound waves have the same wavelength in air. Which of the following characteristics must also be common to both waves? I. Pitch; II. Intensity; III. Amplitude; IV. Frequency. (a) I only (b) I & III only (c) I & IV only (d) I, II & IV only

23. Occurs in the Sun and Occurs in Atomic Bomb. Which of the following describes this arrangement?	12.B
(a) Nuclear fission & Nuclear Fusion	13.C
(b) Artificial Radioactivity & Nuclear Fusion	14.D
(c) Nuclear Fusion & Chain Reaction	15.B
(d) Chain Reaction & Nuclear Fission	16.D
24. Which of the following cells convert light energy to electrical energy?	17.A
I. Leclanche cell; II. Photo cell;	18.C
III. Daniel cell; IV. Solar cell. (a) I only	19.A
(b) IV only (c) I & III only (d) III and IV	20.C
25. At resonance, which of the following is true? I.	21.D
$WL = \omega C$; II. $v_L = v_C$; III. Current is	22.C
minimum; $N.Z = R$. (a) I, II and III (b) I and II	23.C
only (c) I, III and IV (d) II and IV	24.D
	25. I, II, IV

ANSWER

- 1.A
- 2.All
- 3.C
- 4.D
- 5.C
- 6.C
- 7.A
- 8.C
- 9.C
- 10.D
- 11.C

PHYSICS 2021

1. Which of the following gives the widest separation in the spectrum of white light. (a) Green and yellow (b) Red and indigo (c) Red and violet (d) Blue and green
2. A sample of a monatomic ideal gas is originally at 20°C . What is the final temperature of the gas if both the pressure and volume are doubled? (a) 5°C (b) 8°C (c) 1192°C (d) 899°C
3. A man walks towards a mirror at a speed of 0.25m/s . What is the relative motion that the image run toward the man? (a) 0.25m/s (b) 0.50m/s (c) 0.125m/s (d) 0.62m/s
4. Factor affecting speed of sound in air. I. Pressure; II. Loudness; III. Temperature; (a) III only (b) I, II, III (c) I and III
5. To convert temperature from $^{\circ}\text{C}$ to $^{\circ}\text{F}$, the conversion equation is (a) $F=5/9(c-32)$ (b) $F=1.8C+32$ (c) $F=9/5C-32$ (d) $F=5/9(C+32)$
6. The velocity ratio of inclined plane inclined at 30° to the horizontal axis is (a) 2 (b) 3 (c) 0.5 (d) 4
7. A certain radioactive element has a half-life of 20 years. How long will it take the activity to become $1/4$ of its original? (a) 20 years (b) 40 years (c) 60 years (d) 80 years
8. A uniform wave has a speed of 10m/s and a period of 0.5s . The distance between two nearest crests is _____ (a) 0.2 (b) 20m (c) 2m (d) 5m
9. If the cross sectional resistor R ohms is halved area $\propto$ length is doubled, its resistance becomes _____ (a) $4R$ ohms (b) $2R$ ohms (c) R ohms (d) $1/2$ ohms
10. Which of the following is concerning the magnitude of the electric field at a point in space? (a) Measure the total charge of the object (b) Measure the electric force per unit charge on a test charge (c) It is a measure of ratio of the charge on an object to its mass (b) It is a measure of the electric force per unit charge in a test charge
11. A nucleus is unstable if _____ (a) It has two or more isotopes (b) Its binding energy is not sufficient to hold nuclear particles together (c) It is very light with very low density (d) It has no significant mass
12. A motor car moves with a velocity 20m/s on a rough road horizontal road and covers a displacement of 50m . Find the coefficient of dynamic friction between the tyre and the ground ($g = 10\text{m/s}^2$). (a) 0.3 (b) 0.6 (c) 0.5 (d) 0.4
13. The angle of incidence in a denser medium when the angle of refraction in the less dense medium is 90° is called a _____ (a) Critical angle (b) Reflected angle (c) Incident angle (d) Emergent angle
14. Calculate the velocity ratio of screw jack, if the pitch is 5mm and the arm length is 30cm . (a) 37.72 (b) 377.2 (c) 3773 (d) 3.77215

15. A player plays a ball 40mN 300E of N to B who immediately passed 30m 600W of N to C. What is the resultant displacement? (a) 40m (b) 50m (c) 60m (d) 70m
16. Two objects P and Q have the same weight W in air. When completely immersed in a liquid, the apparent weight of P is $\frac{3W}{4}$ while that of Q is $\frac{W}{2}$. It can be concluded that the
I. Volume of P is twice the volume of Q.
II. Volume of Q is twice the density of P.
Which of the above conclusions is/are valid? (a) I only (b) II only (c) I and IV only (d) II and III only
17. A vibrating string has a tension of 40N and produces a note of 200Hz when plucked in the middle. When the length of string is unaltered and the tension is increased to 160N the frequency becomes. (a) 1600Hz (b) 800Hz (c) 400Hz (d) 200Hz
18. A transformer has 400-turns in the primary coil and 200-turns in the secondary coil. If the primary coil is connected to 110V, what will be obtained from the secondary coil? (a) 110V (b) 150V (c) 220V (d) 55V
19. A car 4m long moving at a velocity of 25m/s was beside a lorry 20m long with a velocity 19m/s. At $t=0$, the distance between them was 10m. How long will it take the car to overtake the lorry? (a) 9s (b) 5s (c) 3s (d) 2s
20. A pipe organ X produces a note whose pitch is twice that of the note produces by a second pipe organ Y. If the speed of the wave in X is v , then the speed of the wave in Y is
(a) $\frac{v}{2}$ (b) v (c) $2v$ (d) $4v$
21. Which of the following connections will convert a millimeter into a voltmeter? (a) A high series resistance (b) A high parallel resistance (c) A low series resistance (d) A low parallel
22. 1 farad is equivalent to: (a) 1 coulomb/volt (b) 1 volt/coulomb (c) 1 Joules/coulomb (d) 1 ampere/sec (e) Volts per unit charge
23. A force of 20N applied parallel to the surface of horizontal table just sufficient to make a block of mass 4kg set for motion. Find the acceleration when the force
24. The unit CYCLE/SEC is the same as
(a) Watt (b) Metre/Sec (c) Hertz (d) Radians is doubled
(a) $2\pi/2-3$ (b) $4\pi/s-2$ (c) $5\pi/s-2$ (d) $10\pi/s-$
25. If the length of a simple pendulum is halved, its frequency is
(a) Increased by a factor of 2
(b) Decreased by a factor of
(c) Decreased by a factor of $\sqrt{2}$
(d) Increased by a factor of $\sqrt{2}$

ANSWERS

1.C

2.D

3.B

4.A

5.B

6.A

7.B

8.D

9.A

10.B

11.B

12.D

13.A

14.B

1

5

.

B

:

16. No correct answer

17.X

18.D

19.B

20.B

21.A *Incomplete.*

22.A

23..

24.C

25.D

PHYSICS 2022

1. Which of the following connections will convert a milliammeter into voltmeter? (a) A high series resistance (b) A high parallel resistance (c) A low series (d) A low parallel resistance

2. 1 farad is equivalent to: (a) 1 coulomb/volt (b) 1 volt/coulomb (c) 1 Joules/coulomb (d) 1 ampere/sec (e) Volts per unit charge

3. A force of 20N applied parallel to the surface of horizontal table just sufficient to make a block of mass 4kg set for motion. Find the acceleration when the force

4. If two bodies with masses M and 4M are moving with the same kinetic Energy, the ratio of their velocity is? (a) 2:1 (b) 1:2 (c) 1:4 (d) 4:1

5. If the ice and steam point of a faulty thermometer is -4°C and 104°C respectively. It is used to take a measurement of 32°C what will be its equivalent values on a standard thermometer? (a) 100°C (b) 0°C (c) 33.3°C (d) 89°C

6. If the output voltage of transformer is 120V and the input the voltage is 240V with 500 turns. Find the number of turns in the output voltage. (a) 520 (b) 750 (c) 250

7. If the angle of incidence on a plane mirror is 25° and the mirror is rotated by 15° . Calculate the angle

with which the reflected Ray is turned. (a) 10° (b) 30°C (c) 40° (d) 20°

8. If the reading on a thermometer is $\frac{4}{5}$ of Celsius scale. What is the equivalent reading on Fahrenheit scale? (a) 144 (b) 176 (c) 720 (d) 352

9. When sound travels from air to water _____ I. The velocity decreases; II. The velocity increases; III. The frequency remains constant; IV. The frequency increase. (a) I, II (b) I only (c) IV, then III (d) I, IV, the III

10. A fixed mass of gas is heated at constant volume so that its Celsius temperature is doubled. The pressure of the gas _____ (a) Is halved (b) Is doubled (c) Is more than doubled (d) Increases but is less than doubled

11. The effect of increased pressure on the transformation temperature of water is that the _____. I. Melting point is reduced; II. Melting point is increased; III. Boiling point is reduced; IV. Boiling point is increased. Which of the following statements are correct? (a) I & III only (b) II & IV only (c) II & III only (d) I & IV only

12. Find the electric potential of 1 coulomb charged object placed at the same location. (a) 120J/C (b) 1J/C (c) 12J/C (d) 1.2J/C 13. The following are types of beam except _____ (a) Parallel (b)

Convergent (c) Emergent (d)

Divergent 14. Which of the following produce both transverse and longitudinal wave? (a) A moving pison (b) Sound (c) Water (d) Stretch string

15. In a leclanche cell, the role of granulated carbon mixed with manganese(iv)oxide is to _____. I. Increase the end of the cell; II. Prevent local action in the cell; III. Prevent polarization in the cell; IV. Reduce the internal resistance of the cell. Which of the above is/are correct? (a) II only (b) III only (c) I & II only (d) III & IV

16. A 1.2kWatts electric iron drawing from a 240V mains is to be protected with a fuse. The most suitable fuse rating is _____. (a) 0.2A (b) 5.0A (c) 8.0A (d) 12.0A

17. Which of the following statements are true of an _____

18. The divergence of the leaves of a negatively charged electroscope increase as an object is brought close to its cap. The object is _____

(a) A glass rod (b) A silk cloth (c) Negatively charged (d) Positively charged

19. The net downward force on a metal ball falling through a liquid at the terminal velocity is _____. (a) Zero (b) Its weight (c) The viscous force (d) The upthrust

20. Which of the following are derived quantities? 1. Thrust;

II. 22. Which of the following is a set of Vectors and Scalars, respectively? I. Weight, displacement & moment; 0 upthrust; III. II. Velocity, volume & density; IV. Mass, force & impulse. (a) I & IV (b) IV, II (c) I, III (d) II, III

Temperature; Area;

IV.III.Pressure;(a) 18 IV only (b)

1, III & N (c) I, II, III & IV 21. The

unit of momentum is _____ $J s^{-1}$ (b) $N s$ (c) $N m s$

(d) $N s^{-1}$ (a)

23. Two perpendicular forces have a resultant of 13 N. If one of the forces is 5 N, find the other force. (a) 8 N (b) 9 N (c) 12 N (d) 18 N

24. The assumptions made in a simple pendulum experiment are that _____ I. The suspending string is inextensible; II. The angle of oscillation is small; III. The bob has a very small mass; IV. The length of the string is short. Which of the above statements are correct? (a) I, II (b) I-III (c) I-IV (d) III, IV

Answers

1. A

2. A

3.

Incomplete Q

4. A

5. C

6. C

7. B

8. B

9. II AND III

10. D

11. D

12. B

13. C

14. C

15. D

16. B

17.

Incomplete Q

18. C

19. A

20. C

21. B

22. C

23. C

24. A

UI POST UTME

PHYSICS

1. In addition to a neutron, which of the following particles also exists in the nucleus?
- ☐ A. An electron
 - ☐ B. An α -particle
 - ☐ C. A β -particle
 - ☐ D. A proton
2. Which of the following emit(s) radiant heat energy?
- ☐ I. Forest fire
 - ☐ II. The human body
 - ☐ III. Ice cubes placed on a table
 - ☐ IV. The walls of a room
- ☐ A. I only
- ☐ B. I, II only
- ☐ C. I, II, IV only
- ☐ D. I, II, III, IV
3. A fixed mass of gas is compressed isothermally until its volume is appreciably reduced. Which of the following statements is/are correct about the gas?
- ☐ I. The average kinetic energy of its molecules is increased
 - ☐ II. The rate of molecular collisions is increased
 - ☐ III. The average inter-molecular spacing is reduced
- ☐ A. II only
- ☐ B. I, II only
- ☐ C. II, III only
- ☐ D. I, II, III
4. Which of the following may produce a virtual image?
- ☐ I. Convex Mirror
 - ☐ II. Concave Mirror
 - ☐ III. Convex Lens
 - ☐ IV. Concave Lens
- ☐ A. I, IV only
- ☐ B. I, III only
- ☐ C. I, II, IV only
- ☐ D. I, II, III, IV
5. Light Waves of different frequencies will have different:
- ☐ A. Speeds
 - ☐ B. Amplitude
 - ☐ C. Colours
 - ☐ D. Intensities
6. The Horizontal Component of the Earth's Magnetic field is:
- ☐ I. Zero at the Magnetic equator

o II. Zero at the magnetic poles

o III. Maximum at the magnetic equator

o IV. Maximum at the magnetic poles

o Which of the above

statements is/are correct?

o A. I only

o B. II only

o C. II, III only

o D. I, IV only

7. The following radiations are electrically neutral EXCEPT:

o A. α -rays

o B. γ -rays

o C. x-rays

o D. neutrons

8. The centripetal acceleration of an object moving at constant speed around a circular path is:

o I. directly proportional to the speed of the object

o II. directly proportional to the mass of the object

o III. directly proportional to the square of the speed of the object

o IV. inversely proportional to the radius of the circular path

o Which of the following

statements are correct?

o A. I, II only

o B. II, III, IV only

o C. III, IV only

o D. II, III only

9. The escape velocity of an object from the gravitational field of a planet:

o I. Increases with the mass of the object

o II. Increases with the mass of the planet

o III. Is smaller on the moon than on the earth

o Which of the above statements is/are correct?

o A. I only

o B. I, II only

o C. II, III only

o D. I, II, III

10. Two objects P & Q have the same weight W in air. When completely immersed in a liquid, the apparent weight of P is $3W/4$ while that of Q is $W/2$. It can be concluded that the:

o I. Volume of P is twice the volume of Q

o II. Volume of Q is twice that of P

- o III. Density of P is twice the density of Q
- o IV. Density of Q is twice that of P
- o Which of the above conclusions is/are valid?

- o A. I only
- O B. II only
- o C. I, IV only
- o D. II, III only

11. A beam of light is incident normally on an air/glass interface. Upon passing into the glass, which of the following properties of the light beam will change?

- o I. Frequency
- o II. Speed
- o III. Direction
- o IV. Amplitude
- o A. I, III only
- O B. II, IV only
- o C. I, II, IV only
- o D. None of the above

12. A 12 V battery of internal resistance 0.1Ω is connected across a lamp of resistance R. If a current of 10A flows through the circuit, the value of R is:

- o A. 0.9Ω

- O B. 1.0Ω
 - o C. 1.1Ω
 - O D. 1.2Ω
13. An alternating current with RMS voltage of 50V has a frequency of 60Hz. The A.C can be represented

by the equation:

- o A. $v=50\sin 60\pi t$
- O B. $v=50\sin 120\pi t$
- o C. $v=70.7\sin 60\pi t$
- O D. $v=70.7\sin 120\pi t$

14 The readings on the Celsius and the Fahrenheit scales are numerically equal at a temperature of:

- O A. -80°
- O B. -40°
- o C. 0°
- O D. 40°

15 In an electric kettle, the heating element is usually located near the bottom of the kettle because:

- o I. Cold water is denser than hot water
- o II. Heat is transmitted to all parts of the water primarily by convection
- o III. Heat is quickly conducted to all parts of the water
- o IV. Loss to the surroundings

is minimized

- o A. I, II only
- o B. III, IV only
- o C. I, II, III only
- o D. I, II, IV only

16 Which of the following gives the widest separation in the spectrum of white light? (a) green and yellow

- (b) red and indigo
- (c) red and violet
- (d) blue and green

17. A sample of a monatomic ideal gas is originally at 20 °CC. What is the final temperature of the gas if both the pressure and volume are doubled? (a) 5°C

- (b) 80°C
- (c) 20°C
- (d) 899 °C

18. A man walks towards a mirror at a speed of 0.25 m/s. What is the relative motion of the image towards the man? (a) 0.25 m/s

- (b) 0.50 m/s
- (c) 0.125 m/s
- (d) 0.625 m/s

19. Factors affecting speed of sound in air: i. Pressure
ii. Loudness

iii. Temperature

- (a) iii only
- (b) i, ii, iii
- (c) i and iii

20. To convert temperature from °C to °F, the conversion equation is: (a) $F = 5/9(C-32)$

- (b) $F = 1.8C + 32$
- (c) $F = 9/5C - 32$
- (d) $F = 5/9(C + 32)$

21. The velocity ratio of an inclined plane inclined at 30° to the horizontal axis is: (a) 2

- (b) 3
- (c) 0.5
- (d) 4

22 A certain radioactive element has a half-life of 20 years. How long will it take the activity to become 1/4 of its original? (a) 20 years

- (b) 40 years
- (c) 60 years
- (d) 80 years

23. A uniform wave has a speed of 10 m/s and a period of 0.5 s. The distance between two nearest crests is: (a) 0.2 m

- (b) 20 m
- (c) 2m

- (d) 5 m
24. If the cross-sectional area of a resistor of resistance R ohms is halved and its length is doubled, its resistance becomes: (a) $4R$ ohms
(b) $2R$ ohms
(c) R ohms
(d) $1/2R$ ohms
25. Which of the following is true concerning the magnitude of the electric field at a point in space? (a) It measures the total charge of the object.
(b) It measures the electric force per unit charge on a test charge.
(c) It is a measure of the ratio of the charge on an object to its mass.
(d) It is a measure of the electric force per unit charge on a test charge.
26. A nucleus is unstable if: (a) it has two or more isotopes.
(b) its binding energy is not sufficient to hold nuclear particles together.
(c) it is very light with very low density.
(d) it has no significant mass.
12. A motor car moves with a velocity of 27 m/s on a rough horizontal road

and covers a displacement of 50 m . Find

the coefficient of dynamic friction between the car and the ground ($g = 10 \text{ m/s}^2$): (a) 0.3

- (b) 0.6
(c) 0.5
(d) 0.428 .
28. The angle of incidence in a denser medium when the angle of refraction in the less dense medium is 90° is called: (a) Critical angle
(b) Reflected angle
(c) Incident angle
(d) Emergent angle
29. Calculate the velocity ratio of a screw jack, if the pitch is 5 mm and the arm length is 30 cm : (a) 37.72
(b) 377.2
(c) 3772
(d) 3.772
30. A player plays a ball 40 m 30°E of N to B who immediately passes 30 m 60°W of N to C . What is the resultant displacement? (a) 40 m
(b) 50 m
(c) 60 m
(d) 70 m
31. Two objects P and Q have the same weight W in air. When completely

immersed in a liquid, the apparent weight of P is $3W/4$ while that of Q is $W/2$. It can be concluded that: i.

Volume of P is twice the volume of Q.

ii. Volume of Q is twice the volume of P.

iii. Density of P is twice the density of Q.

iv. Density of Q is twice the density of P.

33 Which of the above conclusions is/are valid? (a) i only

(b) ii only

(c) i and iv only

(d) ii and iii only

34. A vibrating string has a tension of 40N and produces a note of 200 Hz

when plucked in the middle. When

the length of the string is unaltered

and the tension is increased to 160

N, the frequency becomes: (a) 1600

Hz

(b) 800 Hz

(c) 400 Hz

(d) 200 Hz

35. A transformer has 400 turns in the primary coil and 200 turns in the secondary coil. If the primary coil is

connected to 110V, what will be obtained from the secondary coil?

(a) 110 V

(b) 150 V

(c) 220 V

(d) 55 V

36. A car 4 m long moving at a velocity of 25 m/s was beside a lorry 20 m long with a velocity of 19 m/s. At $t = 0$, the distance between them was

10 m. How long will it take the car to

overtake the lorry? (a) 9 s

(b) 5s

(c) 3s

(d) 2 s

37. A pipe organ X produces a note whose pitch is twice that of the note produced by a second pipe organ Y.

If the speed of the wave in X is v ,

then the speed of the wave in Y is:

(a) $v/2$

(b) v

(c) $2v$

(d) $4v$

38. Which of the following connections will convert a milliammeter into a voltmeter? (a) A high series

resistance

(b) A high parallel resistance

immersed in a liquid, the apparent weight of P is $\frac{3W}{4}$ while that of Q is $\frac{W}{2}$. It can be concluded that: i.

Volume of P is twice the volume of Q.

ii. Volume of Q is twice the volume of P.

iii. Density of P is twice the density of Q.

iv. Density of Q is twice the density of P.

33 Which of the above conclusions is/are valid? (a) i only

(b) ii only

(c) i and iv only

(d) ii and iii only

34. A vibrating string has a tension of 40N and produces a note of 200 Hz

when plucked in the middle. When the length of the string is unaltered and the tension is increased to 160 N, the frequency becomes: (a) 1600

Hz

(b) 800 Hz

(c) 400 Hz

(d) 200 Hz

35. A transformer has 400 turns in the primary coil and 200 turns in the secondary coil. If the primary coil is

connected to 110V, what will be obtained from the secondary coil?

(a) 110 V

(b) 150 V

(c) 220 V

(d) 55 V

36. A car 4 m long moving at a velocity of 25 m/s was beside a lorry 20 m long with a velocity of 19 m/s. At $t=0$, the distance between them was 10 m. How long will it take the car to overtake the lorry? (a) 9 s

(b) 5 s

(c) 3 s

(d) 2 s

37. A pipe organ X produces a note whose pitch is twice that of the note produced by a second pipe organ Y. If the speed of the wave in X is V , then the speed of the wave in Y is:

(a) $\frac{V}{2}$

(b) V

(c) $2V$

(d) $4V$

38. Which of the following connections will convert a milliammeter into a voltmeter? (a) A high series resistance

(b) A high parallel resistance

- (c) A low series resistance
- (d) A low parallel resistance
39. 1 farad is equivalent to: (a) 1 coulomb/volt
- (b) 1 volt/coulomb
- (c) 1 joule/coulomb
- (d) 1 ampere/second
- (e) Volts per unit charge
40. A force of 20 N applied parallel to the surface of a horizontal table is just sufficient to make a block of mass 4 kg set in motion. Find the acceleration when the force is doubled: (a) 2 m/s²
- (b) 4 m/s²
- (c) 5 m/s²
- (d) 10 m/s²
41. The unit CYCLE/SEC is the same as: (a) watt
- (b) meters/sec
- (c) hertz
- (d) radians/sec
42. If the length of a simple pendulum is halved, its frequency is: (a) increased by a factor of 2
- (b) decreased by a factor of 2
- (c) decreased by a factor of $\sqrt{2}$
- (d) increased by a factor of $\sqrt{2}$
43. Which of the following is paired correctly? i. Rubbing - Insulator
- ii. Friction - Conductors
- iii. Induction - Neutral bodies
- iv. Conduction - insulator/conductor
- (a) i only
- (b) i and iv only
- (c) ii only
- (d) i and iii only
44. Which of the following radiations cannot be deflected by an electric field or a magnetic field? i. Alpha rays
- ii. Beta rays
- iii. Gamma rays
- iv. X-rays
- (a) i and ii only
- (b) iii and iv only
- (c) i, ii, and iii only
- (d) i, ii, iii, and iv
45. Of the radiations, the one that does not originate from the nucleus is: i. Alpha
- ii. Beta
- iii. Gamma
- iv. X-rays
- (a) i and ii only
- (b) i, ii, and iii
- (c) iv only
- (d) iii only

46. A body accelerates uniformly from rest for a time t . The kinetic energy of the body at t :

- Is directly proportional to the displacement
- Is directly proportional to the square of the instantaneous velocity
- Is directly proportional to t^2
- Depends on the value of the acceleration

- (a) ii only
- (b) i and iii only
- (c) i, ii, and iii only
- (d) i, ii, iii, and iv

47 The kinetic energy of a photoelectron ejected from a metal surface illuminated with radiation depends on the:

- Wavelength of the radiation
- Intensity of the radiation
- Source of the radiation
- Nature of the surface

- (a) i only
- (b) i and iv only
- (c) ii and iii only
- (d) i, ii, and iv only

48. At resonance, which of the following is true?

- $WL = \omega C$
- $\omega L = \omega C$
- Current is minimum

iv. $Z = R$

- (a) i, ii, and iii
- (b) i and ii only
- (c) i, iii, and iv
- (d) ii and iv

49. A fine steel wire mesh can be made to float on water due to:

- The principle of flotation
- The concept of surface tension

(b) The concept of surface tension

50. Which of the following cells convert light energy to electrical energy?

- Leclanche cell
- Photocell
- Daniel cell
- Solar cell

Leclanche cell

ii. Photocell

iii. Daniel cell

iv. Solar cell

- (a) i only
- (b) iv only
- (c) i and iii only
- (d) ii and iv only

51. Which of the following describes the arrangement of occurring in the Sun and an atomic bomb?

- Nuclear fission and Nuclear Fusion
- Artificial Radioactivity and Nuclear Fusion
- Nuclear Fusion and Chain Reaction
- Chain Reaction and Nuclear Fission

fission and Nuclear Fusion

(b) Artificial Radioactivity and Nuclear Fusion

(c) Nuclear Fusion and Chain Reaction

(d) Chain Reaction and Nuclear Fission

52. Two sound waves have the same wavelength in air. Which of the following characteristics must also be common to both waves? i. Pitch
ii. Intensity
iii. Amplitude
iv. Frequency
(a) i only
(b) ii and iii only
(c) i and iv only
(d) i, ii, and iv only

53 Which of the following waves cannot be polarized? (a) Infra-red
(b) Yellow light
(c) X-rays
(d) Sound

54 A rainbow is viewed through a blue sheet of glass. The rainbow will appear: (a) White
(b) Yellow
(c) Blue
(d) Black

55. A positive point charge B is placed near a positively charged metal sphere. As the charge B is moved closer to the sphere: i. Its electric potential energy increases
ii. Its electric potential energy decreases

iii. Work is done by the point charge
iv. Work is done on the point charge

B
(a) i and iv only
(b) ii and iii only
(c) i and iii only
(d) ii and iv only

56 A spring balance measures: i. The mass of the body hung on it
ii. The Earth's gravitational pull on the body hung on it
iii. The weight of the body hung on it
(a) i only
(b) iii only
(c) ii and iii only
(d) i, ii, and iii

57. A body initially at rest falls freely under gravity. $18g = 10m/s^2$, the distance covered by the body in the fourth second of fall is: (a) 35 m
(b) 40m
(c) 45 m
(d) 80m

58 Which of the following connections will convert a milliammeter into a voltmeter? (a) A high series resistance
(b) A high parallel resistance
(c) A low series resistance

(d) A low parallel resistance

59 1 farad is equivalent to:(a) 1

coulomb/volt

(b) 1 volt/coulomb

(c) 1 joules/coulomb

(d) 1 ampere/sec.

(e) Volts per unit charge

60. A force of 20N applied parallel to the

surface of a horizontal table is just

sufficient to make a block of mass

4kg set for motion. Find the

acceleration when the force is

doubled.. (a) 2m/s^2

(b) 4m/s^2

(c) 5m/s^2

(d) 10m/s^2

61. If two bodies with masses M and 4M are

moving with the same kinetic

energy, the ratio of their velocities is:

(a) 2:1

(b) 1:2

(c) 1:4

(d) 4:1

62 If the ice and steam points of a faulty thermometer are -4°C and 104°C

respectively. It is used to take a

measurement of 32°C , what will be

its equivalent value on a standard

thermometer? (a) 100°C

(b) 0°C

(c) 33.3°C

(d) 89°C

63. If the output voltage of a transformer is

120V and the input voltage is 240V with

500turns, find the number of turns in the

output voltage. (a) 520

(b) 750

(c) 250

64. If the angle of incidence on a plane mirror

is 25° and the mirror is rotated

by 15° , calculate the angle with

which the reflected ray is turned. (a)

10°

(b) 30°

(c) 40°

(d) 20°

65. If the reading on a thermometer is

$\frac{4}{5}$ of the Celsius scale, what is the

equivalent reading on the Fahrenheit

scale? (a) 144

(b) 176

(c) 720

(d) 352

66 When sound travels from air to

water: I. The velocity decreases

II. The velocity increases

III. The frequency remains constant

IV. The frequency increases

(a) I,III

(b) I only

- (c) IV, then III
- (d) I, IV, then III
67. A fixed mass of gas is heated at constant volume so that its Celsius temperature is doubled. The pressure of the gas: (a) Is halved
- (b) Is doubled
- (c) Is more than doubled
- (d) Increases but is less than doubled
68. The effect of increased pressure on the transformation temperature of water is that the: I. Melting point is reduced
- II. Melting point is increased
- III. Boiling point is reduced
- IV. Boiling point is increased
- (a) I & III only
- (b) II & IV only
- (c) II & III only
- (d) I & IV only
69. Find the electric potential of a 1 Coulomb charged object placed at the same location. (a) 120 J/C
- (b) 1 J/C
- (c) 12 J/C
- (d) 1.2 J/C
70. The following are types of beams except: (a) Parallel
- (b) Convergent
- (c) Emergent
- (d) Divergent
71. Which of the following produce both transverse and longitudinal waves?
- (a) A moving piston
- (b) Sound
- (c) Water
- (d) Stretched string
72. In a Leclanché cell, the role of granulated carbon mixed with manganese(IV) oxide is to: I. Increase the end of the cell
- II. Prevent local action in the cell
- III. Prevent polarization in the cell
- IV. Reduce the internal resistance of the cell
- (a) II only
- (b) III only
- (c) I & I only
- (d) III & IV
73. A 1.2kW electric iron drawing from a 240V mains is to be protected with a fuse. The most suitable fuse rating is: (a) 0.2A
- (b) 5.0A
- (c) 8.0A
- (d) 12.0A
74. Which of the following statements

are true of an insulated positively charged sphere? I. It contains excess positive charges

II. It has an electric field associated with it

III. It possesses potential energy

IV. It carries an electric current

(a) I, II

(b) I, II, III

(c) II, III

(d) I-IV

75. The divergence of the leaves of a negatively-charged electroscope increases as an object is brought close to its cap. The object is: (a) A

glass rod

(b) A silk cloth

(c) Negatively charged

(d) Positively charged

76. The net downward force on a metal ball falling through a liquid at the terminal velocity is: (a) Zero

(b) Its weight

(c) The viscous force

(d) The upthrust

77. Which of the following are derived quantities? I. Thrust

II. Temperature

III. Area

IV. Pressure

(a) I & IV only

(b) II, III & IV

(c) I, III & IV

(d) I, II, III & IV

78. The unit of momentum is: (a) $J s^{-1}$

(b) Na

(c) Nms

(d) $N s^{-1}$

79. Which of the following is a set of vectors and scalars, respectively? I.

Weight, displacement & moment

II. Velocity, volume & upthrust

III. Density, capacitance & distance

IV. Mass, force & impulse

(a) I & IV

(b) IV, III

(c) I, III

(d) II, III

80. Two perpendicular forces have a resultant of 13N. If one of the forces is 5N, find the other force. (a) 8N

(b) 9N

(c) 12N

(d) 18N

81. The assumptions made in a simple pendulum experiment are that: I.

The suspending string is

inextensible

- II. The angle of oscillation is small
 III. The bob has a very small mass
 IV. The length of the string is short
- (a) I, II
 (b) I-III
 (c) I-IV
 (d) III, IV

82. A certain mass of gas is trapped in a tube of volume V at a temperature of 27°C . If the pressure on the gas is doubled such that its volume is reduced to 75% of its original value, the new temperature of the gas is:

- (a) -73°C
 (b) 40.5°C
 (c) 177°C
 (d) 450°C

83. Which of the units of the following physical quantities are derived? I.

Area

II. Thrust

III. Pressure

IV. Mass

- (a) I-IV
 (b) I-III
 (c) I, II, IV
 (d) I, II

84. Which of the following situations is not possible? (a) A body has zero

velocity and non-zero acceleration

(b) A body travels with a northward

velocity and a northward

acceleration

(c) A body travels with a constant

velocity and a time-varying

acceleration

(d) A body travels with a constant

acceleration and a time-varying

85. A thermos flask has a double-walled

glass container in which heat losses

are minimized by: I. Evacuating the

space between the glass walls

II. Silvering the surfaces on either of

the evacuated space

III. Covering the container with a

cork of low thermal conductivity

(a) I only

(b) I, II only

(c) I, III only

(d) I, II, III

86 Food cooks faster in a pressure

cooker than in an ordinary cooking

pot because: I. The food is not

affected by air convectional currents

II. The boiling point of water is

greater than 100°C in the pressure

cooker

III. Less heat escapes from the

pressure cooker

IV. The vapor pressure is higher in the pressure cooker

(a) I, III only

(b) II, IV only

(c) I-III

(d) II-IV

87. A certain radioactive element has a half-life of 20 years. How long will it take the activity to become 1/4 of its original? (a) 20 years

(b) 40 years

(c) 60 years

(d) 80 years

88. A radioactive substance has a half-life of 4 years. If its activity today is 200 dps, its activity in 8 years from today is: (a) 50 dps

(b) 25 dps

(c) 20 dps

(d) 10 dps

89. A substance has a half-life of 3 min. After 6 min, the count rate was observed to be 400. What was its count rate at zero time? (a) 2400

(b) 200

(c) 1200

(d) 1600

90 The following are not part of the

electromagnetic spectrum: I. Alpha rays

II. Beta rays

III. Gamma rays

IV. Radio Waves

(a) I, II only

(b) III, IV

(c) I, III

(d) II, IV

91. A gold leaf electroscope is to be charged by induction. Which of the following indicates the correct sequence of steps to be taken? I.

Touch the cap with a finger

II. Bring a charged rod near the cap

III. Remove the charged rod

IV. Remove the finger from the cap

(a) I, II, III, IV

(b) II, III, I, IV

(c) II, I, IV, III

(d) II, I, III, IV

92 The direction of heat flow between two objects depends on: (a) Their masses

(b) Their heat contents

(c) Their temperatures

(d) Whether they're in solid, liquid or gaseous state

93. Velocity is the tendency to: (a)

	current flows through them
Be traveling at a speed that is faster than perceived	(d) Two or more resistors are said to be connected in parallel if different
(b) Be traveling at a speed that is currents flow through them slower thanperceived	
(c) Both A & B	
(d) None of the options	Answers
94 An object placed at the center of	1.D
	cur vat ure of a co nca ve mir ror will
	2.D
have its image formed at: (a) A point 3.C	
beyond the center of curvature	4.D
(b) A point between focus and center	5.C
of curvature	6.C
(c) The center of curvature	7.D
(d) The focus	8.C
95. Total internal reflection can take	9.C
place wheh light travels from: (a) Air	10.D
to glass	11.B
(b) Air to water	12.C
(c) Water to glass	13.D
(d) Diamond to glass	14.B
96. Which of the following statements is	15.A
correct? (a) Two or more resistors	
are said to be connected in series if 16.C	
different currents flow through them	17.D

23.D

(b) Two or more resistors are said to	18.B
be connected in series if the same	19.A
current flows through them	20.B
(c) Two or more resistors are said to	21.A
be connected in parallel if the same	22.B

24.A	53.D
25.B	54.C
26.B	55.A
	27.0.7
	356.C
28.A 57.A	
29.B 58.A	
30.B	59.A
31.D	60.C
32..	61.A
33..	62.28°C
34.C	63.C
35.D	64.B
36.B	65.B
37.B	66.A
38.A	67.C
39.A	68.D
40.C	69.B
41.C	70.C
42.D	71.C
43.D	72.D
44.B 73.B	
45.C 74.B	
	75.C
46.D	76.A
47.B	77.C
48.D	78.C
49.B	79.C
50.D	80.C
51.C	81.A
52.C	

82.C

83.B

84.C

85.D

86.B

87.B

88.A

89.D

90.A

91.C

92.C

93.A

94.C

95.D

96.B or D

:

1. What is the difference between a neutron and electron of uranium?(a) 146 (b) 54 (c) 268 (d) 92

2. Which of the following will have the highest disorderliness?
(a) Water 100°C (b) Steam at 100°C
(c) Water at 0°C (d) Water at -50°C

3. During the change of state which of the following remains unchanged.

(a) Temperature (b) Volume (c) Heat content (d) None

4. Assuming the fundamental unit of dimension force F , velocity V , time T . Find the dimension of potential energy. (a) F/T^2 (b) F/T (c) FVT (d) $(FV)/T$

5. A plank of length 4m is inclined to the ground such that its one end is resting on the ground and the other end is 1m above ground level. Calculate the effort that has to be applied to push a load of 48N up the plank. (a) 15 (b) 12 (c) 30 (d) 25

6. If two charged plates are maintained at a potential difference of 2KV, what is the workdone in taking a charge of 500micro-coulomb across the field? (a) 3J (b) 10J (c) 1000J (d) 1J

7. Which of the following electromagnetic waves has the highest speed in a vacuum? I. Infra-red rays; II. Radio waves; III. Light

rays; IV. Ultra violet rays; V. Gamma rays. (a) I and II (b) II and IV

(c) IV only (d) All the ultraviolet rays have same speed in a vacuum

8. The image is to be the same size as the object must be placed in a concave mirror at____(a) F (b) Pole (c) 2F (d) Infinity

9. The image of an object placed at the center of the curvature of a concave mirror is____(a) Inverted and magnified (b) At the principal focus (c) Real and diminished (d) At the center of the curvature

10. The unit of magnetic field is. I. Tesla; II. Newton; III. Coulomb; IV. Weber/m². (a) I and IV (b) I only (c) IV only (d) III

11. If boy P has a mass M and body Q has a mass 16M, the ratio of their frequency if the force constant is the same is____(a) 16:1 (b) 4:1 (c) 1:16 (d) 1:4

12. A spring of force constant 500 N/m is compressed such that its length shortens by 5 cm. The energy stored in the spring is? (a) 0.625 J (b) 6.250 J (c) 62.500 J (d) 625.000 J

13. The d.c. generator has essentially the same components as the a.c. generator except the presence of? (a) Slip ring (b) Carbon brushes (c) Split ring (d) Armature

14. The unit of Planck's constant is (a) Energy time (b) Frequency (c) Power (d) Energy

15. A transformer has 400 turns of wire in the primary coil and 40 turns in the secondary coil. The ratio of the primary voltage to the secondary voltage is

the secondary coil. If the input voltage is 150V. Calculate the magnitude of the output voltage.(a) 150V (b) 55V (c) 36V (d) 15V

16. Given the units of the physical quantities below. $A=L/M$, $B=L/T$

Find the unit of C if $A=B$.(a) T/M (b) L
 $M^2(C)LM^{-3}(d)L/T$

17. Calculate : the torque I'd the distance between the two arms in 20cm and has a force of 40N.
 (a)800J (b) 8000J (c) 160J (d) 8J

18.A substance has a resistivity of 7.1×10^{-6} ohm meter, an increase in Temperature causes the resistivity to increase, the semiconductor is likely_____ (a) Metal
 (b)Semiconductor (c) Gas
 (d)Electrolyte

19._____is not a fundamental parameter of wave power.
 (a)Velocity (b) Frequency
 (c)Wavelength (d) Amplitude

20. If the area between the plates of a capacitor and the distance are both doubled, the capacitance will
 (a) Increase (b) Doubled
 (c)Quadruple (d) Remain unchanged

21. The electric field strength of a body of charge 4Cc as a potential 10MV 5 — (a) 2.5×10^{-12}
 (b) 2.5×10^{-6} (c) 2.5×10^{-9}
 (d) 2.5×10^{-3}

22. if two capacitors of 40micro-farad and 20 micro-farad in parallel is connected to a voltage of 12V.Calculate the total capacitance and the voltage of each capacitor.

(a)60 micro-farad, 12V (b) 60-micro-

farad, 6V (c) 13.3 micro-farad, 6V
(d) 13.3 micro-farad, 12V

23. A car 4m long moving at a velocity of 25m/s was beside a lorry 20m long with a velocity 19m/s. At $t=0$, the distance between them was 10m. How long will it take the car to overtake the lorry? (a) 2s (b) 9s (c) 5s (d) 3s

24. If the refractive index of glass is 1.25, what is the critical angle at the air-glass interface? (a) $\sin^{-1} \frac{1}{2}$ (b) $\sin^{-1} \frac{2}{3}$ (c) $\sin^{-1} \frac{4}{5}$ (d) $\sin^{-1} \frac{3}{5}$

25. If the angle of incidence is 50° . During total internal reflection the critical angle is equal to θ . Which of the following is correct? (a) θ is greater than 50° (b) θ is less than 50° (c) θ is equal to 50° (d) θ is equal to 50°

26. A gold leaf electroscope is to be charged by induction. Which of the following indicates the correct sequence of steps to be taken?
I. Touch the cap with a finger;
II. Bring a charged rod near the cap;
III. Remove the charged rod;
IV. Remove the finger from the cap.
(a) I, II, III, IV (b) II, III, I, IV (c) II, I, IV, III (d) II, I, III, IV

Answers

1.B

2. B

3. A

4.C

5. B

6. D

7. D

8. C

9. D

10.A

11.B

12.A

13.C

14.A

15.D

16.A

17.D

18.A

19.A

20.D

21.. *Incomplete*

22.A

23.C

24.C

25.B

26.C

UI POST UTME **PHYSICS 2024**

1. An image is said to be virtual if
- A. It appears to diverge from a point
 - B. It actually diverges from a point
 - C. It appears to converge at a point
 - D. It actually converges at a point

Answer;

Option A

Explanation;

An image is said to be virtual if ***the light rays appear to diverge from a point*** but do not actually pass through that point. Instead, the image is formed by the apparent extension of diverging rays when traced backward.

This happens, for example, in plane mirrors and convex lenses when the object is within the focal point.

2. Dimension of plank constant is the same as
- A. Energy
 - B. Power
 - C. Angular Momentum.
 - D. None of the above

Answer;

C. Angular Momentum

Explanation;

The Planck constant (denoted as h) is a that relates the energy of a photon to its frequency. The dimensions of the Planck constant can be derived from its definition:

$$E = hf$$

Where:

- E is energy,
- f is frequency.

The dimensions of energy E are: ML^2T^{-2}

Dimension for frequency is: T^{-1}

Therefore, the dimensions of the Planck constant are:

$$ML^2T^{-2}/T^{-1} = ML^2T^{-1}$$

Now, let's compare this with the dimensions of the given options:

A. ***Energy***: The dimensions of energy are: ML^2T^{-2}

B. ***Power***: Power is the rate of doing work, and its dimensions are: ML^2T^{-3}

C.*Angular Momentum*: The dimensions of angular momentum are:
 $M \times L \times LT^{-1} = ML^2T^{-1}$

* C.0km/hr*

These match the dimensions of the Planck constant.

3. The phenomenon where two waves combine to form a wave with larger amplitude is called

- A. Standing wave
- B. Destructive interference
- C. Constructive interference
- D. Superpositioning

Answer;

C. Constructive interference.

Explanation;

Constructive interference occurs when two waves of the same frequency and phase meet, resulting in a wave with an amplitude that is the sum of the individual amplitudes. This leads to a wave with a larger amplitude.

4. A man moves from X to Y 50km and again from Y to X 50km. Given that the time spent in both motion is 2hrs, calculate the average velocity

- A. 50km/hr
- B. 100km/hr
- C. 0km/hr
- D. 2km/hr

Answer;

Explanation;

To calculate the average velocity, we need to consider both the displacement and the total time taken.

Displacement is the change in position from the starting point to the ending point. Since the man moves from X to Y and then back to X, his net displacement is zero because he ends up at the same point where he started.

Average velocity is defined as the total displacement divided by the total time taken. Mathematically, it is given by:

Average velocity = Total displacement/time

Given:

- Total displacement = 0 km (since he returns to the starting point)
- Total time = 2 hours

Plugging in the values:

Average velocity = $0/2$
 $= 0 \text{ km/hr}$

5. The following is/are true about evaporation and boiling point of water

- I. Evaporation occurs only on the surface
- II. Boiling occurs through out the water body
- III. Boiling and evaporation are not simultaneous

- A. I and III
- B. I and II
- C. II and III
- D. I, II and III

Answer;
 D-I, II & III

Explanation;

I. *Evaporation occurs only on the surface*: This is true. Evaporation is a surface phenomenon where molecules at the surface of a liquid gain enough energy to transition into the gas phase.

II. *Boiling occurs throughout the water body*: This is also true. Boiling occurs when the vapor pressure of the liquid equals the atmospheric pressure, causing bubbles of vapor to form throughout the liquid, not just at the surface.

III. *Boiling and evaporation are not simultaneous synced*: This is true as well. Evaporation can occur at any temperature below the boiling point and only at the surface, while boiling occurs at a specific temperature (the boiling point) and

involves the entire volume of the liquid.

6. In a photoelectric effect experiment, the stopping potential is
- A) kinetic energy to stop energetic electrons emitted
 - B) potential energy to stop energetic electrons emitted
 - C. Kinetic energy to emit electrons
 - D. Potential energy to emit electrons

Answer;
 B. potential energy to stop energetic electrons emitted

Explanation;

In a photoelectric effect experiment, the *stopping potential* is the voltage required to stop the most energetic electrons emitted from the metal surface. This stopping potential is directly related to the maximum kinetic energy of the emitted electrons.

- The stopping potential is the potential energy (voltage) required to stop the most energetic electrons emitted from the metal surface.
- It is not the kinetic energy itself but the potential energy that counteracts the kinetic energy of the electrons.

7. Which of the following is not a characteristic of electric lines of forces
- A. They cross each other
 - B. They emerge from the positive charge

C. They enter into the negative charge

D. B and C

Answer:

Option A

Explanation;

A. *They cross each other*: This is *not* a characteristic of electric lines of force. Electric field lines never cross each other. If they were to cross, it would imply that at the point of intersection, there are two different directions for the electric field, which is not possible.

B. *They emerge from the positive charge*: This is a characteristic of electric lines of force. Electric field lines originate from positive charges.

C. *They enter into the negative charge*: This is also a characteristic of electric lines of force. Electric field lines terminate at negative charges.

8. If p is pressure and ρ density what's the unit of p/ρ

A. N/m^2

B. Kg/m^3

C. N/kg

D. m^2/s^2

Answer;

Option D

Explanation;

Let's tackle this with dimensions.

Dimension for pressure = $\text{ML}^{-1}\text{T}^{-2}$

Dimension for density = ML^{-3}

$$\frac{\text{Pressure}}{\text{density}} = \frac{\text{ML}^{-1}\text{T}^{-2}}{\text{ML}^{-3}} = \text{L}^2\text{T}^{-2}$$

In units, that is m^2/s^2 .

9. Which of the following is not basic wave motion quantity

I. Frequency

II. velocity

IV. Amplitude

IV. wavelength

A. I only

B. II only

C. III only

D. IV only

Answer;

Best answer: B -Velocity

Explanation;

The basic wave motion quantities are:

Frequency (I) - the number of oscillations per second.

Velocity (II) - the speed at which the wave propagates.

Amplitude (III) - the maximum displacement from the equilibrium position.

Wavelength (IV) - the distance between two successive crests or troughs.

All of these are fundamental properties of wave motion.

However , if the question intends to ask for a quantity that is not always essential in defining a wave (depending on context), *velocity*(option B) might be considered less fundamental since it depends on the medium through which the wave propagates. *But generally, all four are crucial in describing wave motion.*

10. An atomic bomb is an example of

- A. uncontrolled fission reaction
- B. controlled fusion reaction
- C. fusion
- D. all of the above

Answer;

Option A

Explanation;

In an atomic bomb, the energy is released reaction. This involves the splitting of heavy atomic nuclei (such as uranium-235 or plutonium-239) into smaller nuclei, releasing tremendous amount of energy in the process

11. The following are scalar quantities only

- I. Workdone
- II. Magnetic flux density
- III. Displacement
- IV. Pressure
- V. Impulse

- A. I, III, IV and V
- B. I, III and IV
- C. II, III and V
- D. I and IV

Answer;

D-I and IV

Explanation;

- 1. *Workdone*: Work is a scalar
- II. *Magnetic flux density*: Magnetic quantity flux density (also known as magnetic field strength) is a vector quantity
- III. *Displacement*: Displacement is a vector quantity as it has both magnitude and direction,

representing the change in position of an object.

IV. ***Pressure***: Pressure is a scalar quantity

IV. ***Impulse***: Impulse is a vector quantity

12. Which of the following has the same unit as impulse

A. Momentum

B. pressure

C. power

D. Workdone

Answer;

Option A

Explanation;

Impulse is defined as the change in momentum of an object when a force is applied over a time interval. The unit of impulse is the same as the unit of momentum, which is kilogram meter per second ($\text{kg}\cdot\text{m/s}$).

Let's analyze each option:

A. ***Momentum***: Momentum is the product of mass and velocity, and its unit is kilogram meter per second ($\text{kg}\cdot\text{m/s}$), which is the same as the unit of impulse (Ns).

B. ***Pressure***: Pressure is force per unit area, and its unit is Pascal (Pa) or Newton per square meter (N/m^2)

C. ***Power***: Power is the rate of doing work, and its unit is Watt (W) or Joule per second (J/s)

D. ***Workdone***: Workdone is the product of force and displacement, and its unit is Joule (J) or Newton meter ($\text{N}\cdot\text{m}$)

13. The frequency of a beam A is twice that of

B. The photon energy ratio, E_A/E_B is?

A. 2

B. $1/2$

C. $3/4$

D. $\%s$

Answer;

Option A

Explanation;

The energy of a photon is given by the equation:

$$E = hf$$

Where:

E is the photon energy,

H is Planck's constant,

F is the frequency of the photon.
 15. Two inductances of 2H and 1H are connected in series. What's the effective inductance?

Given that the frequency of beam A is twice that of beam B, we can write:

- A. 3H
- B. 1H
- C. 4H
- D. 5H

Thus, the energy ratio is:

$$\begin{aligned} E_a/E_b &= \\ h f_a/h f_b &= \\ =2 f_b/f_b &= \\ \text{Answer} &= 2; \\ \text{*Option A*} \end{aligned}$$

So, the correct answer is: 2

- A. Diffraction
- B. Reflection
- C. Refraction
- D. Dispersion

Answer;
 Option C
 Mirages are caused by refraction of light.

Explanation;
 14. Mirages are caused by? total (effective) inductance is the sum of the When inductances are connected in series, the individual inductances:

$$L_{total} = L_1 + L_2$$

Given:

$$\begin{aligned} L_1 &= 2H \\ L_2 &= 1H \\ L_{total} &= 2H + 1H = 3H \end{aligned}$$

Explanation; 16. Power can be expressed in each of the following ways except

$$\begin{aligned} \text{C. } P &= I^2 R \\ \text{A. } P &= N \\ \text{B. } P &\equiv V^2 \end{aligned}$$

When light passes through layers of air with $1^2/R$ different temperatures, its speed changes due to varying air densities. This bending of light,*known as

refraction,* creates the illusion of water or distorted images on hot surfaces like roads or deserts.

Answer;

$$*B.P = I^2/R*$$

Explanation:

Power in an electric circuit is gotten from the multiplication of the current I in the circuit by the voltage V.

$$*P=NI$$

Different variants of the formulae are gotten from ohm's law which is represented as $V=IR$.

$$\text{Thus } I = V/R$$

By substituting I and V in the initial formula for power, we can get all the following;

$$*P = I^2R*$$

$$*P = V^2/R*$$

However, we cannot get $P = I^2/R$, thus making it the odd one.

17. If the angle of incidence is 25° . During total internal

Reflection the critical angle is equal to θ . Which of the following

Is correct

A. θ is greater than 25°

B. θ is less than 25°

C. θ is equal to 25°

D. θ is equal to 50°

Answer;

*B. θ is less than 25° *

Explanation:

Total internal reflection occurs when light travels from a denser medium to a less dense medium and the angle of incidence exceeds the critical angle (θ)

The critical angle is the minimum angle of incidence at which total internal reflection occurs.

For total internal reflection to happen, the angle of incidence must be greater than the critical angle or, the critical angle must be less than the angle of incidence

18. What does peak current of an AC circuit NOT depend on

A. Frequency

B. Angle

C. Peak voltage

D. All of the above

Answer;

D.All of the above

Explanation;

The peak current of an AC circuit depends on all the parameters given

Frequency : Since the current in the circuit is periodic, it's value depends on its frequency.

Angle : A periodic current is dependent on the phase angle of its motion

Peak voltage : From ohm's law, there's a direct variation between the current and the voltage thus they are dependent on each other.

19. Human eyes are most sensitive to ____

- A. All colors in equal proportion
- B.Red
- C. Blue
- D.Green

Answer;

D.Green.

Explanation:

The human eye is most sensitive to *green light*, which has a wavelength of around 555

nanometers under normal daylight conditions. This is due to the way the photoreceptor cells (cones) in the retina are structured they respond to different wavelengths of light

20. "E" in LASER means ____

- (a) Emission
- (b) Emissivity
- (c) Extortion
- (d) Emitted

Answer;

A. Emission

Explanation:

- LASER stands for *Light Amplification by Stimulated Emission of Radiation*.
- The "E" in LASER refers to *Emission*, specifically the process of stimulated emission, which is a key principle behind how lasers work.

21. "S" in the word LASER stands for ____

- (a) Stimulation
- (b) simplifier
- (c) stimulated
- (d) simplification.

Answer;

C. stimulated.

Explanation:

LASER stands for *Light Amplification by Stimulated Emission of Radiation*,

The "S" in LASER refers to *Stimulated*, as in *Stimulated Emission*, which is the process by which photons are emitted in a coherent manner to produce laser light.

22. Which of the following best describes the phenomenon occurred when a rainbow is visible

- I. Refraction
- II. Reflection
- III. Diffraction
- IV. Opacity

- A. I only
- B. II only
- C. III only
- D. IV only

Answer;

A-I only

The correct answer is:

1. Refraction

When a rainbow is visible, sunlight undergoes refraction as it enters a raindrop, bending and dispersing into different colors. Then, *internal reflection* occurs inside the droplet, followed by another refraction as the light exits, forming the rainbow.

Note that total internal reflection is not directly under reflection, rather it's under refraction.

That's why refraction is the most suitable

23. The focal length of a convex mirror is 12cm, the object distance is 6cm what's the image distance

- A. 12cm in front of the mirror
- B. 12cm behind the mirror
- C. 4cm in front of the mirror
- D. 4cm behind the mirror

$$* \frac{1}{f} = \frac{1}{u} + \frac{1}{v} *$$

Sign Conventions

* For a convex mirror, the focal length (f) is negative.

* The object distance (u) is positive when the object is in front of the mirror.

* The image distance (v) is positive when the image is real (in front of the mirror) and negative when the image is virtual (behind the mirror).

Applying the Equation

D. Decrease conductivity

Given:

F=-12cm (convex mirror)

U=6cm

Plug the values into the mirror equation:

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

Solve for 1/v:

$$\frac{1}{v} = \frac{1}{-12} - \frac{1}{6}$$

$$\frac{1}{v} = -\frac{1}{12} - \frac{2}{12}$$

$$\frac{1}{v} = -\frac{3}{12}$$

$$\frac{1}{v} = -\frac{1}{4}$$

$$v = -4\text{cm}$$

Interpreting the Result

* The negative sign for v indicates that the image is virtual and located behind the mirror.

* The image distance is 4cm.

Therefore, the correct answer is D. 4cm behind the mirror.

24. In semiconductors the addition of acceptor electrons.....

- A. Increase resistivity
- B. Increase holes in the valence band
- C. Increase electrons in the conduction band

Answer;

B. Increase holes in the valence band

Explanation:

- In semiconductors, the addition of acceptor atoms* (such as boron in silicon) creates *holes* in the valence band. Acceptor atoms have fewer valence electrons than the semiconductor material, so they "accept" electrons from the valence band, leaving behind holes (positive charge carriers).
- This process increases the number of holes in the valence band, which is characteristic of *p.type semiconductors*.
- The resistivity may decrease (not increase) because holes contribute to conductivity.
- Electrons in the conduction band are not significantly increased by acceptor atoms; instead, holes in the valence band are the primary effect.
- Conductivity typically increases (not decreases) due to the presence of holes.

25. Which of the following has the highest frequency.

- A. Microwave
- B. Ultraviolet
- C. Visible light
- D. Infrared ray

Answer:

B. Ultraviolet

In the electromagnetic spectrum, frequency increases in the following order:

RadioWaves<Microwaves<Infrared<Visible light<Ultraviolet<Xray<Gamma ray.

Ultraviolet(UV) rays have a higher frequency than visible light, infrared rays, and microwaves. Higher frequency corresponds to shorter wavelengths and higher energy

PHYSICS 2025

1. If $A=BC$ and $A=MLB=(MN-2LT) \wedge -$

1) What is C

2. Tricky questions came out from momentum and KE like, the KE of a ball is 5×10 the momentum is 7×10 what is the mass what is the velocity

3. The critical angle required for when the angle of incidence is 25

A. =25

B. <25

C. >25

D. 25/1

4. Two object P and Q, have the same momentum. Q has more kinetic energy than P if it:

A) weighs more than P

B) weighs same as P

C) is moving slower than P

D) is moving faster than P

5. Two object P and Q, have the same momentum. Q has more kinetic energy than P if it:

A) weighs more than P

B) weighs same as P

C) is moving slower than P

D) is moving faster than P

6. The mass of a ball is 160g and extension 0.4m. what is the force

A) 4 constant

B) 400

C) 40

D) 640

7. On a hot day, water on the floor is due to

A. Refraction

B. Dispersion

C. Reflection

D.

8. A mass of 2.5kg, time of 4 seconds (Falls to the earth)

What is the momentum?

A. 0

B. 98

C. 48

D. 28

9. Electric lines of force

Which is false?

A. Travels from positive to negative

B. It can be used to sketch an electric field

C. It can be used to determine the path that a charge will follow

D. When they are closest, the field is smaller

10. Peak current doesn't depend on

- A. Frequency
- B. Angle
- C. Capacitance
- D. Peak voltage

11. A mass of 2.5kg, time of 4seconds (Falls to the earth)

What is the momentum?

- A. 0
- B. 98
- C. 48
- D. 28

12. What is the radius of curvature of an object placed 20cm in front of a concave mirror whose image is 2 times the size of the object?

- A. 27cm
- B. 20cm
- C. 10cm
- D.

13. The mass of a body is 3kg and the energy is 600J($g = 10m/s^2$) calculate the height 14. A boy holds an object of 40N for 10sec the height of his hand is 1.5m calculate the work done before he dropped the body

15. Plank constant is the same as a. Energy
b. velocity C. angular momentum

Arrange the following in single, double and triple bond chlorine, oxygen and nitrogen

16. If the Planck's constant is constant, $E =$

i. $h/2$

ii. h/π

iii. $3h/2\pi$

iv. $h/2\pi$

17. A stone thrown at 30m/s reached the ground 4s later.

18. Another stone is released from a vertical height, at what time will it reach the ground?

If a moving trolley A hit a stationary trolley B

Which of the following is correct

- i. The kinetic energy is equal to momentum
- ii. The momentum is equal before and after the collision
- iii. the momentum increases
- iv

19. S in laser stands for?

20. E in laser stands for?

21. An electrolyte of mass 6.4g with electrochemical equivalence of $1.6 \times 10^{-4}g/c$ was passed through current of 40A. What is the time it takes to decompose

A. 1000s

B. 40000s

- C.0.0001s
- C.glucacoma
- D.riverblindness

22. The following is/are true about evaporation and boiling point of water

- I. Evaporation occurs on the surface
- II. Boiling takes place through out the water body

III. Boiling point of water at 1atm is 100°C and evaporation occurs at no specific temperature surfacing

23. A convex mirror has focal length 12cm. An object is placed 6cm in front of the mirror,what is the image distance?

A. 6cm in front of the,mirror

C.4cm in front of the mirror

D.4cm behind the mirror

24. Acceptor electrons when added to a pure what is the semiconductor

D. Decreases conductivity

26. the carrier in electrolyte

- A.mobile ion
- B.mobile electron
- C.mobile holes

27. which of the following is used as road

- A.disel
- B.kerosene
- C.bitumen
- B.6 cm behind the mirror
- D.lubricating oil

28. A convex mirror has a focal length of 12cm image height.and the object height is 6cm

- A.4cm behind the mirror
- A. Increases holes in valence band
- B.12 cm behind the mirror
- B. Increases resistivity
- C.4cm in front of the mirror
- C. Increases electrons in conduction band
- D.12cm in front of the mirror

25. which of the following is eye defect

a.longsightedness 29. which are associated with hearing

b.shortsightedness

I.iris

li.cornea

lvi.ossicle

lvii.pinna

30. which are associated with eye vision

A.cornea

B.iris

C.pinna

D.ossicle

31. .which is not correct

A.difference between node and adjacent antinode is half of wavelengths

B.difference btw node and adjacent is quarter of wavelengths

C.difference btw two node is half wavelength

D.difference btw two antinode is half of wavelength

32. A positively charged glass that was charged with silk, what will happen if brought close to the insulating rod

I.Loss of electron

II. Gain of electron

III. Gain of proton

IV. Loss of proton

33. Index refraction for diamond is 2.5 which of the following is correct. a. $\sin a/\sin c=2.5$ b. $a/c=2.5$ c. $\sin c/\sin a=2.5$ d. $\sin b/\sin d =2.5$ 34. . The smallest particle of any chemical element that can exist by itself and retain qualities that distinguish it as an element is

known as a. Proton b. Atom c. Neutron

D.Molecule

35. If a machine is lubricated with oil then a.Both its mechanical advantage and efficiency increases b. The mechanical advantage increases c. The mechanical efficiency increases d. Mechanical advantage decreases efficiency increases

36. A radioactive element has a mass of 200g and its half-life is 10 days calculates its mass after 30 days a. 25 b. 75 c. 50 d. 100

37. A string as a resultant of two pulses acting in opposite direction to it what is later observed. a. The string is straightened b.The pulses disappears c. The pulses lose energy d. The string becomes slicked

38. Which of the following is not a mechanical wave a. Radio waves b. Sound waves c.Rope waves d. Water waves

39. If the focal length is multiplied by N to get the radius of Curvature

N is

A $1/2$

B2

C $1/4$

D4

40. Peak current doesn't depend on

A. Frequency

B. Angle

C. Capacitance

D. Peak voltage

41. The Dimension for power is A

ML^2T^{-2}

B ML2T-3

C MLT2

D ML-1T2

42. Which of the following is correct

Sound wave is Transverse

Sound waves can be polarized

Light is a longitudinal wave

Transverse waves can be polarized

II. Capacitance is proportional to the distance between the surface of the plates

III. Capacitance is inversely proportional to the distance between the two plates

A. I-III

B. III only

C. I-II

D. I and III

E. II only

43. A mirror is rotating at 100 rpm, if a light is incident on it, what is the rotation of the light?

A. 40 rpm

B. 100 rpm

C. 50 rpm

D. 200 rpm

46. The focal length of a circular concave mirror is 12cm, if an object is placed at the 18cm mark

A. The image is at infinity

B. The image is at 36cm in front of the mirror

C. At the radius of the circular mirror

D. 7.8cm from the mirror

44. The electron of an atom has;

A. No mass but charge

B. Mass but less than that of proton

C. Equal mass with the proton

D. No mass and charge

47. The half-life of an element of 200g is 10 days. Find the remaining mass after 30 days

A. 75g

B. 50g

C. 100g

D. 25g

45. Choose the odd one for a capacitor

I. Capacitance is proportional to the Area of plates

48. Arrange the following in the order of their frequency;radio waves(r),visible light (v),infrared light(i),x-ray(x), ultraviolet light(u)

- a.r xvul
- b.vxuir
- c.rivux
- d.rviux.

E.is moving at the same speed as P.

49. The units of planck's constant are those

- of?a.momentum
- b.Energy times time
- c.speed
- d.wavelength

50. .The rainbow seen after a rain shower is caused by?

- A.diffraction
- B.Polarization
- C.Interference
- D.Absorption
- E.Refraction

51. Two objects ,P and Q,have the same momentum.Q has more kinetic energy than P if it.

- A.Weighs more than P
- B.is moving faster than P
- C.weighs the same as P
- D.is moving slower than P

52. Suppose $A=BC$, where A has the dimension L/M and C has the dimension L/T . Then B has the dimension:

- A. T/M
- B. L^2/TM
- C. TM/L^2
- D. L^2T/M
- E. M/L^2

53. A body of mass m and a body has a mass $16m$ the ratio of their frequency if the force constant is the same is

- A. 4:1
- B. 16:1
- C. 1:16
- D. 1:4

54. if the critical angle is 25° during total internal reflection the angle of incidence is θ which of the following is correct

- A. θ is greater than 25°
- B. θ is less than 25°
- C. $\theta = 25^\circ$
- D. θ equals 50°

55. if the refractive index of glass is 1.25 what is the sine of the critical angle as the ear glass interface

- a 0.5
- b 0.2

- c 0.3
- d 0.25
- e 0.67

56. The DC generator is essentially the same component as the AC generator except the presence of

- a carbon brushes
- b commutators
- c. Armature
- d. split rings

57. The unit of magnetic field is

- A. Tesla
- B Weber
- C coulomb
- D. Joules

58. How many number of turns will be required to produce an output voltage of 120 volts if the input voltage is 240 volts and the number of turns is 500

- a. 750
- b. 250
- c. 1000
- d. 500

59. A material of mass 200g has a half life of 10 days. What's of it would remain after 30 days.

- I. 100g II. 75g III. 50g IV. 25g

60. A pendulum oscillates 20 times in 10 seconds. What is the period.

- I. 0.5 Hz II. 0.5 s III. 2 Hz IV. 2 s

61. Which one is impossible

- I. A body having velocity east and east acceleration
- II. A body having zero velocity and non zero acceleration.
- III. A body having constant acceleration and variable velocity.
- IV. A body having variable acceleration and constant velocity.

The number 4 question is a diagram so I can't type it. Pls check the picture I sent.

62. Calculate the time taken for an object to be dropped from an height of 45m ($g=10$)

- A 3
- B 9
- C 4
- D 2

63. Calculate the relative atomic mass of an element that contains two isotopes

One of mass 184.953 which is 37.4% and another one of 62.6% of mass 186.963

I can't remember the options though

64. Calculate the time of flight for an object of velocity 20 ms^{-1} ($whg = 10 \text{ m/s}^2$)

I have forgotten the option though

65. In intrinsic semiconductor

A. The number of holes are greater than the number of electrons

B. Less

C. Same

66. The p. d at the centre of a hollow metal is

A. V

B. -V

C. Zero

67. Two projectile are in flight at the same time, the acceleration of one relative to another is

A. 9.8

B. 19.6

C. Zero

68. Which of this is equal to planks constant h

I. Momentum

II. Energy

III. Energy x time

IV force

69. Through which substance does the current enters and leaves the electrolyte

A Electrode

B Voltmeter

C Voltmeter

D Electrolyte

70. Which of the following types of waves cannot travel through a vacuum?

A. Sound waves

B. Light waves

C. Infra-red waves

D. X-radiation

E. Radio waves

Answer: A. Sound waves

71. The temperature at which water vapor in the air saturates the air and begins to condense is known as:

A. Boiling point

B. Melting point

C. Triple point

D. Dew point

E. Critical temperature

Answer: D. Dew point

72. The units of capacitance are equivalent to:

A. J/C

B. V/C

C. J²/C

D. C/J

73. A moving charge in a magnetic field experiences: I. Force perpendicular to velocity and field. II. Force along the

velocity. III. Force along the field. IV. No force if moving parallel to the field. A) I and IV only · B) I only · C) II and III only · D) I, II and IV only · E) All of the above

74. Ohmic conductors: I. Obey Ohm's law. II. Have constant resistance. III. Show linear V-I graph. IV. Resistance varies with constant temperature. A) I, II and III only · B) I and IV only · C) II and IV only · D) I, II, III and IV · E) All of the above

75. The core of an electromagnet is made of soft iron because soft

I. small susceptibility iron has

III. large retentivity

II. small retentivity

IV. Small density

V. Large susceptibility

A. I & III only

B. IV & V only

D. III & V only

C. II & V only

E. I & II

76. CO₂ is a..

A. Mixture

B. Compound

C. Element

D.

77. Arrange the following in the order of frequency; radio waves (r), visible light (v), infrared light (i), x-ray (x), ultraviolet light (u)

a. rxvul

b. vxuir

c. rivux

d. rviux.

78. The units of Planck's constant are those of

a. momentum

b. Energy times time

c. speed

d. wavelength

79. The rainbow seen after a rain shower is caused by?

A. diffraction

B. Polarization

C. Interference

D. Absorption

E. Refraction

80. Two objects, P and Q, have the same momentum. Q has more kinetic energy than P if it.

A. Weighs more than P

B. is moving faster than P

C. weighs the same as P

D. is moving slower than P

E. is moving at the same speed as P.

81. Suppose $A=BC$, where A has the dimension L/M and C has the dimension L/T . Then B has the dimension:

- d 0.25
- e 0.67

- A. T/M
- B. L^2/TM
- C. TM/L^2
- D. L^2T/M
- E. M/L^2

82. A body of mass m and a body has a mass 16 m the ratio of their frequency if the force constant is the same is

- A. 4:1
- B. 16:1
- C. 1:16
- D. 1:4

83. if the critical angle is 25° during total internal reflection the, angle of incidence is θ which of the following is correct

- A. θ is greater than 25° B. θ is less than 25°
- C. $\theta = 25^\circ$
- D. θ equals 50°

84. if the refractive index of glass is 1.25 what is the sine of the critical angle as the glass interface

- a 0.5
- b 0.2
- c 0.3

85. The DC generator is essentially the same component as the AC generator except the presence of

- a carbon brushes
- b commutators
- c. Armature
- d. split rings

86. The unit of magnetic field is

- A. Tesla
- B Weber
- C coulomb
- D. Joules

87. How many number of turns will be required to produce an output voltage of 120 volts if the input voltage is 240 volts and the number of turns is 500

- a. 750
- b. 250
- c. 1000
- d. 500

88. If the mass of a sphere is 60 grams and it is floating such that two-thirds of its volume is immersed in water of density 1 g/cm^3 , what is the volume of the sphere?

89. What is the terminal velocity of an object falling through a fluid?
- The velocity at which the force of gravity equals the force of friction
 - The velocity at which the force of gravity equals the force of buoyancy
 - The maximum velocity an object can reach while falling
 - The velocity at which the object stops accelerating
90. A car travels half the distance l with speed v and the other half with speed v . What is the average speed for the entire trip.....
91. What is the work done in moving a charge q through a potential difference v
92. .What is the principle behind the operation of a transformer?
- Electromagnetic induction
 - Magnetic resonance
 - Electric conduction
 - Thermal expansion
93. The persistence hearing of sound after the source has been removed is ----- a reflection b. Echo c. Reverberation d . refraction
94. Which of the following is correct
- Transverse wave cannot be polarised
 - Transverse wave can be polarised
 - Mechanical waves are transverse
 - Transverse waves cannot be reflected
95. The focal length is N times the radius of curvature when N is
- a. $1/4$ b. $1/2$ c. 1 d 2
96. which of the following will change when wave move from a dense to denser medium.
- a frequency b. velocity c.amplitude d.97. Which of the following indicate the energy stored in a capacitor
- $Q^2/2C$
 - C/V^2
 - Q/V ,
 - QV^2
98. Which of the following is incorrect for a capacitor
- $C=fO$
 - $C=1/d$
 - $C=d$
 - $C=A$
99. Which of the following is required for a nuclear reaction to occur
- Temperature
 - Catalyst
 - Pressure
 - Heat

A None of I-iv

BI-IV

C I, iii,iv

D iv only

100. The SI standard of length is based on:

A. the distance from the north pole to the equator along a

meridian passing through Paris

B. wavelength of light emitted by Hg

C. wavelength of light emitted by Kr86

D. a precision meter stick in Paris

E. the speed of light

101. A ball thrown vertically upward from an height will

A. Travel in a Straightline

B parabolic

C.Curved

D. I've forgotten

102. Capacitance of capacitor connected in parallel can be increased by

A. Decreasing the distance apart the plate

B. Decreasing the area of the plate

C increasing the potential difference across the plate.

D. I've forgotten

103. Half life of radioactive element is:

A. Time taken for the radioactive element to disintegrate

B. Time take for half of the element to decay

C. Time taken for half of the original/initial element present to decay

D. I've forgotten

104. The donor replacement of silicon number is

A.3

B.4

C.5

D.6

There were calculations under capacitance. We were given values for charges and voltage then asked to calculate work done/energy.

Calculations on wavelength given speed of light and frequency. I can't remember the numerical value for each parameters given.

105. In which of the following areas is radioactive used.

- I. Agriculture
- II Weather forecast
- III industry
- IV medicine

108. A femtometer is

- A. 10^{-9}m
- B. 10^{-12}m
- C. 10^{-15}m
- D...less than 10^{-18}m

I've forgotten how the Roman numerals were paired, however there was no I-IV In the options.

ANSWERS

106. A
bimetallic/thermostat (one of them was used but can't remember which) made of two different metals will be different in

- A. mass
- B. Length
- C linear expansivity. :
- D. I don forget

107. A knife is made of metal and woody handle, the metallic part of the knife was cold to touch on a cool day. Which of the following is true

- A. The wood absorb heat than the metal
- B The metal is a good conductor
- C. The....

N:B for this question, I've forgotten the options but the question is still on point.

- 1. $C = M^3T$
- 2..
- 3. B
- 4. D
- 5. D
- 6. A
- 7. A
- 8. B
- 9. D
- 10. B
- 11. B
- 12. A
- 13. 20m
- 14. 0J
- 15. C
- 16. IV
- 17. 4secs
- 18. II ONLY
- 19. Stimulated
- 20. Emission
- 21. A
- 22. 1, II, III
- 23. D
- 24. A
- 25. ALL
- 26. A
- 27. C
- 28. A

29. III& IV	73.A
30. A and B	74.A
31.A	75.C
32. II ONLY	76.B
33.A	77.C
34.B	78.B
35.A	79.E
36.A	80.B
37.A	81.A
38.A	82.A
39.B	83.A
40.B	84.0.8
41.B	85.B
42.D	86.A
43.D	87.B
44.B	88.90cm ³
45.E	89.A
46.D	90..
47.D	91.W=qV
48.C	92.A
49.B	93.C
50.E	94.II ONLY
51.B	95.B
52.A	96.B
53.A	97.A
54.A	98.C
55.0.8	99.A
56.B	100. E
57.A	101. A
58.B	102. A
59.IV	103. C
60.II	104. C
61.IV	105. I,III,&IV
62.A	106. C
63.186.21	107. B
64..	108. C
65.C	
66.C	
67.C	
68.11I	
69.A	
70.A	
71.D	
72. C/V or C2/J	

UI POST UTME 2024

ENGLISH

1 which of these is correct

- A.bizarre
- B.bizzarre
- C.bizarre
- D.bizzare

2. which of these is correct

- A.Weird
- B.Wierd
- C.Weid
- D.Wied

3.which of these is correct

- A. Onomatopiea
- B. Onomatopia
- C. Onomatopoeia
- D. Onomatopoaie

4. Which of these syllable is the word stressed

- A.celeBRAtion
- B CElebration
- C.ceLEbration
- D.celebraTION

5. Which of following follows the pattern of the below word

CONFIDENTIAL

- I obsolete ii conspicuous iii pluralism
iv.community v. Purify
- Aionly

B ii and iii

C. lii and iv

Dland v

E li only

6. Hardly had she gone they started fighting

- A.And
- B.Than
- C.When
- D.Then

7. My sibilings..... in the cinema

I is ii was iii were iv are

- A. lii only
- B. li and iv
- C. Iv and iii
- D.li only

8. Choose the onefrom the given sentence that is the appropriate answer

My mother's FRIEND hates pen

- A. Does your mother boss hate pen?
- B. Does your mother's friend love pets?
- C. Does your mothersfriend hate toys?
- D. Does your mothers friend hate pets?

9.Choose the one from the given sentence that is the appropriate answer

Most candidates found that the problem was PRETTY difficult

- A. Did most candidates find the problem somewhat difficult?
- B. Did a few candidate find that problem pretty difficult?

iv

c Did most teachers find that problem pretty difficult?

D. Did most candidates find that problem pretty easy?

10. Find the antonym of the capitalised word

Most politicians are FRUGAL in their spending

I Miser

li Wasteful

lii Extravagant

Iv Parsimonious

A. I only

B. li and iv

C. lii only

D. I and iv

11. Identify the synonym of the capitalised word

My father lived a BAROQUE lifestyle

I Fancy

li Ornate

lii Bizarre

Iv Elegant

A. lii only

B. lii and iv

C. I-iv

D. None of the above

12. Now that Adumu had seen his Achilles heel, woman, he has revealed the whole secret

Which idiom BEST describes this

I Adami has spilled the milk

li Adamu had spilled the beans

lii Adamu had told the gospel truth

If Adamu has let the cat out of the bag

A. lii only

B. I and iv

C. Iv only

D. li and iv

13 Choose the vowel sound that is represented below

birth

I heart ii swirl iii soldier iv journey

A. ii and iv

B. I and iv

C. li only

D. None of the above

14. Which of these spelling is correct

A. Supercede

B. Supersede

C. Suppercede

D. Suppersede

15. The two boys stared at....

A. One another

B. each other

C. Themselves

D. Their selves

16. The PASSENGER boat docked at Accra yesterday

A. Did the passenger boat docked at Accra last week

B. Did the cargo boat docked at Accra yesterday

C. Did the passenger boat docked at tema yesterday

D. Did the cargo boat sail from Accra yesterday

17 The.....collection of her book has just been published

A. Postmortem

B. Posthumous

C. A and B

D. Ludicrous

18. No sooner had he left..... It started raining

A. When

B. Than

C. That

D. Then

19. The principal, with his family members.....here yesterday

A. Are

B. Were

C. Is

D. Was

20. Choose the word with the same sound says

I contain

li leopard

lii wicked

A. I and ii only

B. li only

C. lii only

D. I only

21. Fjord

Choose the options that corresponds with the above word

I Ford

li chord

lii cord

Iv Cob

A. I, ii and iii

B. I and iii

C. I and iv

D. I only

22. Many a student..... Rice

A. Likes

B. Like

C. Liked

D. A and C

23. Only the girls did the entire work

A. Didn't they

B. Don't they

C. Won't they

D. Wouldn't they

24 Choose the word that rhymes with the below word

Suit

A. Sweet

B. Soothe

C. Suite

D. Soot

25. Choose the word that rhymes with the below word

Uncle

I sun ii son iii correct iv body v orange

A. I, iv, v

B. lii and iv

21.D

C.I and ii

22.A

D. I,ii
and v

23.A

E. li
and

ili
only

24.D

25.A

Answers

1.C

2.A

3.C

4.A

5.E

6.C

7.C

8.A

9.A

10.D

11.D (Because the supposed answer I and II are not
in the option)

12.D

13.A

14.B

15.B

16.B

17.B

18.B

19.D

20.B

UI POST UTME

ENGLISH

Choose the option that is NEAREST IN
MEANING to the underlined word.

1 They are notorious for their
DIABOLICAL plans.

- o A. diametrical
- o B. evil
- o C. practical
- o D. theoretical

2 His INDUSTRY saw him through in
the foreign land.

- o A. hard work
- o B. factory
- o C. commerce

o D. wealth

3 A sot is the same as:

- o A. a cheat
- o B. analcoholic
- o C. a simpleton
- o D. an urchin

4 There is nodoubt that INDOLENCE
is the cause of the man's
predicament.

- o A. slothfulness
- o B. indulgence
- o C. indiscipline
- o D. indigestion

5 Are you SCEPTICAL about the

woman's promise to finance your
education in the United Kingdom?

- o A. worried
- o B. nervous
- o C. doubtful
- o D. thankful

6 The subject is a SINE QUA NON.

- o A. a compulsory
- o B. an optional one
- o C. a very difficult one
- o D. a simple one

7 Obika is too TEMPERAMENTAL to
be the captain of the team.

- o A. moody
- o B. timorous
- o C. unstable in character
- o D. gentle

8 The demonstrators ran
HELTHER-SKELTER when they
sighted the mobile policemen.

- o A. bare-footed
- o B. fast
- o C. into a shelter
- o D. in disordered haste

9 Choose the option that is
OPPOSITE IN MEANING to the
underlined word.

10 The man is PRISSY but the woman
is:

- o A. careful
- o B. careless
- o C. lazy
- o D. prayerful

11 James is COLD-HEARTED but

Bantu is:

- o A. sympathetic
- o B. hot-hearted
- o C. warm-minded
- o D. fearless

12 It is either their efforts to check corruption in our society are ...or EULOGIZED by many people.

- o A. emphasized
- o B. manipulated
- o C. denigrated
- o D. justified

13 While the girls COUNTENANCED the proposal, the boys ... it.

- o A. counterbalanced
- O B. demurred
- o C. estimated
- o D. viewed

14 Although the woman said the killing of her goat was PREMEDITATED, the accused maintained that the killing was:
A. postmeditated
O B. inadvertent

- o C. studied
- O D. preconditioned

15 Is the man's contribution to the programme ... or STUPENDOUS?

- o A. wise
- O B. idiotic
- o C. inconsequential
- o D. horrendous

16 While Simbo is in the VANGUARD of the struggle, Samba is in the:

- o A. motorguard
- O B. armoured car
- o C. safeguard
- o D. rearguard

17 The girls are EXUBERANT but the boys are:

- o A. tolerant
- o B. melancholic
- o C. aberrant
- o D. foolish

18 The sitting-room is DINKY.

- o A. small and dirty
- o B. large and neat
- o C. large and dirty
- o D. small and neat

19 The girls are ... but the boys are UPBEAT.

- o A. defeated
- o B. victorious

- o C.moody
- o D. cheerful

20. Neither of the presidential candidates.....here.

- o (a) are
- o (b) is
- o (c) were
- o (d) will

21.I knew David.....let us down.

- o (a) will
- o (b) might
- o (c) can
- o (d) shall

22. He who plays the piper.....the time.

- o (a) dictates
- o (b) dictate
- o (c) dictated
- o (d) dictating

23. You... me last week.

- o (a) ought to seen
- o (b) ought to had seen
- o (c) ought to has seen
- o (d) ought to have seen

24 Two wrongs.....a right.

- o (a) does not makes
- o (b) does not make
- o (c) do not makes

o (d) do not make 25.....is added to a letter after it
has been printed in case of mistake.

- o (a) addendum
- o (b) postscript
- o (c) prelude

26 The decision is.....

- o (a) yours
- o (b) your's
- o (c) yours'
- o (d) your

27..... house was burnt.

- o (a) Whose
- o (b) Who
- o (c) Whom
- o (d) Who's

28 Chief Thomas served..... the Board of Education last year.

- o (a) on
- o (b) at
- o (c) in
- o (d) for

29. The counsel to the accused.....the court to adjourn the case.

- o (a) prayed
- o (b) advised
- o (c) begged
- o (d) requested

Choose the word that is spelt

correctly in the following options:

30 (a) Generocity

o (b) Generrosity

o (c) Generosity

o (d) Generousity

31 (a) Proffesion

o (b) Proffession

o (c) Profescion

o (d) Profession

32(a) Barbeque

o (b) Barbeque

o (c) Babecue

o (d) Barbecue

33(a) Guarantee

o (b) Guarantte

o (c) Guaranttee

o (d) Guarantee

34(a) Diarhoae

o (b) Diarhoea

o (c) Diarrhoea

o (d) Diarrhoae

35(a) Millennium

o (b) Milenium

o (c) Millenium

o (d) Millennium

36 (a) Connoisseur

o (b) Conoisieur

o (c) Connoiseur

o (d) Conoiseur

37(a) Occurrence

o (b) Occurence

o (c) Ocurrence

o (d) Ocurence

38 (a) Geurillas

o (b) Geurrillas

o (c) Guerrillas

o (d) Guerillas

39(a) Etiquete

· (b) Ettiquette

· (c) Ettiquete

· (d) Etiquette

Choose the word that is nearest in

meaning to the underlined word:

40 The vice chancellor always looks at
his political colleagues

SUPERCILIOUSLY.

o (a) amazingly

o (b) disdainfully

o (c) lovingly

o (d) carefully

41 Many people look into the future with
TREPIDATION.

o (a) certainty

o (b) uncertainty

o (c) fear

o (d) faith

42. The arrival of Mensah made the
party become ANIMATED.

o(a) rough

03/06/14/100
24/06/12/0
06/06/14/08

o (b) chaotic

o (d) anointed

43 One of the interviewees

demonstrated STUPENDOUS

intelligence.

o (a) astonishing o (d) forgery

o (c) stupid

o (d) idiotic

PECUNIARY consideration.

o (b) power

o (c) monetary

o (d) chieftaincy

45 Johnson has just written a MEMOIR.

o (a) biography

o (b) a story

o (c) an autobiography

o (d) a memory verse

46 Chief Ajete is a VENAL politician.

o (a) famous

o (b) powerful

o (c) weak

o (d) corrupt

47. Syphilis is a VENEREAL disease.

o (c) lively 48. The man was jailed
for PERJURY.

o (a) falsehood

o (b) stealing

o (c) murder

o (b) foolish 49. The meal is okay for a
SEDENTARY worker.

44. He
did-what
he did
for his o
(a) lazy
community last
year for o (b)
hardworking

o (c) sitting

o (a) political Choose the word
that is opposite in meaning to the
underlined word:

50. The drug made the man
LETHARGIC.

o (a) insane

o (b) enthusiastic

o (c) asthmatic

o (d) tired

51. The boy looks like a MENDICANT
as he sits.

o (a) medical student

o (b) ruffian

o (c) benefactor

o (d) sailor

o (a) widespread 52. Is the woman
selling fruits under that

baobab tree really COMPOS

MENTIS or?

- o (a) insane
- o (b) poor
- o (c) dirty
- o (d) considerate

53. Sabeto WVese is a young person of

FLIGHTY behaviour but Mamakoko

is ever.....?

- o (a) fickle
- o (b) arrogant
- o (c) gentle
- o (d) steady

54 While William made FLIPPANT

comments at the occasion, Tambo

made remarks.

- o (a) detailed
- o (b) polite
- o (c) rude
- o (d) flimsy

55 The writer of the controversial article

is ANONYMOUS but the writer of

the open letter to the President is:

- o (a) callous
- o (b) known
- o (c) brilliant
- o (d) lettered

56 The children say they enjoy from

their mother while they hate the

FORCE MAJEURE their father

demonstrates from time to time.

- o (a) love
- o (b) fearlessness
- o (c) laissez-faire
- o (d) carte blanche

57 The FAMILIARITY of your formula

makes it uninteresting just as the

.....of mine makes many people

admire it.

- o (a) facility
- o (b) uniqueness
- o (c) clarity
- o (d) visibility

58. While Peter's theory is said to be

DUD, Paul's is considered.....

- o (a) bright
- o (b) flawless
- o (c) drab
- o (d) exotic

59 In your opinion, are all earthly

phenomena STATIC or ? ;

- o (a) dynamic
- o (b) stable
- o (c) magnetic
- o (d) sadistic

Choose the option that sounds the

same with the given word:

60.KEY

- o (a) kiss o (d) intenTIONal
- o (b) quay 66 management
- o (c) keyed o (a) MAnagement

- 61.TWO
- o (a) tool
 - o (b) too
 - o (c) to
 - o (d) tow

- 62.KNOW
- o (a) now
 - o (b) nose
 - o (c) known
 - o (d) no

- o (a) list o (d) blue
- o (c) wright
- o (d) rim

Choose the most appropriate o (c) bath

64. congratulation

o (c) intentionAL

o (d) kissed o (b) maNagement

o (c) manaGEment

o (d) manageMENT

Choose the option that has the same sound as the one represented by the letter underlined:

67.brEW

o (a) sew

o (b) build

63 RITE
o (c) guilt

o (b) wit
68.healTHY

o (a) worthy

o (b) with

stress
pattern
from the
options:o
(d) breathe

o (a) congraT Ulation
69.cUnny
o (b) congratu LAtion o
(a) van
o (c) CONgrat

ulation o
(b) does
o (d)
conGRAt
ulations
o (c)
float

65. intentional o (d) bully

o (a) inTENTional 70. Which of these is correctly spelt? (a)

o (b) INTentional Brouhaha

(b) Bruhaha

(c) Brohaha

(d) Brouhaha

71 She always prides herself____her beauty. (a) on

(b) with

(c) of

(d) at

72. Handpick the grammatically right option. (a) Your room is in a shambles

(b) Your room is in shamble

(c) Your room is a shamble

(d) Our room is in a shamble

73 The seller will not take a____from any customer. (a) promissory note

(b) promisory note

(c) promise note

(d) promisary note

74. He____left before I came.I. Has

II. Would

III. Had

IV. Have

(a) I, III

(b) II, III

(c) II

(d) III

75 Having won the cup three times, my club has won the cup. (a) for keeping

(b) by keep

(c) for keeps

(d) for keeping

76 Choose the correct sentence. (a)

Pulip is a moneybags

(b) Pulip is a moneybag

(c) Pulip is a money - bag

(d) Pulip is a money - bags

77. The woman has gone to inform her husband _the police station. (a)

against

(b) about

(c) of

(d) for

78 Those____field are in a position to right the wrongs. I.in

II.on

III. on the

IV. in the

(a)I

(b)1I,III

(c) IV

(d) I, IV

79. After the meeting, he was____I, applause

II. applaud

III. applauded

IV. applausd

(a) I,IV

(b) 1I

(c) II,III

(d) 1, II, IV

80. In the question below choose the option opposite in meaning to the

word in capital letters: He is loved for

his ALTRUISM. (a) benevolence

(b) sincerity

(c) selfishness

(d) selflessness

(e) kindness

81 Did the mechanic handle the repair

of your car engine in à MALADROIT

manner? (a) Skilful

(b) Simple

(c) Costly

(d) Complex

82 The boy's OBEISANCE contrasts

with the girl's____(a) Laziness

(b) Obedience

(c) Recalcitrance

(d) Recklessness

83 It is a QUIRKY, not a____event in

my own opinion. (a) Strange

(b) Fair

(c) Funny

(d) Familiar

84. While I live in a CAPACIOUS room

in the village, Dimka lives in____

room in the city. (a) a capricious

(b) a rough

(c) a poky

(d) an attractive

85. Choose the option nearest in

meaning to the italicizedwords in

the following sentence: He is a

LOUT and can't be relied upon at all.

(a) vagabond

(b) thug

(c) unserious

(d) liar

86 Lorenzo may be your____he is my

ADVERSARY. (a) Crony

(b) Adviser

(c) Customer

(d) Foe

87 The bully wanted to COW me into

submission but I remained adamant.

(a) Dehumanize

(b) Beg

(c) Intimidate

(d) Animalize

88 His explanation of that point was

LUCID. (a) Lengthy

(b) Judicious

(c) Clear

(d) Complicated

89. The evidence of the second witness

CORROBORATES that of the

accused. (a) Contradicts

(b) Reinforces

(c) Cancels

(d) Coordinates

90 Choose the most appropriate option

to complete the gap in the following

sentence: I am sure all those who

mistook President Buhari's___

demeanour for cowardice will soon

know him for who he is. (a) stealthy

(b) sloppy

(c) snaky

(d) succulent

91 Choose the correct spelling. (a)

Guerrilla

(b) Guerila

(c) Guerrila

(d) Guerilla

92. Choose the correct spelling. (a)

Trauma

(b) Trauma

(c) Truama

(d) Trauma

93 Choose the correct spelling. (a)

Lieutenat

(b) Lieutenant

(c) Leiutenants

(d) Leiutenant

94. Choose the correct spelling.(a)

Proclame

(b) Proclaime

(c) Proclam

(d) Proclaim

95. Choose the correct spelling. (a)

Noodles

(b) Nodles

(c) Noodlees

(d) Nodels

96 "I loved the way the lightning danced

in the sky". This expression is__

(a) Metaphor

(b) Personification

(c) Simile

(d) Irony

97 "The stick is a machine". This

expression is___I. Simile

II. Personification

III. Euphemism

IV. Apostrophe

(a) I only

(b) I & III only

(c) IV only

(d) II & IV only

98. "Mowing the lawn was a walk in the

park". This expression is a/an__

(a) Epigram

(b) Metaphor

(c) Simile

(d) Oxymoron

99."The fall leaves danced across the lawn". The correct figure of speech is____(a) Alliteration

(b) Ephemeral

(c) Personification

(d) Metaphor

Answers

1 DIABOLICAL - Nearest in meaning: B. evil

2 INDUSTRY - Nearest in meaning: A. hard work

3 A sot - Nearest in meaning: B. an alcoholic

4 INDOLENCE - Nearest in meaning: A.slothfulness

5 SCEPTICAL - Nearest in meaning: C. doubtful

6 SINE QUA NON - Nearest in meaning: A. a compulsory

7 TEMPERAMENTAL - Nearest in meaning: A.moody

8 HELTER-SKELTER - Nearest in meaning:D.in disordered haste

9 PRISSY - Opposite in meaning: B. careless

10 COLD-HEARTED - Opposite in meaning: A.sympathetic

11 EULOGIZED - Opposite in meaning: C.denigrated

12 COUNTENANCED - Opposite in meaning: B.demurred

13 PREMEDITATED - Opposite in meaning: B.inadvertent

14 STUPENDOUS-Opposite in meaning:c inconsequential

15 VANGUARD- Opposite in meaning: D.rearguard

16 EXUBERANT-Opposite in meaning: B.melancholic

17 DINKY-Opposite in meaning: B. large and neat

18 UPBEAT-Opposite in meaning: C. moody

19 Neither of the presidential candidates..Correct option: B. is

20 i knew David... let us down- Correct option:B.might

21 He who plays the piper... the time - Correct option: A. dictates

22 You... me last week - Correct option: D.ought to have seen

23 Two wrongs... a right - Correct option: D.do not make

24... is added to a letter after it has been printed - Correct option: B. postscript

25 The decision is ... - Correct option: A. yours

26... house was burnt - Correct option: A.Whose

27 Chief Thomas served ... the Board of Education - Correct option: A. on

28 The counsel to the accused ... the court to adjourn - Correct option: A. prayed

29 Correct spelling: C. Generosity

30 Correct spelling: D. Profession

31Correct spelling: D. Barbecue

32 Correct spelling: A. Guarantee

33 Correct spelling: C. Diarrhoea

34 Correct spelling: D. Millennium

- 35 Correct spelling: A. Connoisseur
- 36 Correct spelling: A. Occurrence
- 37 Correct spelling: C. Guerrillas
- 38 Correct spelling: D. Etiquette
- 39 SUPERCILIOUSLY- Nearest in meaning:
B. disdainfully
- 40 TREPIDATION - Nearest in meaning: C. fear
- 41 ANIMATED - Nearest in meaning: C. lively
- 42 STUPENDOUS - Nearest in meaning: A.
astonishing
- 43 PECUNIARY - Nearest in meaning: C.
monetary
- 44 MEMOIR - Nearest in meaning: C. an
autobiography
- 45 VENAL - Nearest in meaning: D. corrupt
- 46 VENEREAL - Nearest in meaning: B. sex-
related
- 47 PERJURY - Nearest in meaning: A. falsehood
- 48 SEDENTARY - Nearest in meaning: C. sitting
- 49 LETHARGIC - Opposite in meaning: B.
enthusiastic
- 50 MENDICANT - Opposite in meaning: C.
benefactor
- 51 COMPOS MENTIS - Opposite in meaning: A.
insane
- 52 FLIGHTY - Opposite in meaning: D. steady
- 53 FLIPPANT - Opposite in meaning: B. polite
- 54 ANONYMOUS - Opposite in meaning: B.
known
- 55 FORCE MAJEURE - Opposite in meaning: C.
laissez-faire
- 56 FAMILIARITY - Opposite in meaning: B.
uniqueness
- 57 DUD - Opposite in meaning: B. flawless
- 58 STATIC - Opposite in meaning: A. dynamic
- 59 KEY - Sounds the same: B. quay
- 60 TWO - Sounds the same: B. too
- 61 KNOW - Sounds the same: D. no
- 62 RITE - Sounds the same: C. wright
- 63 Congratulation - Stress pattern: D.
conGRAtulations
- 64 Intentional - Stress pattern: A. inTENTional
- 65 Management - Stress pattern: B.
maNAgement
- 66 brEW - Same sound: D. blue
- 67 healTHY - Same sound: A. worthy
- 68 cUnny - Same sound: D. bully
- 69 Correct spelling: D. Brouhaha
- 70 She always prides herself ... her beauty-
Correct option: A. on
- 71 Correct sentence: A. Your room is in a
shambles
- 72 The seller will not take a ... from any
customer - Correct option: A. promissory note
- 73 He... left before I came - Correct option: D.
I'll (Had)
- 74 Having won the cup three times, my club has
won the cup... - Correct option: C. for keeps
- 75 Correct sentence: B. Pulip is a moneybag
- 76 The woman has gone to inform her husband
... the police station - Correct option: B. about
- 77 Those... field are in a position to right the
wrongs - Correct option: D. I, IV (in, in the)
- 78 After the meeting, he was ...- Correct
option: C. II, III (applaud, applauded)

- 79 ALTRUISM - Opposite in meaning:
C.selfishness
- 80 MALADROIT - Opposite in meaning: A. Skilful
- 81 OBEISANCE - Opposite in meaning:
C.Recalcitrance
- 82 QUIRKY - Opposite in meaning: D. Familiar
- 83 CAPACIOUS - Opposite in meaning: C. a poky
- 84 LOUT - Nearest in meaning: C. unserious
- 85 ADVERSARY - Nearest in meaning: D. Foe
- 86 COW - Nearest in meaning: C. Intimidate
- 87 LUCID - Nearest in meaning: C. Clear
- 88 CORROBORATES - Nearest in meaning:
B.Reinforces
- 89 President Buhari's ... demeanour -
Correct option: A. stealthy
- 90 Correct spelling: A. Guerrilla
- 91 Correct spelling: A. Trauma
- 92 Correct spelling: B. Lieutenant
- 93 Correct spelling: D. Proclaim
- 94 Correct spelling: A. Noodles
- 95 "I loved the way the lightning danced in the
sky" - Figure of speech: B. Personification
- 96 "The stick is a machine" - Figure of speech:
A.I only (Simile)
- 97 "Mowing the lawn was a walk in the park" -
Figure of speech: B. Metaphor
- 98 "The fall leaves danced across the lawn"
-Figure of speech: C. Personification

Choose the option that is NEAREST IN MEANING to the underlined word.

They are notorious for their DIABOLICAL plans. (a) Diametrical (b) Evil (c) Practical (d) Theoretical

1. His INDUSTRY saw his through in the foreign land. (a) Hardwork (b) Factory (c) Commerce (d) Wealth

2. A SOT is the same as____(a) A cheat (b) An alcoholic (c) A simpleton (d) An urchin

3. There is no doubt that INDOLENCE is the cause of the man's predicament.(a) Slothfulness (b) Indulgence (c)Indiscipline (d) Indigestion

4. Are you SCEPTICAL about the woman's promise to finance your education in the United Kingdom? (a) Worried (b) Nervous (c) Doubtful (d) Thankful

5. The subject is a SINEQUANION. (a) Compulsory (b) An optional one (c) A very difficult one (d) A simple one

6. Obika is too TEMPERAMENTAL to be the caption of the team. (a) Moody (b) Timorous (c) Unstable in character (d) Gentle

7. The demonstrators ran HELTER-SKETER when they sighted the mobile policemen. (a) Barefooted (b) Fast (c) Into shelter (d) In disordered haste

8. Choose the option that is **OPPOSITE IN MEANING** to the underlined word.

9. The man is PRISSY but the woman is (a) Careful (b) Careless (c) Lazy (d) Prayerful

10. James is COLD-HEARTED but BANTU is____(a) Sympathetic (b) Hot hearted (c) Warm-minded (d) Fearless

11. It is either their efforts to check corruption in our society are____or EULOGIZED by many people. (a) Emphasized (b) Manipulated (c) Denigrated (d) Justified

12. While the girls COUNTENANCE changed the proposal, the boys_ it.

(a) Counterbalanced (b) Demurred (c) Estimated (d) Viewed

13. Although the woman said the killing of her goat was PREMEDITATED, the accused maintained that the killing was (a) Post-meditated (b) Inadvertent (c) Studied (d) Preconditioned

14. Is the man's contribution to the programme_____or STUPENDOUS? (a) Wise (b) Idiotic (c) Inconsequential (d) Horrendous

15. While Simbo is in the VANGUARD of the struggle, Samba is in the____(a) Motor guard (b) Armored (c) Safeguard (d) Rearguard

16. The girls are EXUBERANT but the boys are____(a) Tolerant (b) Melancholic (c) Aberrant (d) Foolish

17. The sitting-room is DINKY. (a) Small and Dirty (b) Large and Neat (c) Large and Dirty (d) Small and Neat 18. The girls are____but the boys are UPBEAT.

(a) Defeated (b) Victorious (c) Moody (d)
Cheerful

:
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1
9
F
9
F

		Answers
Choose the word that is SPELT CORRECTLY in the following options;		1.B
19. (a) Occurrence(b) Ocuurrence		2. A
(c)		3. B
20.(a) Geurillas Ocurrencie		4. A
(d) Ocurence		5. D
(b) Geurrillas		6. A
(c) Guerrillas (d)		7. C
Guerillase		8. D
21. (a) Etiquete (b) Ettiquette		9.B
(c) Ettiquete (d)		10.A
Etiquette		11.C
Choose the option that SOUNDS THE SAME with the given word;		12.B
22.KNOW (a) Now		13.B
(b) Nose (c) Known		14.C
(d) No		15.D
23. RITE (a) List (b)		16.B
Wife (c) Wright (d) Rim		17.B
Choose the most appropriate STRESS PATTERNS from the options;		18.C
24.Intentional : (a) inTENTional		19.A
(b)INtentional (c) intentionAL		20.C
(d)intenTIONal		21.D
25. Management (a) MAnagement (b)		22.D
maNAgement (c) manaGEment (d)		23.B&C
management		24.A
		25.A

ENGLISH 2021

In the options lettered A-D, choose the **CORRECT** **OPTION** that fits in the expression.

1. Neither of the presidential candidates_here. (a) Are (b) Is (c) Were (d) Will
2. I knew David_let us down. (a) Will (b) Might (c) Can (d) Shall
3. He who plays the piper_____the time. (a) Dictates (b) Dictate (c) Dictated (d) Dictating
4. You_____me last week.(a) Ought to seen (b) Ought to had seen (c) Ought to has seen (d) Ought to have seen
5. Two wrongs____a right. (a) Does not makes (b) Does not make (c) Do not makes (d) Do not make
6. is added to a letter after it has been printed in case of mistake. (a) Addendum (b) Postscript (c) Prelude
7. The decision is_____(a) Yours (b) Your's (c) Yours' (d) Your
8. house was burnt. (a) Whose (b) Who (c) Whom (d) Who's
9. Chief Thomas served_____the Board of Education last year. (a) On (b) At (c) In (d) For
10. The counsel to the accused_____ the court to adjourn the case. (a) Prayed (b) Advised (c) Begged (d) Requested

Choose the work that is **NEAREST** IN **MEANING** to the underlined word.

11. The vice chancellor always look at his political colleagues **SUPERCILIOUSLY**. (a) Amazingly (b) Disdainfully (c) Lovingly (d) Carefully

12. Many people look into the future with **TREPIDATION**. (a) Certainty (b) Uncertainty (c) Fear (d) Faith
13. The arrival of Mensah made the party become **ANIMATED**. (a) Rough (b) Chaotic (c) Lively (d) Anointed
14. One of the interviewees demonstrated **STUPENDOUS** intelligence. (a) Astonishing (b) Foolish (c) Stupid (d) Idiotic
15. He did what he did for his community last year for **PECUNIARY** consideration. (a) Political (b) Power (c) Monetary (d) Chieftaincy
16. Johnson has just written a **MEMOIR**. (a) Biography (b) A story (c) An autobiography (d) A memory verse
17. Chief Ajeteisa **VENAL** politician. (a) Famous (b) Powerful (c) Weak (d) Corrupt

Choose the word that is **OPPOSITE** IN **MEANING** to the underlined word.

18. The drug made the man **LETHARGIC**. (a) Insane (b) Enthusiastic (c) Asthmatic (d) Tired
19. The boy looks like a **MENDICANT** as he sits. (a) Medical student (b) Ruffian (c) Benefactor (d) Sailor
20. Is the woman selling fruits under that baobab tree really **COMPOS MENTIS** or (a) Insane (b) Poor (c) Dirty (d) Considerate
21. Sabeto Wese is a young person of **FLIGHTY** behaviour but Mamakoko is ever_____ (a) Fickle (b) Arrogant (c) Gentle (d) Considerate
22. While William made **FLIPPANT** comments at the occasion, Tambo

ANSWERS

made remarks. (a) Detailed (b) Polite
(c) Rude (d) Flimsy

- 1.B
- 2. B
- 3. A

Choose the option that SOUNDS THE SAME with
the given word.

- 4.D
- 5. D

23. KEY (a) Kiss (b) Quay
(c) Keved (d) Kissed
24. TWO (a) Tool (b) T00
(d) To (d) Tow

- 6.B
- 7.A
- 8.A
- 9.A
- 10.A
- 11.B

Choose the most appropriate stress pattern from
the options.

- 12.C
- 13.C

25. Congratulation (a) congraTulation
(b) congratuLation (c)
CONgratulation (d)
conGRAtulations

- 14.A
- 15.C
- 16.A
- 17.D
- 18.B
- 19.C
- 20.A
- 21.D
- 22.B
- 23.B
- 24.A/B
- 25.B

ENGLISH 2022

Choose the word that is NEAREST IN MEANING to the underlined word.

1. The vice chancellor always look at his political colleagues SUPERCILIOUSLY.
(a) Amazingly (b) Disdainfully
(c) Lovingly (d) Carefully
2. Many people look into the future with TREPIDATION. (a) Certainty
(b) Uncertainty (c) Fear (d) Faith
3. The arrival of Mensah made the party become ANIMATED. (a) Rough
(b) Chaotic (c) Lively (d) Anointed
4. One of the interviewees demonstrated STUPENDOUS intelligence. (a) Astonishing (b) Foolish
(c) Stupid (d) Idiotic
5. He did what he did for his community last year for PECUNIARY consideration.
(a) Political
(b) Power (c) Monetary (d) Chieftaincy
6. Johnson has just written a MEMOIR. (a) Biography (b) A story (c) An autobiography
(d) A memory verse
7. Chief Ajete is a VENAL politician.
(a) Famous (b) Powerful (c) Weak
(d) Corrupt
8. Syphilis is a VENERAL disease. (a) A wide spread (b) A sex-related (c) A realistic (c) An undisclosed
9. The man was jailed for PERJURY.
(a) Falsehood (b) Stealing (c) Murder
(d) Forgery
10. The meal is okay for a SEDENTARY worker. (a) Lazy (b) Hardworking (c) Sitting

Choose the word that is OPPOSITE IN MEANING to the underlined word.

11. The writer of the controversial article is ANONYMOUS but the writer of the open letter to the President is ____ (a) Callous (b) Known
(c) Brilliant (d) Lettered
12. The children say they enjoy from their mother while they hate the FORCE MAJEURE their father demonstrates from time to time.
(a) Love (b) Fearlessness (c) Laissez-faire (d) Carteblanche
13. The FAMILIARITY of your formula makes it uninteresting just as the of mine makes many people admire it. (a) Facility (b) Uniqueness (c) Clarity (d) Visibility
14. While Peter's theory is said to be DUD, Paul's is considered
(a) Bright (b) Flawless (c) Drab
(d) Exotic
15. In your opinion are all earthly phenomena STATIC of
(a) Dynamic (b) Stable (c) Magnetic
(d) Sadistic

Choose the option that has the SAME SOUND as the represented by the letter underlined.

16. brEW (a) Sew (b) Build (c) Guilt (d) Blue
17. HealTHY (a) Worthy (b) With (c) Bath
(d) Breathe
18. cUnny (a) Van (b) Does (c) Float (d) Bully
19. (a) Noodles (b) Nodles (c) Noodless (d) Nodels

20. I loved the way the lightening danced in the sky'.This expression is (a) Metaphor (b)Personification (c) Simile (d) Irony 21.The stick is a machine 'This expression is. I. Simile; II.Personification: III. Euphemism; IV;Apostrophe. (a) I only (b) I and III only (c) IV only (d) II & IV

22. Mowing the lawn was a walk in the park 'This expression is a/an (a)Epigram (b) Metaphor (c) Simile (d)Oxymoron

23.The fall leaves danced across the laws. 'The correct figure of speech is_____ (a) Alliteration (b)Ephemeral (c) Personification (d)Metaphor

ANSWER

1.B

2.C

3.C

4.A

5.C

6.A

7.D

8.B

9.A

10.C

11.B

12.C

13.B

14.B

15.A

16.D

17.C

18.B

19. No phonetic symbol

20.B

21.No Answer

22.B

23.C

ENGLISH 2023

1. Which of this is correctly spelt? (a) Brohaha
(b) Bruhaha (c) Brohaha (d) Brouhaha
2. She always prides herself ____ her beauty. (a)
On (b) With (c) Of (d) At

HAND PICK the grammatically right option.

3. (a) Your room is in a shamble
(b) Your room is in shamble
(c) Your room is a shamble
(d) Our room is in a shamble
4. The seller will not take a ____ from any customer. (a) Promissory note (b) Promisory note (c) Promise note (d) Promisary note
5. He _____ left before I came. I. Has;
II. Would; III. Had; IV. Have
6. Having room won the cup three times, my club has won the cup. (a) For keeping
(b) By keep (c) For keeps (d) For keeping

Choose the **CORRECT SENTENCES**.

7. (a) Pulip is a money bags
(b) Pulip is a money bag
(c) Pulic is a money-bag
(d) Pulip is money bags
8. The woman has gone to inform her husband at the police station.
(a) Against (b) About (c) On (d) For
9. Those ____ field are in position to right the wrongs. I. In; II. On; III. On the; IV. In the. (a) I (b) II, III (c) IV (d) I, IV

10. After the meeting, he was ____ I. Applause;
II. Applaud; III. Applauded; IV. Applauded.
(a) I, IV (b) III (c) II, III (d) I, II, IV

In the question below choose the option **OPPOSITE IN MEANING** to the word in capital letters.

11. He is loved for his **ALTRUISM**.
(a) Benevolence (b) Sincerity (c) Selfishness
(d) Selflessness
12. Did the mechanic handle the repair of your car engine in a **MALADROIT** manner? (a) Skillful (b) Simple (c) Costly (d) Complex
13. The boy's **OBEISANCE** contrast with the girls _____. (a) Laziness (b) Obedience
(c) Recalcitrance (d) Recklessness
14. It is a **QUIRKY**, not a _____ event in my own opinion. (a) Strange (b) Fair (c) Funny
(d) Familiar
15. While I live in a **CAPACIOUS** room in the village Dimka lives in _____ room in the city.
(a) Acapricious (b) A rough (c) A poky (d) An attractive

Choose the option **NEAREST IN MEANING** to the italicized words in the following sentence.

16. He is a **LOUT** and can't be relied upon at all. (a) Vagabond (b) Thug (c) Unserious
(d) Liar
17. Lorenzo may be your _____ he is my **ADVERSARY**. (a) Crony (b) Adviser
(c) Customer (d) Foe
18. The bully wanted to **COW** me into submission but I remained adamant. (a) Dehumanize (b) Beg (c) Intimidate
(d) Animalize

19. His explanation of that point was LUCID.(a) Lengthy (b) Judicious (c) Clear (d) Complicated
20. The evidence of the second witness **CORROBORATES** that of the accused.(a) Contradicts (b) Reinforces (c) Cancels (d) Coordinates

Choose the MOST APPROPRIATE option to complete the gap in the following sentence.

21. I am sure all those who mistook President Buhari's_____demeanour for cowardice will soon know him for who he is.
(a) Stealthy (b) Sloppy (c) Snaky(d)Succulent

Choose the **CORRECT SPELLING**.

22. (a) Guerrilla (b) Guerila
(c) Guerrila (d) Guerilla
23. (a) Trulma (b) Trumua
(c) Truama (d) Trauma
24. (a) Lieutenat (b) Lieutenant
(c) Leiutenants (d) Leiutenant
25. (a) Proclame (b) Proclaime
(c) Proclam (d) Proclaim

ANSWERS

- 1.D
2. A
3. No answer
4. A
- 5.II
6. Incorrect question
- 7.A
- 8.C
- 9.C
- 10.B
- 11.C
- 12.A
- 13.C
- 14.D
- 15.C
- 16.B
- 17.D
- 18.C
- 19.C
- 20.B
- 21.A
- 22.A
- 23.D
- 24.B
- 25.D

pick the option that shows the correct spelling of the word.1. (a) diarrhea (b) diarheoa (c) diarrhoea(d)diarhea

2. (a) Catarh (b)Cattarh (c) Catarrh (d)Cattarrh

3. (a) supersede (b) superseed (c)superceed (d)

4. (a) Vaacu (b) Vaacum (c) Vacuum (d)Vacumm

5. (a) rithm (b) rythm (c) rhythm (d) rhydhm

6. (a) ocassion (b) occasion (c) occassion

7. (a) comitment (b) comittment

(d)gargoyle (c) gargoyle (e) gargoyle

(d)gargoyle (c) gargoyle (e) gargoyle

Choose the option that has the correct stress pattern of the word.

9. (a) ACcommodating (b) acCOmmodating (c)

accoMMOdating (d) accommoDAting 10. (a)

COmmunualism (b) coMMUnualism (c)

commuNUALism (d) communuaLIsm

11. (a)COLonialism (b) coLONialism

12. (a) REgulate (b) reGulate (c) reguLAt

(e) reGulate 14. (a) ACceptate (b)

(c) Photocopy (d) Photocopy (e) Photocopy

(c) Photocopy (d) Photocopy (e) Photocopy

15. (a) (COM)MUNISociality (b) (b)pmMUNISociality

(c) hospitaNtity (d) hospitaBility

16. (a) COMMUNISociality (b) (b)pmMUNISociality

(c) hospitaNtity (d) hospitaBility

(c) hospitaNtity (d) hospitaBility

(c) hospitaNtity (d) hospitaBility

(c) hospitaNtity (d) hospitaBility

(c) hospitaNtity (d) hospitaBility

(c) hospitaNtity (d) hospitaBility

(c) hospitaNtity (d) hospitaBility

(c) hospitaNtity (d) hospitaBility

(c) hospitaNtity (d) hospitaBility

(c) hospitaNtity (d) hospitaBility

19. (a) INtermittent (b) inTERmittent

(c)interMITtent (d) intermitTENT

20. (a) Universal (b) uNlversal (c)

uniVERsal (d) univerSAL

21. (a) Opinion (b) oPInion (c) opiNlon

(d)opiniON

22. Many people go to conference from

WORK. (a) Do many people go to

conference fromm work? (b) Do many

people go to conference from house?

(c)Do many people go to house from

work?(d) How many people go from

conference to work?

Choose the correct interpretation of the following idioms.

23. Everyone knew he had to either

improve his job or quit. (a) He was

expected to shape in or ship out (b) He

was expected to cave in or cave out (c)

He was expected to get better or start

going

24. The boy finished six plates of food ;it

was simply an unpleasant sight. (a) He

ate like a pig (b) He ate like a horse (c)

He ate like a hen

25. Tolu told a big far cry from what

happened. (a) Told another version of the

story (b) Cried when she

wanted to spill out the beans (c) Was crying

26. You need to show me the ropes. (a) Show

me where to hang myself (b) Teach me how

to do something (c) tell me how to tie the

27. Went bananas (a) become extremely

happy (b) to buy bananas in bulk (c) to

choose from a difficult situation 28. The

new director's head hit the ceiling when he

heard the news. (a) he jumped up and hit

his head (b) he became very angry (c) he

died

Pick the correct option in each of the

following.29. One of the women who

the car _____died. (a) bought/have (b)

bought/has (c) buys/have (d) buys/has

Answer

30. The man, along with his children and wife, _____ here.

i. is ii. are iii. was iv. were (a) i only (b) ii and iv (c) i and iii (d) ii only

_____ foot.

31.1 _____ to go to school _i) used, on ii) use, by iii) use, on iv) used, by

(a) i only (b) iii only (c) i and iv (d) iii only

32. Pick the word that rhyme with 'fjord' i)

sword ii) flood iii) lord (a) i only (b) i and ii (c) i and iii

(d) i, ii and iii

33. Choose the antonym. The man showed

complete audacity. (a) weakness (b) reticence

(c) cowardice (d) bravery

34. The friction between siblings in movies

are exaggerated. The synonym is (a) struggle

(b) antagonism (c) hatred

35. Synonym of miniature is i) Compressed ii)

Smaller iii) Decreased iv) Abridged (a) i and ii (b) ii

alone (c) ii and iii (d) ii and iv

20.C

21.B

22.B

23. A

24. A

25. A

26.B

27. A

28. B

29.B

30. D

31.A

32.C

33.B

34. 名

B

35. D

spellings questions on USE OF ENG that was asked:

I. Diarrhoea (double¹¹*r*¹¹ and 'hoea')

II. Catarrh (double r)

III. Supersede (('sede' ending)

IV. Vacuum (double ' u' and one 'm')

V. Rhythm

VI. Occasion (double 'c' and one 's')

VI. Gargoyle ('oyle' ending)

VII. Commitment (double ' m')

IX. Millennium (double ' l' and double 'n')

X. Liaison (' i' before and after letter a).

XI. Siege ('ie' before letter 'g')

XII. Communication (" double " m")

XIII. Caribbean (one ' r' and double ' bb')

XIX. Maintenance ('tenance' ending)

XX. Camouflage (there is no letter'r')

XXI.Abhorrence (double 'r')

XXII. Mischievous ("l " first before 'e' and "ous 'ending).

Stress Pattern

1. A. INtravenous. B.inTRAvenous
C.intraVENous D.intraveNOUS

2. A. statisTlcal B. statistiCAL C. staTISTical D. STATistical

3. Which of the following has its stress on the first syllable? I. Scintillating II.

Bouquet III. Wardrobe IV. Exercise A. I,III, IV B. I, IV C. I, II D. II, IV

4. Which of these is stressed on the first syllable? (i) Principled (ii) Political (iii)Gargoyle (iv) Grotesque (A) ii and iii (B) i and ii (C) iii and iv (D) i and iii

5. Which of the following is in appropriate stress

a.coLOnialism b.COlonialism

c.colonIalIsm d.colonialISM

6. a.acCOModating b.accoMODating

c.ACcomodating d.accomODating

7. A. PREposterous B. prePOsterous
C.prepoSTERous D. preposterOUs

8. A.commuNlcation B. COMMunication
C.communiCAtion D. comMUnication

9. A. DEMocracy B. deMOcracy
C.demoCRAcY D. democraCY

10. A. ACcentuate B. acCENTuate
C.accentUJate D. accentuATE

11. A. Universal B. uNlversal C. uniVERsal
D.univerSAL

12. A. preEminent B. preeMInent
C.PreemiNENT D. PREeminent

13. a HOSpitality b hosPItality c hospiTAlity
d hospitalITY

14. which word is stress correctly A. EN-vi-ron-ment
B. en-vi-RON-ment C. en-vi-ron-MENT
D.en-VI-ron-ment

15. which word is stress correctly A. i-den-ti-CAL
B. i-DEN-ti-cal C. i-den-TI-cal
D. I-den-ti-cal

16. A. HORTiculture B. horTlIculture
C.hortiCULTure D. horticultURE

17. A. PRiVilege B. priVllege C. priviLEge
D.privILEGE

18.A. PHOTography B. phoTOgraphy
C.photoGRaPhy D. photograPHY

19. A. REfrigerate B. reFRigerate
C.refriGERate D. refrigeRATE

20. Which of these is stressed on first syllable? i) Sniper ii) Phantom iii)

Commendable iv) arrive A. I, IV B. I, II
C.1, II & IV D.II,III 21. In the word import,
stress falls on the first syllable when it is
used as a...A)Noun B) Verb C) Adjective D)
Adverb

22. Everyone should take care of their

ENVIRONMENT

A. Should everyone take care of their
house?

B. Should anyone take care of their
environment?

C. Does anyone take care of their
environment?

D No one should take care of their
environment?

23. Phones are NECESSARY these days.

A) Phones are necessary tomorrow?

B) Phones are optional these days?

C) Cars are necessary these days?

D) Telephones are necessary these days?

24. Do many people WORK here? i.
Many people meet here ii. Little people
work here iii. Many child work here iv.
Many people work there A. i only b. i
and ii c. ii and iii d. iii and iv

SPELLING

25. A. Milenium B. Milllenium C.
Milliniem D.Millennium

26. A. Wierd B. Weird C. Wieird D. Weeird

27. a. catarrh b. catarh c. cattarrh d. cattarh

28. A. diarrhea B. diarheoa C. diarrhoa
D.diarrhoea

29. A) Siege B) Seege C) Sieige D) Seige

30. A. Supercede B. Supersede C.
Supercide D. Suppercide

31. A. Vacuum B. Vaccuum C. Vaccum
D.Vacum.

32. A. Occasion B. Ocasion C. Ocassion
D.Occassion

33. I. Interpret II. Interpret III. Interprite
IV.Interpreete

34. I. Privilegell.PrivyledgeIII. □ fNied %三

Privilege

35. A. Bizzare B. Bizare C.Bizarre0.Brrtarre

36. A. CommitmentB. Comittment

37. A. Gargyle B. Gargyle C GargoYelD
Gargyole

38. A.rhythm B. rithym C. rhthym D.rythm

39. A. lacquer B. Laquer C. Lacqel D. Lacque

40. A. Enterprenuer B. Entrepreneur c
Entrepreneur D. Entorpreneur

SENTENCE INTERPRETATION

41. To give somebody up is to a. give up
a ghost b. scorn the person c. criticize
the person d. keep away from the
person

42. Tunde is a bull in a china shop. a. a
very careful but rough person b. a very
careful person c. a very greedy person d.a
very powerful person

43. Adele confided in Selena but Selena
exposed her to the public.

I. Selena spilled the milk

II. Selena spilled the beans

III. Selena sold her out

IV. Selena let the cat out of the bag

A) II and IV

B) I and IV

C) All of the above

D) none of the above

44. The team found the project to be a
piece of cake a.it is baked with cake b.it is
a easy task c.it broken into piece of cake

45. He's advised not to put too many irons
in the fire.

(i) to become a blacksmith

(ii) chew more than what can
fit his mouth

(iii) harm himself with the
many irons in the fire

(iv) commit himself to too
many tasks

- (A) i and iii
- (B) iv only
- (C) i and iv
- (D) ii and iv

46. The man did a lot of mistake in the company, he is expected to improve or start going a. he should cave in and cave out b. he should shape in or shape out c. he should improve or be going

47. His Excellency "hit the ceiling" when he heard about the huge estimate for the new road construction. This statement means _____

- I. His Excellency was joyous
- II. His Excellency was suspicious
- III. His Excellency was angry
- IV. His Excellency threw an object to the ceiling

48. Toluwase told Susan is plans to move out of the country but she failed him

- I. Susan let the cat out of the bag
- II. Susan spilled the milk
- III. Susan spilled the beans
- IV. Susan sold him out.

A. 1

B. 1&II

C. All of the above

D. I & IV

E. I, II & IV

49. You have to brush up your communication skills before the oratory competition tomorrow

- I. improve in your communication skill
- II Change your brush

III buy a new brush 50. His excellency hit the ceiling when he was presented with a huge estimate for road construction.

- I. the excellency became angry
- II. excellency became suspicious
- III. The excellency became excited
- IV. the excellency threw and object at the ceiling

- A. I only B. I and II C. III only D. IV only

51. Justin is a straight arrow everyone knows that.

- I. He is rigid and difficult to convince
- II. he is honest and upright
- III. He is ethical and moral
- IV. He is inaccessible and unaccommodating

A. I and II only B. I and III only C. I and IV only D. IV only.

52. After her examination her results was not unfavourable. A. She failed her exam B. Her results was inconclusive C. She passed her examination D.

53. Dapo finished all six dishes set before him. This means that

- I. He ate like a pig
- II. he ate like a bird
- III. he ate like a bird
- IV. he ate like a horse

54. Idiom of "break the ice" I. to do something reassuring li. to create a more relaxed environment lii. To have fun Iv. To act funny A. I, II B. I only C. II only D. I, II and III

55. Idiom of "hit the ceiling" I. To get angry li. To be surprising lii. To be straight forward Iv. To lose temper A. I, IV B. I only C. IV only D. I, II and III

56. The men were not pawns in someone else's political game.

I. The action they executed was their idea.

II. The men used someone else's plan.

III. They were used by someone's political game.

IV. The men were not being manipulated by someone else in a political scheme or strategy.

A. i & ii. B. i & ii. C. i & iii. D. i & iv.

57. The woman had a nodding acquaintance with the doctor.

- a. the woman was an old friend of the doctor
 b. the woman had a casual relationship with the doctor
 c. the woman and the doctor were lovers
 d. the woman and the doctor were enemies

SYNONYMS

58. The friction between siblings in movies are exaggerated (i) struggle (ii) antagonism (iii) disfavour (iv) anger
 a. I & li b. ii c. li & iv d. iv

59. Frugality is one important quality to imbibe during hard times.

- I) Thriftiness
- II) Disposition
- III) Parsimony
- IV) Managing

A. I, II B. I, IV C. I, III D. I only

60. The FEROCITY of a bull.

- I. Fierceness
- li. Viciousness
- lii. Angst
- Iv. Apprehension

A. I, II B. I, IV C. I, III D. I only

61. You should consider the prospects of the job before taking it

- I. Potential
- li. Availability
- lii. Announcement
- Iv. Simplicity

62. Weddings are known to be GLAMOROUS events

- I. Sophisticated
- II. Wasteful
- III. Unnecessary
- IV. Extravagant

A. I and IV B. IV only C. II and III D. None of the above

63. Synonyms of frivolous

- I. Unserious
- II flippant

III. Sympathetic

IV. Unwise A. I, II & III B. C..WD.1only 64.

Synonyms of penurious.

- I. miserly
- li. happy
- lii. parsimonious

Iv. aggressive. A. I, II B. I, IV C. I, III D. I only

65. synonyms of tenacious

- I. Unyielding
- li. Determined
- lii. Persistent

Iv. Dogged A. I, II B. I, IV C. I, III D. All of the above

66. Oli chose only SELECT schools abroad I FEW

- II EXCELLENT
- III LARGE
- IV FAMOUS

67. The lady was completely nonplussed when her suitor rejected her gifts?

I. Unpuzzled. 1I. Perplexed. III Unconscious. IV. Bewildered. A. ii and iv. B. None of i-iv. C. i only. D. iii only.

68. the union leader's aggressive approach may jeopardize the success of the negotiations

- I. affect II. make dangerous III. endanger
- IV. nullify
- a. II only
- b. I and IV only
- c. I and III only
- d. IV only

ANTONYMS

69. Antonym of audacity (i) cowardice (ii) reticence (iii) resolve (iv) weakness A. I, II B. I, IV C. I, II & IV D. I only 70. Antonym of educated 1 lost (ii) misunderstood (iii) gauged (iv) repelled A. I, II B. II, III C. I only D. IV only

71. The man displayed a sense of audacity a.cowardice b.hope c.confidence d.weakness

72. The relationship between the brothers has been frigid

- I) warm
- II) Frugal
- III) Cordial
- IV) Tonic

A. I, II B. I, IV C. I, III D. I only

73. These politicians are SPENDTHRIFTS

- I. Frugal
- li. Wasteful
- lii. Carefree
- IV. Parsimonious

A. I only B. II & III C. II D. IV E. None of the

above

74. His new house is quite BAROQUE

- I. Extravagant
- li. Plain
- lii. Without flowers
- IV. Unpainted

75. antipathy

- I. Sympathy
- II. Pity
- III passion
- IV. enmity

76. Antonyms of Ominous

- I. auspicious
- li.Favorable
- lii.preserving

iv. Resilient A. I, II B. I, IV C. I, III D. I only

ORAL ENGLISH

77. Word that rhymes with Fjord a. Sword b.

Flood c. Lop d. Blood

- A.I only
- B.I,II IV
- C.IV
- D.I& III

78. Words that rhyme with gnarl: (i) snail (ii)snarl (iii) gnome (iv) knob (A) ii and iii (B) i and iv (C) ii and iv (D) ii only

79. Word that rhymes with school

- I.Shawl
- II. Tulle
- III. Scout
- IV. Cool

A. I, II B. II, IV C. I, III D. I only

80. mail

- I: gale
- li: gall
- lii: male
- IV: save

A. I, II B. I, IV C. I, III D. I only

81. Same sound Learn I. Birth ii. Swirl III.Search IV. Silt a. III only b. I and II only C.IV only D. II only

82. Pleasure (i) Seize (ii) Pressure (iii) Genre (iv) Fusion

(A) i and iii (B) ii and iii (C) iii only (D) all

except i 83. The word that has the same sound as Birch (i) Cappuccino (ii) Chalet (iii) Sachet (iv) Pitch (A) i and iv (B) ii, III and iv (C) iv only (D) iii and iv

84. bridge

- I.Silk
- li.Beach
- lii.Grinch
- IV. Search

85. The vowel sound in ship is...

- A) /i:/
- B)/I/
- C)/e/
- D)/ə/

86. Let I. Calm II. Yolk III. Colonel

IV. Black

87. Flood I. Floozy II. Sun III. Brood IV. Come 88. Push I)sugar li)chalet lii) gourd 89. Same sound art I) Artsy ii) Hearth

iii)Heart iv) Verse

90. pressure I.) Champagne ii) Measure
iii) shut iv.) Picture
91. coup I. whoop II. hoot III. couple
IV. scout

a. III and IV b. I and II c. II only d. IV only

GRAMMAR

92. One of the women who _____ the car died.
(i) bought/have (ii) bought/has
(iii) buy/has (iv) buys/have
(A) ii and iii (B) i and iv (C) i and iii (D) ii and iv
93. A number of protesting students _____ here (i) was (ii) would be (iii) where (iv) were
94. My wife and treasure _____ a good source of encouragement to me (i) are (ii) is (iii) are both (iv) were
95. A number of people _____ there a. is b. was c. were d. are
96. 30 percent of pupils _____ the examination
a. fails b. fail c. failling d. will fail
97. I have known him _____ fifteen years.
I) since
II) From
III) For
IV) Throughout
98. The team _____ playing well today.
I) are II) is III) were IV) be A. I, II B. I, IV C. I, III D. I only
99. Neither of the candidates qualified. A) are B) is C) have D) were
100. If I _____ rich, I would travel abroad. A) am B) were C) was D) will be
101. He hardly works, _____? A) does he B) doesn't he C) is he D) isn't he
102. The classroom was a beehive of activities. The figure of speech is..
A) Simile B) Metaphor C) Personification D) Irony

103. Choose the correct sentence:
A) He is senior than me. B) He senior than me. C) He is senior 6 四 30 te is senior over me. D) 香
104. Many a family _____ to square. ea
r

- I. Struggles, meals
II. Struggle, meal
III. Struggled, meals
IV. Struggle, meals

105. _____ is the man _____ we told you about

- I. Here, whom
II. There, who
III. There, whom
IV. Here, who

106. We will follow our parents wherever they go, _____? I. Wouldn't we?

- II. Won't we?
III. Shouldn't we?
IV. Should we?

107. The man together with his wife and children _____ present at the party. I. Is

- II. Are
III. Was
IV. Were

108. the teacher told the principal what she did,?

- I) didn't she ii) didn't he. iii) didn't they IV) did they v) don't she
a) I and ii b) I and ii c) I only d) IV and ii e) I and v

109. Which of the following is correct grammatically

- I) rice and beans is my favourite food
II) rice and beans are my favourite food
III) rice and beans are my favourite foods IV) rice and beans is my favourite foods. A) I and ii. b) I and ii c) ii and IV d) IV and iii

110. It is high time _from your job. I. You resigned ii. that you resigned III. You are resigning iv. You have resigned A. I and iii B. I and li C. I, li and III D. I only.

111. _____ Tunde _____ bola _____ to be blame 1. _____ neither/nor/is II. Either/and/are III neither/nor /am IV. either/or/am

a I and iv b ii and III c iv only d li only

112. I would rather you _____ up your room

- A. Tidy
- B. Tidied
- C. Will tidy
- D. Tiedy

113. It is high time you _____ today. A cooked B. Have cook C. Cook D has cooked

114. The teacher disagreed when she told him, _____?

- I. Didn't she
- II. Did she
- III. Didn't he
- IV. Isn't he

A. I, II B I only C. I and III D. IV only 115.

Neither of the presidential candidates suitable.

A. are B. is C. is been D. are been suitable

116. If I were the president, I ask for the money.

A. will B. should C. would D. can 117. “

You _____ now see the director.” the secretary told the visitors. Neither of the boys _____ guilty of the offence.

- I. are
- II. Is
- III. were
- IV. be A. I only B. II only C. III only D. IV only

119. The plural of crisis is:

I. crises

II. crises

III. crisi

IV. crisis

120. Paper is made _____ wood pulp I. on II. of III. from IV. with

a. III and IV b. III only c. I only d. IV only

		40.B
Answers		41.D
1.C		42.A
2.C		43.A
3. A		44.B
4. D		45.B
5. A		46.B
6. A		47.C
7. B		48.D
8.C		49.A
9.B.		50.A
10.B		
11.C		51.B
12.A		52.C
13.C		53.D
14.D		54.A
15.B		55. A
16.A		56.D
17.A		57.B
18.B		58.A
19.B		59.C
20.B		60.A
21.A		61.A
22.A		62.A
23.B		63.A
24.A		64.C
25.D		65.D
26.C		66.B
27.A		67.A
28.D		68.C
29.A		69.C
30.B		70.A
31.A		71.A
32.A		72.C
33.B		73.A
34.A		74.B
35.C		75.A
36.A	:	76.A
37.A		77.A
38.A		78.D
39.A		79.B
		80.C

81.B
82.D
83.A
84.1&111
85.B
86. III & IV
87. II & IV
88.1&11
89.1,II & III
90. 1,III, & IV

91.B
92.A
93.IV
94.11
95.C
96.B
97.111
98.A
99.B
100. B
101. A
102. B
103. C
104.1
105. I&III
106.1I
107. I&II
108.C
109. B
110. D
111. A
112. B
113. A
114.C
115.B
116.C
117. B
118.B
119.II
120. B

1 Which of the following will produce nitrogen(IV) oxide and oxygen when heated strongly?

- ☐ I. NaNO_3
- ☐ II. KNO_3
- ☐ III. $\text{Mg}(\text{NO}_3)_2$
- ☐ IV. $\text{Pb}(\text{NO}_3)_2$

- ☐ A. I-IV
- ☐ B. I-III
- ☐ C. III-IV
- ☐ D. IV only

2 Which of the following statements is/are correct of ΔH , the enthalpy of reaction?

- ☐ I. The heat change accompanying a chemical reaction
- ☐ II. Negative for exothermic reactions
- ☐ III. Positive for exothermic reactions

CHEMISTRY

☐ IV. Greater than ΔS for reversible reactions

- ☐ A. I-IV
- ☐ B. I-II
- ☐ C. III-IV
- ☐ D. IV only

3 In which of the following reactions are the underlined species oxidized?

- ☐ I. $2\text{Fe}^{2+} + \text{Cl}_2 \rightarrow 2\text{Fe}^{3+} + 2\text{Cl}^-$
- ☐ II. $\text{S}^{2-} + \text{Cl}_2 \rightarrow \text{S} + 2\text{Cl}^-$
- ☐ III. $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
- ☐ IV. $2\text{Al} + 6\text{H}^+ \rightarrow 3\text{H}_2 + 2\text{Al}^{3+}$

- ☐ A. I-IV
- ☐ B. I-III
- ☐ C. I, III, IV
- ☐ D. IV only

4 Which of the following decrease(s) from left to right across the period but increase(s) down the group in the periodic table?

- ☐ I. Ionization energy
- ☐ II. Electron affinity

o III. Electronegativity

electrons

o IV. Atomic radius



O B.I-III

o C.III-IV

o D. IV only

5 A homologous series is one in

which:

o I. All compounds can be

represented by one general

formula

o II. Successive members differ

from one another by CH₃

o III. Physical properties differ

only in terms of the number

of carbon atoms per

molecule

o IV. The members have the

same physical and chemical

properties

O A.I-IV

O B.I,III

O C.III-IV

o D. IV only

6 Which of the following is/are correct

of the ammonia molecule?

o I. It has a tetrahedral shape

o II. It has a lone pair of

☐ III. The H-N-H bond angle is 180°

☐ IV. It is a non-polar molecule

☐ A.I-IV

☐ B.1-III

☐ C.III-IV

☐ D. None of the above

7 What reaction(s) occur at the electrodes during the electrolysis of concentrated sodium chloride solution using carbon electrodes?

☐ I. Oxidation takes place at the anode

☐ II. Chlorine is evolved at the anode

☐ III. Hydrogen is evolved at the cathode

☐ IV. Oxygen is evolved at the cathode

☐ A.I-IV

☐ B.I-III

☐ C.I only

☐ D.IV only

8 Which of the following statements is/are correct of the gaseous state?

☐ I. Intermolecular spaces are very large

☐ II. Molecules move very fast

☐ III. Intermolecular forces are

negligible

O IV. Intermolecular spaces can

be compressed

O A.I-IV

O B.I-III

o C.III-IV

o D. IV only

9 Which of the following is/are
properties of ethanol?

o I. It is colorless

o II. It is miscible with water

o III. Its boiling point is 78°C

o IV. It is a strong electrolyte

OA.I-IV

O B.I-III

o C.III-IV

o D. IV only

10 Which of the following statements
is/are correct about equilibrium?

o I. Chemical equilibrium is
attained when the rates of
forward and backward
reactions are equal

o II. Changes in concentration
of reactants will alter
equilibrium concentration

o III. A catalyst alters equally
the rates of both the forward
and the backward reactions

o IV. Temperature affects
equilibrium concentrations

O A.I-IV

O B.I-III

o C.III-IV

o D. IV only

11 Aluminum is above iron in the
electrochemical series, yet iron
corrodes easily on exposure to air
while aluminum does not. This is
because aluminum:

o A. Has a lower density than

iron

o B. Is a better conductor than
iron

o C. Does not corrode
spontaneously

o D. Forms a thin layer of inert
oxide in moist air

12 Which of the following compounds
will undergo addition reactions?

o A. Ethyne

o B. Butane

o C. Pentane

o D. Tetrachloromethane

o E. Ethanol

13 Which of the following processes
is/are suitable for investigating
whether or not a liquid is pure

o I. Testing it with anhydrous copper (II) tetraoxosulphate

IV

o II. Testing it with cobalt(II) chloride paper

o III. Testing it with litmus

o IV. Determining its boiling

point

O A.I-IV

O B.I-III

o C.III-IV

o D. IV only

14 Molecules which have the same shape as methane include:

o I. CCl_4

o II. Cl_4

o III. C_2H_4

o IV. C_2H_2

OA.I-IV

O B.I-III

o C. III-IV

o D.I-II only

15 Which of the following statements is/are correct of the compound with the formula $\text{K}_4\text{Fe}(\text{CN})_6$?

o I. Its IUPAC name is

potassium

hexacyanoferrate(II)

O II. It has 6 ligands

O III. It is a complex salt of a transition metal

o IV. Its solution gives a green precipitate with aqueous sodium hydroxide

OA.I-IV+

O B.I-III

O C.III-IV

o D. IV only

16 The factors that affect the rate of chemical reactions include:

o I. Concentration

o II. Temperature

o III. Presence of a catalyst

o IV. Collision frequency of particles

OA.I-IV

OB.I-11

O C.III-IV

o D. IV only

17 Which of the following statements is/are correct?

o I. A primary cell cannot maintain a steady current over a long period

o II. Polarisation is said to occur in a cell when

hydrogen bubbles form on

the plate.

III. The Daniel cell is an

electrochemical cell

o IV. A primary cell can be

recharged

o C. III-IV

have the highest rate of diffusion

under the same conditions? (H=1,
C=12, O=16, S=32, Cl=35.5)

OA. O_2

OB. Cl_2

OC. HCl

OD. H_2S

O

OE. C_2

19 The ionic radii of metals are usually:

o A. Greater than their atomic

radii

o B. Unaffected by the charge

on the ion

o C. Less than their atomic

radii

o D. Greater than their atomic

radii

20 Which of the following can be

obtained from its oxide by heating

with carbon?

ol. Lead

O A. I-IV

O B. I-III

O B. I-III

O C. III-IV

18 Which of the following gases
will undergo selective discharge of ions

electrolysis depends on the:

o I. Position of ions in the
electrochemical series

o II. Nature of the electrodes

o III. Charge on the ions

o IV. Quantity of electricity
passed through the

electrolyte

O A. I-IV

OB. I-II

o C. III-IV

o D. IV only

22 Oxidation is a reaction which can
involve:

o I. Loss of electrons

o II. Increase in oxidation
number

o III. Gain of oxygen

o IV. Loss of hydrogen

8 — IV

O B.1-1II

o C. III-IV

o D. IV only

23 Which of the following solids has a network structure?

o A. Diamond

o B. Iodine

o C. Sulphur

o D. Graphite

24. Alums are classified as:

o A. Simple Salts

o B. Acid Salts

o C. Anhydrous Salts

o D. Double Salts

o E. Basic Salts

25 Which of the following is an Alloy of Mercury?

o A. Stainless steel

o B. Soft Solder

o C. Coinage Bronze

o D. Amalgam

o E. Duralumin

26. A positive brown ring test indicates the presence of:

O A. NO₂

O B. Fe³⁺

o C. SO₃²⁻

o D. Cu²⁺



27 Which of the following methods is/are used for the extraction of the alkali metals?

o I. Reduction by the use of carbon

o II. Roasting of the ore in air

o III. Thermal reduction

o IV. Reduction by Electrolysis

O A.I-IV

O B.I-III

O C.III-IV

o D. IV only

28 How many resonance structure are there in Co³⁺-

O A.3

O B.2

o C.4

OD.5

29 Which of the following reactions at equilibrium will be affected by an increase in pressure?

o I. $N_2 + 3H_2 \rightarrow 2NH_3$

o II. $2SO_2 + O_2 \rightarrow 2SO_3$

o III. $N_2 + O_2 \rightarrow 2NO$

ON. $CH_4 + H_2O \rightarrow CO_2 + H_2$

A.1-11

O B.I-III

o C.III-IV

o D. IV only

27d

28a

ANSWER

29a

1c

2b

3c

4d

5b

6d

7b

8a

9B

10a

11d

12 a

13d

14d

15a

16a

17 B

18 A

19c

20b

21b

22a

23a

24d

25d

26a

CHEMISTRY 2019

1 The oxidation states of S in

$\text{Ca}(\text{HSO}_3)_2$ is:

O(a)+2

o (b)-2

O(c)+4

o(d)-4

O(e)+6

2. The oxidation state of P in $\text{H}_2\text{P}_2\text{O}_7$ is:

O(a)-3

O(b)+3

O(c)+1

o (d) -2

O(e)+5

3. Potassium ethanoate is formed

when:

o (a) Methanoic acid reacts

with KOH

o (b) Ethanoic acid with KOH

o (c) Methanol with K_2CO_3

o (d) Ethanol with CH_3COOH

4. Calculate the mass of oxygen

produced when 3g of KClO_3

decomposes according to the

equation $3\text{KClO}_3 \rightarrow 3\text{KCl} + 3\text{O}_2$.

5. The compound used in weathering
of mortar is:

o (a) $\text{Ca}(\text{OH})_2$

o (b) $\text{Ca}(\text{HCO}_3)_2$

o (c) CaCO_3

o (d) CaO

6. 0.16g of methane when burnt

increases the temperature of 100g of water by
 40°C . What is the heat of

combustion of methane if the heat

capacity of water is $4.2\text{J/g}^\circ\text{C}$

($\text{CH}_4=16$)?

o (a) 1,160 kJ/mol

o (b) 1,180 kJ/mol

o (c) 1,560 kJ/mol

o (d) 1,600 kJ/mol

o (e) 1,680 kJ/mol

7. Another name for heavy water:

o A Deuterium

B Protium

C Tritium

D Chloroform

8. Uses of chlorine in water purification:

OA To kill germs

B To aerate

C To sieve particles

D for sedimentation

9. The particle with the highest
penetrating power is:

☐ (a) γ

☐ (b) β

☐ (c) α

☐ (d) X-ray

10. Alkaline pyrogallol increases in weight when it absorbs:

☐ (a) Nitrogen

☐ (b) Oxygen

☐ (c) CO_2

☐ (d) Rare gases

11. Liquid oxygen is obtained by:

☐ (a) Fractional distillation of purified air

☐ (b) Crystallisation

☐ (c) Fractional crystallization

☐ (d) Fractional distillation of liquefied air

12. Ripening of fruit is by:

☐ (a) Ethyne

☐ (b) Ethane

☐ (c) Propene

☐ (d) Ethene

13. Which is the constituent of soap?

☐ A NaOH/KOH and

animal fat & oil

☐ B Na and H_2O

☐ C NaOH and Protein

☐ D Oils

14. Hydrogen is used in oxy-hydrogen

flames for melting metals because it:

☐ (a) Is a rocket fuel

☐ (b) Evolves a lot of heat

when burnt

☐ (c) Combines explosively

with oxygen

☐ (d) Is a very light gas

15. Pure solvent is extracted by:

☐ (a) Evaporation

☐ (b) Distillation

☐ (c) Crystallization

☐ (d) Decantation

16. Hard water is turned to soft water by:

☐ (a) Adding CaCO_3

☐ (b) Ion-exchange resin

☐ (c) Passing water over

heated charcoal

☐ (d) Boiling

17. Which is not a mixture:

☐ (a) Vulcanizer solution

☐ (b) Petroleum

☐ (c) Carbon(IV) sulfide

☐ (d) Rubber latex

18. Two important sources of detergent are:

☐ (a) Fat/oils and hydrocarbons

☐ (b) Coal and cement

☐ (c) Pulp and wood

☐ (d) Water and gas

19. Methane gas can be made from

molecules

gives: carbon (II) oxide gas according to the equation $2\text{CO(g)} + \text{CH}_4\text{(g)} \rightarrow \text{CH}_4\text{(g)} + \text{CO}_2\text{(g)}$. Calculate the

mass of CO required to produce 8.75×10^{23} molecules of CH_4 (At

masses: C=12.011, H=1.008,

O=15.999, Avogadro's no: 6.022×10^{23} molecules/mol):

o (a) 8140g

o (b) 4070g

o (c) 1600g

o (d) 32.00g

o (e) 20.35g

20. What is the name of the reaction between benzene and chloroethane in the presence of AlCl_3 :

o (a) Addition reaction

o (b) Substitution reaction

o (c) Hoffman's reaction

21. Sodium aluminate is used in the treatment of water because it:

o (a) Kills germs

o (b) Filters water

o (c) Coagulates the solid particles

o (d) Gives water a refreshing taste

22. Condensation of two glucose

23. When benzene reacts with

chloroethane, it is an example of:

o (a) Electrophilic Aromatic Substitution

o (b) Electrophilic Addition

o (c) Nucleophilic Aromatic Substitution

o (d) Nucleophilic Addition

24. If the cost of electricity required to deposit 1g of magnesium is #5.00, how much would it cost to deposit 10g of aluminium:

o (a) #10.00

o (b) #27.00

o (c) #44.44

o (d) #66.67

25. Determine the mass of copper

deposited by 4.0 moles of electrons

in the reaction represented by the equation below. $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu(s)}$

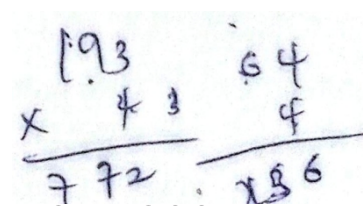
o (a) 32g

o (b) 64g

o (c) 127 g

o (d) 256g

$$4 = \frac{Q}{2 \times 96500}$$



$$\frac{m}{64} = \frac{9 + 200}{2 \times 96500}$$

26. Which of the following can be explained by the kinetic theory:

- o I. The physical state of matter
- o II. Diffusion of gases
- o III. Melting of solids
- o IV. Evaporation of liquids
- o (a) IV only
- o (b) I, II, III & IV
- o (c) III & IV
- o (d) I, II & III
- o (e) I & II

27. Which of the following methods is/are applicable in preventing corrosion of metals:

- o I. Storing in oil
- o II. Allowing an inert oxide surface layer to form
- o III. Coating with paint or tar
- o IV. Plating with another metal
- o (a) I & II
- o (b) IV only
- o (c) I, II and III
- o (d) All of the above

28. Hydrogen bonding exist in

- I HCl
- II HF
- III HBr
- IV HI

A II only B all of the above C i, ii D I-III

29. Which of the following pairs illustrates isotopy:

- o (a) But-1-ene & But-2-ene
- o (b) Ortho-hydrogen
- o (c) Oxygen & Ozone
- o (d) Hydrogen & Deuterium
- o (e) Alpha-particle & beta-particle

30. What does X stand for in the following expression: $X = -\log_{10}(H^+)$:

- o (a) Enthalpy Change
- o (b) Equilibrium Constant
- o (c) Solubility Product
- o (d) Degree of acidity

31. Which of the following indicates the correct increasing order of oxidizing power of the Halogens:

- O(a) $F < Cl < Br < I$
- O(b) $I < Br < Cl < F$
- O(c) $I < Cl < Br < F$
- O(d) $Br < I < Cl < F$
- O(e) $Cl < F < I < Br$

32. Which of the following is a peroxide:

- o I. H_2O_2
- o II. CuO
- o III. Na_2O_2
- o IV. Al_2O_3

o V. Pb304

o (a) I & II only

o (a) I, IV, V <

0(b)1,111,N×

o (c) I only

o(d)1,111.

33. When water is added to burnt ashes

of plants, the resulting solution

would be:

o (a) Acidic

o (b) Basic

o (c) Neutral

o (d) Amphoteric

34. Which of the following is a gaseous

pollutant:

o I. Smoke

o II. Soot

o III. Acid Rain

o IV. Carbon(IV) Oxide

o (a) I & II

o (b) III & IV

o (c) I, II & III

o(d) I, II & IV

35. Tetraoxosulphate(VI) acid will react

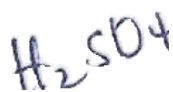
with which of the following metals:

o I. Fe

o II. Mg

o III. Cu

O IV. Pb



☐ (b) II only

☐ (c) I, II, III & IV

☐ (d) I, II & IV

36. The heat evolved when a solute dissolves in water depends on:

☐ I. Solubility

☐ II. Concentration

☐ III. Mass of solute

☐ IV. Nature of solute

☐ (a) I only

☐ (b) II only

☐ (c) III & IV

☐ (d) I, II, III & IV

37. Which of the following indicates that the volume of a gas is directly proportional to the number of moles:

☐ (a) Boyle's Law

☐ (b) Charles' Law

☐ (c) Dalton's Law

☐ (d) Avogadro's Law

38 Which of the following solutions is neutral:

☐ (a) NaOH

☐ (b) HCl

☐ (c) KOH

☐ (d) NaCl

☐ (e) CuSO₄

39. Which of the following processes is a physical change:

o (a) Fermentation	14 B
o (b) Condensation	15b
o (c) Hydrolysis	16b
o (d) Decomposition	17c
40. Which of the following is/are not	18a
endothermic:	19
o I. Neutralization	20b
o II. Evaporation	21c
o III. Photosynthesis	22a
o IV. Combustion	23a
o (a) II only	24d
o (b) II& III	25c
o (c) I & IV	26b
o (d) III & IV	27d
	28a
ANSWER	29d
1c	30d
2e	31b
3B	32d
	433
	b
	5B
	34
6e 35d	
	7a
	36d
8a 37d	
9a 38d	
	10
	B
	39b
	11d
	40c
12d	
13a	

(c) $3.08 \times 10^{-3} \text{ mol/dm}^3$

CHEMISTRY 2021

1 Entropy and enthalpy are: I. Degree of orderliness

II. Heat evolved

III. Heat contents

IV. Degree of dissociation

(a) I and II

(b) I and III

(c) II and III

(d) III and IV

2. Which is used as an anaesthetic?

(a) CH_3OH

(b) CHCl_3

(c) CH_3COOH

(d) $\text{C}_2\text{H}_5\text{OH}$

3. The alkene for ripening of fruits is:

(a) Ethene

(b) Propene

(c) Butene

(d) Pentene

4. Solubility products of

$\text{Cu}(\text{IO}_3)_2$ solution is $1.08 \times 10^{-7} \text{ mol}^3/\text{dm}^6$,

calculate the solubility. (a) 1.08

$\times 10^{-3} \text{ mol/dm}^3$

(b) $2.08 \times 10^{-3} \text{ mol/dm}^3$

(d) $4.08 \times 10^{-3} \text{ mol/dm}^3$

5. The abnormal high boiling point of water is due to: (a) Ionic bonds

(b) Covalent bonds (c) Hydrogen bonds

(d) Metallic bonds

6. Which of the following is/are air pollutants? I. H_2S

II. SO_2

III. CO

IV. Smoke

(a) I only

(b) I and II only

(c) I, II, and III only

(d) I-IV

7. The oxidation state of the following transition metal increases in which order: (a) $\text{V}_2\text{O}_5 < \text{K}_2\text{Cr}_2\text{O}_7 <$

KMnO_4

(b) $\text{KMnO}_4 < \text{K}_2\text{Cr}_2\text{O}_7 < \text{V}_2\text{O}_5$

(c) $\text{K}_2\text{Cr}_2\text{O}_7 < \text{V}_2\text{O}_5 < \text{KMnO}_4$

(d) $\text{K}_2\text{Cr}_2\text{O}_7 < \text{KMnO}_4 < \text{V}_2\text{O}_5$

8. The most stable shell is: (a) s-shell

(b) p-shell

(c) d-shell

(d) f-shell

9. Commercial production of chlorine:

(a) Electrolysis of brine

(b) Heating of NaCl

(c) Decomposition of HCl

- (d) Reaction of NaCl with H_2SO_4
10. Which of the following is false of ethanedioic acid? (a) It is an oxidizing agent
(b) It is a reducing agent
(c) It is a weak acid
(d) It is used as a cleaning agent
11. Which of the following is false of benzoic acid? (a) It has a formula of $\text{C}_6\text{H}_6\text{COOH}$
(b) It is a weak acid
(c) It is used as a preservative
(d) It is an aromatic compound
12. Which of the following is a pollutant in water even when present in traces? (a) Na^+
(b) K^+
(c) Hg^+
(d) Ca^{2+}
13. When ketone reacts with hydroxylamine, what is formed? (a) Aldehyde
(b) Oxime
(c) Ester
(d) Alcohol
14. Air heated over phosphorus was successively passed through KOH and H_2SO_4 . The gases remaining are: (a) Oxygen and nitrogen
(b) The rare gases and nitrogen
(c) Carbon dioxide and water
(d) The rare gases and oxygen
15. When petrol dropped on plastic material is burnt, the gases produced are: (a) CO, CO_2 & HCl
(b) SO_3 , CO & CO_2
(c) HCl, SO_3 & CO_2
(d) HCl, CO & SO_3
16. Iron and sulfur can be separated by dissolving it in: (a) NaOH
(b) Benzene
(c) HCl
(d) H_2SO_4
17. Nitrogen and oxygen are separated from each other based on their: (a) Density
(b) Boiling point
(c) Solubility
(d) Color
19. How many resonance structures does SO_3 have? (a) 1
(b) 2
(c) 3
(d) 4
20. Organic compounds: (a) Contain HC and complexes
(b) Have sweet smelling
(c) A and B

(d) Contain Hydrogen and Oxygen

21. An example of transition metals is:

(a) Mn

(b) Li

(c) Sr

(d) Ca

22. Which of these contribute to water pollution? (a) Salt and pesticide

(b) Aeronautics

(c) Trace elements

(d) Fossil fuels

23. The reaction between butanol and butanoic acid gives: (a) Ester

(b) Aldehyde

(c) Ketone

(d) Alcohol

24. Which of the following is correct? (a)

Fe^{2+} is more stable than Fe^{3+}

(b) Fe^{3+} is more stable than Fe^{2+}

(c) Fe^{4+} is more stable than Fe^{5+}

(d) Fe^{5+} is more stable than Fe^{4+}

25. The presence of a catalyst in a chemical reaction affects the: (a)

Rate of reaction

(b) Equilibrium constant

(c) Product yield

(d) Activation product

26. Smoke is an example of: (a) Liquid dispersed in liquid

(b) Solid/liquid dispersed in gas

(c) Solid dispersed in liquid

(d) Gas dispersed in gas

27. Which of the following is used in welding? (a) Oxyacetylene flame

(b) Hydroacetylene flame ;

(c) Ethene flame

(d) Oxygen flame

28. Iron fillings can be separated from sand by: (a) Magnetization

(b) Distillation

(c) Centrifugation

(d) Crystallization

29. Protein and cellulose are: (a) Polymers

(b) Aldehyde

(c) Ketones

(d) Monomers

30. Calculate the pH of 0.001 mol/dm³ of H₂SO₄. (a) 1 (b) 2

(c) 3

(d) 4

31. The elements that make up amine are: (a) C, H, O

(b) C, H, N

(c) C, H, S

(d) C, H, P

32. The shape of H₂S is: (a) Linear
(b) Tetrahedral

(c) Bent

(d) Trigonal planar

33. A system is in equilibrium when

change $\Delta G =$ (a) 0

(b) 1

(c) -1

(d) 100

34. Flame used in cutting metals is: (a)

oxyacetylene flame

(b) Hydroacetylene flame

(c) Ethene flame

(d) Oxygen flame

35. In water purification, alum is used

for: (a) Sedimentation

(b) Filtration

(c) Chlorination

(d) Ion exchange

36. Thermometer whose thermometric

property depends on change in color

is: (a) Alcohol thermometer

(b) Mercury thermometer

(c) Liquid crystal thermometer

(d) Infrared thermometer

37. The energy required to start a

reaction is known as: (a) Activation

energy

(b) Kinetic energy

(c) Potential energy

(d) Thermal energy

38. Smoke is colloid of: (a) Solid in gas

(b) Liquid in gas

(c) Gas in liquid

(d) Solid in liquid

39. The most malleable metal is: (a)

Gold

(b) Silver

(c) Copper

(d) Aluminum

40. What is the property used to

separate nitrogen and oxygen?

(a) Density

(b) Boiling point

(c) Solubility

(d) Color

Answer

1b

2b

3a

4a

5c

6d

7a

8a

9a

10b

11a

12c

13b
14b
15a
16c
17
18B
19c
20a
2 1 a
22a
23a
24b
25a
2 6 B
27a
2 8 a
29a
30c
31b
32c
33a
34a
35a
36c
3 7 a
38a
39a
40b

CHEMISTRY 2022

1. What is the empirical formula of a hydrocarbon containing 0.160 moles of carbon and 0.640 moles of hydrogen?

- ☐ A. CH₂
- ☐ B. CH₃
- ☐ C. CH₄
- ☐ D. C₂H₄

2. A compound contains 40.0% carbon, 6.7% hydrogen, and 53.3% oxygen.

If the molar mass of the compound is 180, what is the molecular formula? [H=1, C=12, O=16]

- ☐ A. CH₂O
- ☐ B. C₃H₆O₃
- ☐ C. C₆H₁₂O₆
- ☐ D. C₆H₁₂O₃

3. What is the percentage of sulphur in Sulphur (IV) oxide?

- ☐ A. 66%
- ☐ B. 25%
- ☐ C. 40%
- ☐ D. 50%

4. What is the percentage of oxygen in C₂H₅COOH?

- ☐ A. 53.3%

☐ B. 50%

☐ C. 25%

☐ D. 106.6%

5. Benzene produces more soot than ethene on burning because benzene:

- ☐ I. has a higher molar mass
- ☐ II. has a higher degree of unsaturation

☐ III. undergoes substitution and addition reactions

☐ IV. is a liquid at room temperature

☐ A. I and II

☐ B. II only

☐ C. III only

☐ D. II-IV

6. C₈H₁₈ will undergo the following reactions except:

- ☐ I. cracking
- ☐ II. combustion
- ☐ III. substitution
- ☐ IV. addition

☐ A. I only

☐ B. III only

☐ C. IV only

☐ D. II only

7. A given fuel has an octane number of 100. This means that:

o A. the knocking tendency of

the fuel is low

o B. the fuel is a mixture of

several hydrocarbons

o C. a large amount of alkanes

is present in the fuel

o D. the fuel needs to be

cracked before use

8. Ethyne can be prepared from which

of the following?

o I. H_2O

o II. $CaCO_3$

o III. CaO

o IV. CaC_2

o A. I and II

o B. I and III

o C. II and IV

o D. I and IV

9. The following can be used to remove

oil spillage on a water surface

except:

o I. Ammonia

o II. Kerosene

o III. Detergent

o IV. Perfume

o A. I only

o B. III only

o C. I, II and IV

o D. III and IV

10. A type of water that lathers with

soap easily is:

o A. Distilled water

o B. Pure water

o C. Soft water

o D. Hard water

11. Octane number is highest in petrol

containing a high proportion of:

o A. Heptane

o B. Octane

o C. 2-methylpentane

o D. 2,2,4-trimethylpentane

12. A gas will be ideal at:

o I. High temperature and low

pressure

o II. Low pressure and low

temperature

o III. Low pressure and low

temperature

o IV. High-temperature and

high pressure

o A. I only

o B. IV only

o C. II only

o D. III only

13. In which of the following processes

is NaOH a raw material?

o I. Extraction of iron

o II. Extraction of sulphur

o III. Purification of bauxite

o IV. Extraction of copper

o A. II only

o B. III only

O C. III and IV

o D.I only

14. Permanent hardness of water can be removed by:

o I. Activated charcoal

o II. Adding slaked lime

o III. Ion exchange resin

o IV. Boiling

o A.I and II

O B. II and III

o C. III and IV

o D. III only

15. Temporary hardness of water is caused by the presence in water of:

o I. CaHCO_3

o II. CaCO_3

o III. CaSO_4

o IV. MgHCO_3

o A.I and V

O B. II and III

o C.I and IV

o D. II only

16. A balanced chemical equation obeys the law of:

O A. Conservation of mass

o B.Definite proportions

o C. Multiple proportions

o D. Conservation of energy

17. In any chemical reaction, the total mass of products is always equal to that of the reactants. This is a statement of the law of:

o A. Conservation of matter

o B. Constant composition

o C. Definite proportions

o D. Multipleproportions

18. A hydrocarbon has an empirical formula C_2H_5 and a vapour density of 29. Determine its molecular formula.[$\text{C}=12, \text{H}=1$]

o A. C_2H_6

O B. C_3H_8

o C. C_4H_{10}

O D. C_6H_6

19. Which of the following theories states that particles collide during a reaction?

o A. Collision theory

O B. Activation theory

o C. Kinetic theory

o D. Rate theory

20. In the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ $\Delta H = -90 \text{ kJ}$, the yield of

ammonia can be decreased by:

- o A. Increasing the pressure
 - o B. Removing the ammonia
- as it is formed
- o C. Increasing the temperature

- o D. Adding a catalyst

21. The decomposition of hydrogen peroxide can be reduced by:

- o I. Decrease in pressure
- o II. Addition of MnO_2
- o III. Carrying out the reaction in the dark area
- o IV. Heating the chemical

reaction

o A. I and II

O B. I, II and IV

o C. III only

O D. I-IV

22. Which of the following is a toxic pollutant to an aquatic environment?

- o I. n-heptane
- o II. n-hexane
- o III. Toluene
- o IV. Mercury

o A. I and II

O B. III only

o C. IV only

o D. I, II and III

23. In thermodynamics, the internal

energy is affected by:

o I. Work done

o II. Heat evolved

o III. Heat absorbed

o IV. Velocity of the particles

O A. III only

O B. I and II

o C. I, II and III

o D. II and III

24. An unsaturated compound is said to be in:

o A. Dynamic

O B. Static

o C. Thermometric

o D. None

25. When the Gibbs free energy is positive, the reaction is:

o I. Non-spontaneous

o II. At equilibrium

o III. ΔH is endothermic

o IV. Spontaneous

o A. II and III

o B. I and IV

o C. IV only

o D. I only

26. In a chemical reaction, the standard heat of neutralization, combustion, reaction, and formation is

represented by:

o A. I and II

O B. III and IV

- o C. III only
 - o D. I, II, and IV
27. When energy is absorbed in a system, the system is said to be:
- O A. Endothermic
 - O B. Exothermic
 - o C. Isothermal
 - o D. Isobaric
28. The rate of a reaction can be increased by all the factors except:
- o A. Increasing temperature
 - o B. Increasing the concentration of the reactants
 - o C. Increasing the activation energy
 - o D. Use of catalyst
29. The addition of a catalyst in a chemical reaction:
- o A. Increases the yield of the products
 - o B. Increases the surface area of the reactant
 - o C. Does not increase the yield of the product
 - o D. Increases the temperature of the reaction
30. The unit of the rate of reaction is:
- O A. $\text{mol dm}^{-3} \text{s}^{-1}$
 - OB. $\text{mol dm}^{-3} \text{s}^{-1}$
 - OC. $\text{mol dm}^{-3} \text{s}^{-1}$

- o D. $\text{mol dm}^{-3} \text{s}^{-1}$. The law which states that energy can neither be created nor destroyed is:
 - o A. Law of definite proportion
 - o B. Law of multiple proportion
 - o C. Law of conservation of mass
 - o D. Law of reciprocal proportion
32. The valence shell electron pair repulsion theory (VSEPR) is used to predict the geometry of molecules. Which of the following is not a postulate of the theory?
- o A. Valence shell electrons are paired
 - o B. Valence shell electron pairs repel each other
 - o C. Lone pairs and bond pairs repel each other
 - o D. Bond pairs do not repel each other
33. Which of the following reactions does not produce chlorine gas?
- o A. Action of concentrated HCl on MnO_2
 - o B. Electrolysis of brine
 - o C. Reaction of KMnO_4 with concentrated HCl
 - o D. Action of concentrated

HCl on PbO₂

34. Chlorine is used in the production of:

- o I. Bleach
- o II. Plastics
- o III. Pharmaceuticals
- o IV. Paints
- o A. I and II
- o B. III and IV
- o C. I, II and III, IV
- o D. II and III

35. The substance used as a drying agent in the preparation of chlorine gas is:

- o A. H₂SO₄
- o B. CaCl₂
- o C. PCl₅
- o D. NaOH

36. Anhydrous calcium chloride can be used to dry:

- o A. NH₃
- o B. H₂S
- o C. HCl
- o D. CO₂

37. The industrial preparation of sulphuric acid involves the following processes except:

- o A. Contact process
- o B. Lead chamber process
- o C. Ostwald process

o D. Electrolysis process

38. Which of the following is not an amphoteric oxide?

- o A. ZnO
- o B. Al₂O₃
- o C. PbO₂
- o D. MgO

39. Bleaching powder can be used as:

- o I. Deodorant
- o II. Disinfectant
- o III. Oxidizing agent
- o IV. Drying agent

o A. I and III

o B. I and IV

o C. II and III

o D. III and IV

40. Which of the following is used in the preparation of ethanoic acid?

o A. CH₃CH₂OH

o B. CH₃CHO

o C. C₂H₆

o D. CH₃OH

41. Which of the following is the molecular formula of ethyne?

o A. CH₃CH₃

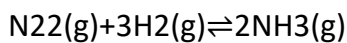
o B. C₂H₄

o C. C₂H₂

o D. CH₄

42. The Haber process is used to

manufacture ammonia. The reaction is represented by the equation:



What will shift the equilibrium position to the right?

- ☐ A. Decrease in pressure
- ☐ B. Decrease in temperature
- ☐ C. Addition of a catalyst
- ☐ D. Removal of NH_3 as it is formed

43. The concentration of H^+ ions in a solution of pH 4.5 is:

- ☐ A. 4.5
- ☐ B. $10^{-4.5}$
- ☐ C. 4.5×10^{-1}
- ☐ D. 4.5×10^1

44. The bond angle in a molecule of water is approximately:

- ☐ A. 104.5°
- ☐ B. 90°
- ☐ C. 120°
- ☐ D. 180°

45. A substance that changes the rate of a chemical reaction without undergoing permanent chemical change is called:

- ☐ A. A catalyst
- ☐ B. An enzyme
- ☐ C. A reagent
- ☐ D. A product

46. The process of removing a substance from a solution by evaporating the solvent is called:

- ☐ A. Filtration
- ☐ B. Distillation
- ☐ C. Crystallization
- ☐ D. Precipitation

47. The process of separating a mixture into its individual components based on differences in boiling points is called:

- ☐ A. Filtration
- ☐ B. Distillation
- ☐ C. Chromatography
- ☐ D. Crystallization

48. In electrolysis, the substance that is broken down by the electric current is called:

- ☐ A. Anode
- ☐ B. Cathode
- ☐ C. Electrolyte
- ☐ D. Electrode

49. The electrode where oxidation occurs during electrolysis is called the:

- ☐ A. Anode
- ☐ B. Cathode
- ☐ C. Electrolyte
- ☐ D. Electrode

50. The electrode where reduction

occurs during electrolysis is called	24a
the:	25d
o A. Anode	26
o B. Cathode	27a
o C. Electrolyte	28c
O D. Electrode	29c
	30c
Answer	31c
1c	32d
2c	33d
3d	34c
5B	35 a
	6c
	36
	a
	37c
7a	
	38d
8d	
	39c
10c	
	40a
11d	
	41c
	12a
	42d
13b	43b
14d	44a
15c	45a
16a	46c
17a	47b
18c	48c
19a	49a
20c	50b
21c	
22c	
23c	

CHEMISTRY 2024

1. Which of these is correct when copper II tetraoxosulphate VI is electrolysed using copper electrode

- A. The anode decrease
- B.The cathode decreàse
- C. The anode increases
- D. The cathode increases

2. Which of these are monoprotic

- I.Hcl
- II.CH3COOH
- III.H2SO4
- IV. HNO3
- A. I, II, III and IV
- B.I, II, and IV
- C.I, II, and III
- D. II, IIIand IV

3. What is the pH of 0.001mol/dm3concentration in a solution

- A.1
- B.4
- C.3
- D.5

4.ion that is pollutant to drinking water even in trace amount is

- A.Ca2+

- B. Mg^{2+}
- C. Cu^{2+}
- D. Hg^{2+}

5. The drying effect of CaCl2 is due to which property of the compound

- A. Delinquent
- B Efflorescent
- C. Hygroscopic
- D.Reactive

6. Which of the followingwill not be used in the laboratory in the process of evaporation

- I evaporating dish
- II. Burrete
- III. Wire gauze
- IV. Filter funnel

- A.IV only
- B.II only
- C. II and Iv only
- D.i and III only

7. Arrange the following according to single bond double bond and triple bond

- I. Chlorine gas
- II. Nitrogen gas
- III. Oxygen gas

- A.1, lii and II

B. I, II and III

C.IIIIII

D.IIIII1

8. Which of the the following separation techniques is used to separate mixture of common salt and iodine

I. Chromatography

II. Precipitation

III. Fractional distillation

IV. Sublimation

A.I only

B.II only

C.III only

D.IV only

9. Which of the following is or are correct about graphite

I. Shape of graphite is hexagonal

II. Graphite is hard as diamond

lii. Graphite is a good conductor

IV. Graphite is octahedral im

Shape

A.II and IV

B.II only

C.I and III

D.IV only

10.The reaction occurring in Daniel cell is

A. Atomization

B. Neutralization

C. Double decomposition

D. Redox

11. C₈H₁₈ undergoes the following except

I. Addition

II. Substitution

III. Combustion

IV. Cracking

A.Ionly

B. I-IV only

C.II only

D. III and IV only

12. Charge to mass ratio was by

A Joseph Proust

B JJ Thompson

C Ernest Rutherford

D Neil Bohr

13.Which of the following can be used to determine the number of moles contained in a particle?

A. Ideal gas equation

B. Charles' law

C. Graham's law

D. Boyle's law

14. Which correctly indicates boyle's law?

I. P is directly proportional to v with T being constant

II. P is inversely proportional to v with T being constant

III. $PV = K$ with T being constant

IV. V is directly proportional to T with P being constant

A. I only

B. II only

C. II and III only

D. IV only

B. III only

C. III and IV only

D. II only

17. The temperature of a gas is 27°C . If the volume is doubled and the pressure is constant, what's the new temperature?

A. 900K

B. 600K

C. 300K

D. 150K

15. Fats and oils are soluble in

I. Trichloromethane

II. Benzene

III. Tetrachloromethane

IV. Water

A. IV only

B. I, II and IV only

C. I, III and II only

D. II only

18. A water contaminated with spilled oil can best be treated

With I. Kerosene. II. Ammonia III. Detergent.

IV. Dispersants

(a) I-IV

(b) I & II

(c) I, II & IV

(d) II only

(e) II & IV

16. Which of the following undergoes both substitution and addition reactions?

I. Pentane

II. Hexene

III. Benzene

IV. Ethene

A. I only

19. What are the similarities between acid bases and salts

A. They dissociate exothermically

B. They liberate OH^-

C. They are electrolytes

D. They liberate H_3O^+

20. What is the definition of a reaction that can go backwards and forward

- A. Isothermal
- B. Isobaric
- C. Reversible
- D. Isochoric

21. Alkanols undergo the following reaction except

- I oxidation
- li saponification
- lii esterification
- Iv fermentation

- a. I only
- b. ii and iii only
- c. iii and iv only
- d. II only

22. The atomic number of an element is equals to

- A. The number of protons in its nucleus
- B. The number of electrons in its nucleus
- C. The approximate mass of it's nucleus
- D. A and B

23. If 5 drops of 2mol dm^{-3} of HCl is added to 10cm^3 of a solution of pH 5. The resulting solution will be

- A. Greater than 5
- B. Less than 5

- C. Equal to 5
- D. Will not change

24. K which is a stable shell is

- A. $n=1$
- B. $n=2$
- C. $n=3$
- D. $n=\infty$

25. Which one of these convert red litmus paper blue

- A. NH_4Cl
- B. Na_2CO_3
- C. CH_3COOH
- D. H_2SO_4

Answers

1 a and d

2b

3c

4
d
.

5a

6c

7a

8d

9c

10d

11a

12b

13a

14c

15c

16b

17b

18 iii and iv

19c

20c

21c

22a

23b

24a

25b

CHEMISTRY 2019

1. Underline species oxidized? I. $2\text{Fe} + \text{Cl}_2 \rightarrow 2\text{FeCl}_3$; II. $\text{S} + 2\text{Cl}_2 \rightarrow \text{S}_2\text{Cl}_2$; III. $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$; IV. $2\text{Al} + 6\text{H}^+ \rightarrow 2\text{Al}^{3+} + 3\text{H}_2$
2. Which of the following statements is/are correct of ΔH , the enthalpy of reaction, H , is? I. The heat change accompanying a chemical reaction; II. Negative for exothermic reaction; III. Positive for exothermic reactions; IV. Greater than ΔS for reversible reaction. (a) I - IV (b) I-III (c) III-IV (d) IV only
3. Which of the following take place during the electrolysis of concentrated sodium chloride solution using carbon electrodes? I. Oxidation takes place at the anode; II. Chlorine is evolved at the Anode; III. Hydrogen is evolved at the cathode; IV. Oxygen is evolved at the cathode. (a) I-IV (b) I-III (c) III-IV (d) IV only
4. Which of the following statements is/are correct of the gaseous state? I. Intermolecular spaces are very large; II. Molecules move very fast; III. Intermolecular forces are negligible; IV. Intermolecular spaces can be compressed. (a) I-IV (b) I-III (c) III-IV (d) IV only
5. Which of the following is/are properties of ethanol? I. It is colourless; II. It is miscible with water; III. Its boiling point is 78°C ; IV. It is a strong electrolyte
6. Which of the following is/are properties of ethanol? I. It is colourless; II. It is miscible with water; III. Its boiling point is 78°C ; IV. It is a strong electrolyte. (a) I-IV (b) I-III (c) III-IV (d) IV only
7. Which of the following statements is/are correct about Equilibrium? Chemical Equilibrium is, attained when the rates of forward and backward reactions are equal; II. Changes in concentration of reactants will alter equilibrium concentration; III. A catalyst alters equally the rates of both the forward and the backward reaction; IV. Temperature affects equilibrium concentration. (a) I-IV except III (b) I-II except II (c) III-IV (d) IV only
8. Aluminum is above iron in the electrochemical series, yet iron corrodes easily on exposure to air while aluminum does not. This is because aluminum (a) Has a lower density than iron (b) Is a better conductor than iron (c) Does not corrode spontaneously (d) Forms a thin layer of inert oxide
9. Which of the following will undergo addition reactions? (a) Ethane (b) Butane (c) Pentane (d) Tetrachloromethane
10. Which of the following processes is/are suitable for investigating whether or not a liquid water is pure? I. Testing it

anhydrous copper(II) tetraoxosulphate (IV); II. Testing is with cobalt(II) chloride paper; III. Testing is with litmus; IV. Determining its boiling point. (a) I-IV (b) I-III (c) III-IV (d) IV only

11. Molecules which have the same shape as methane include I. CCl_4 ; II. C_2H_4 ; III. C_2H_2 . (a) I-IV (b) I-III (c) III-IV (d) I only

12. Which of the following statements is/are correct of the compound with the formula $\text{K}_4\text{Fe}(\text{CN})_6$? I. Its IUPAC name is potassium hexacyano ferrate (II); II. It has 6 ligands; III. It is a complex salt of transition metal; IV. Its solution gives a green precipitate with aqueous sodium hydroxide. (a) I-IV (b) I-III (c) III-IV (d) IV only

13. The factors that affect the rate of chemical reactions include; I. Concentration; II. Temperature; III. Presence of a catalyst; IV. Collision frequency of particles. (a) I-IV (b) I-III (c) III-IV (d) IV

14. Which of the following statements is/are correct? I. A primary cell cannot maintain a steady current over a long period; II. Polarization is said to occur in a cell when hydrogen bubbles form on the plates; III. The Daniel cell is an electrochemical cell; IV. A primary cell can be recharged

15. Which of the following gases will have the highest rate of diffusion under the same conditions? ($H = 1$, $C = 12$, $O = 16$, $S = 32$, $Cl = 35.5$). (a) O_2 (b) Cl_2 (c) HCl (d) H_2S (e) CO_2

16. The ionic radii of metals are (a) Greater usually than their atomic radii (b) Unaffected by the charge on the ion (c) Less than their atomic radii (d) Greater than their atomic radii

17. Which of the following can be oxidized by heating obtained from its reaction with carbon? I. Lead; II. Zinc; III. Iron; IV. Calcium. (a) I-IV (b) I-III (c) III-IV (d) IV only

18. Selective discharge of ions during electrolysis depends on the (a) Position of ions in the electrochemical series; II. Nature of the electrodes; III. Charge on the ions; IV. Quantity of electricity passed through the electrolyte. (a) I-IV (b) I-II (c) III-IV (d) IV only

19. Oxidation is a reaction which can involve (a) Loss of electrons; II. Increase in oxidation number; III. Gain of oxygen; IV. Loss of hydrogen. (a) I-IV (b) I-III (c) III-IV (d) IV only

20. Which of the following solids has a network structure? (a) Diamond (b) Iodine (c) Sulphur (d) Graphite

21. Alums are classified as (a) Simple Salts (b) Acid Salts (c) Anhydrous Salts (d) Double Salts (e) Basic Salts

1. Fe^{2+} , S^{2-} , mg and Al

8

3.B

4. A

5. 1-11l

6.B

7. A

8. D

9..

10.D

11.D

12.B

13.A

14.1-111

15.A

16.C

17.B

18.B

19.A

20.A

21.D

CHEMISTRY 2021

1. The oxidation states of S in $\text{Ca}(\text{HSO}_3)_2$ is _____ (a) +2 (b) -2 (c) +4 (d) -4
2. The oxidation state of P in $\text{H}_2\text{P}_2\text{O}_7^{2-}$ is _____ (a) -3 (b) +3 (c) +1 (d) -2
3. Potassium ethanoate is formed when _____ (a) Methanoic acid reacts with KOH (b) Ethanoic acid with KOH (c) Methanol with KCO_3 (d) Ethanol with CH_3COOH
4. Calculate the mass of oxygen produce when 3g of KClO_3 decompose according to the equation $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
5. The compound used in weathering of mortar is _____ (a) $\text{Ca}(\text{OH})_2$ (b) $\text{Ca}(\text{HCO}_3)_2$ (c) Ca
6. 0.16g of methane when burnt increases the temperature of 100g of water by 40°C , what is the heat of combustion of methane if the heat capacity of water is $4.2\text{Jg}^{-1}\text{ }^\circ\text{C}^{-1}$? ($\text{CH}_4 = 16$). : (a) $1,160\text{kJmol}^{-1}$ (b) $1,180\text{kJmol}^{-1}$ (c) $1,560\text{kJmol}^{-1}$ (d) $1,600\text{kJmol}^{-1}$ (e) $1,680\text{kJmol}^{-1}$
7. Another name for heavy water is _____
8. One of the use of chlorine in water purification is _____
9. The particle with the highest penetrating power is _____ (a) γ (b) β (c) α (d) x-ray
10. Alkaline pyrogallol increases in weight when it absorbs _____ (a) Nitrogen (b) Oxygen (c) CO_2 (d) Rare gases
11. Liquid oxygen is obtained by _____ (a) Fractional distillation (b) purified air (c) Crystallization (d) Fractional distillation of liquefied air
12. Ripening of fruit is by _____ (a) Ethyne (b) Ethane (c) Propene (d) Ethene
13. Which is the constituent of soap?
 14. Hydrogen is used in oxy-hydrogen flames for melting metals because is (a) Is a rocket fuel (b) Evolves a lot of heat when burnt (c) Combines explosively with oxygen (d) Is a very light gas
 15. Pure solvent is extracted by (a) Evaporation (b) Distillation (c) Crystallization (d) Decantation
 16. Hard water is turn to soft water by _____ (a) Adding CaCO_3 (b) Ion-exchange resin (c) Passing water over-heated charcoal (d) Boiling
 17. Which is not a mixture (a) Vulcanizer solution (b) Petroleum (c) Carbon(iv) sulphide (d) Rubber
 18. Two important sources of detergent are: (a) Fat/oils and hydrocarbons (b) Coal and cement (c) Pulp and wood (d) Water and gas
19. Methane gas can be made from carbon(ii)oxide gas according to the equation $2\text{CO}(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g}) + \text{CO}_2(\text{g})$. Calculate the mass of CO required to produced 8.75×10^{23} molecule of CH_4 ? (At masses: C=12.011, H=1.008, O=15.99, Avogadro's Sn: 6.022 10^{23} molecules/mole). (a) 8140g (b) 4070g (c) 1600g (d) 32.00g (e) 20.35g
20. What is the name of reaction between benzene and chloro ethane in the presence of AlCl_3 ? (a) Addition

reaction (b) Substitution reaction
(c)Hoffman's reaction

21. Sodium aluminate is used in treatment of water because it (a) Kills germ (b) Filters water (c) Coagulates the solid particle (d) Gives water its refreshing water

22. Condensation of two glucose molecules gives _____

(a)Maltose (b) Fructose
(c) Sucrose (d)Starch

23. When benzene reacts with chloroethane, it is an example of

(a) Electrophilic Aromatic Substitution (b) Electrophilic Addition

(c) Nucleophilic Aromatic Substitution (d) Nucleophilic addition

24. If the cost of electricity required to deposit 1g of magnesium is \$5.00, how much would it cost to deposit 10g of aluminum? (a) \$10.00 (b)\$27.00 (c) \$44.44 (d) \$66.67

25. Determine the mass of copper deposited by 4.0 moles of electrons in the reaction represented by the equation below. $Cu^{2+} + 2e^{-} \rightarrow 2Cu(s)$. (a) 32 (b) 64 (c) 128 (d) 256

Answers

1.C

2..

3.B

4. 1.18g

5. B

6. E

7. Deuterium D₂O

8. Kill germs

9.A

10.B

11.D

12.D

13. Sodium or potassium salts of long-chain fatty acids (e.g., sodium stearate).

14.B

15.B

16.B

17.C

18.A

19.81.8g

20.B

21.C

22.A

23.A

24.D

25.C

UI POST UTME CHEMISTRY 2024

1. Which of these is correct when copper II tetraoxosulphate VI is electrolysed using copper electrode

- A. The anode decrease
- B. The cathode decrease
- C. The anode increases
- D. The cathode increases

Answer;

Options A and D :

Explanation;

When copper(II) tetraoxosulphate(VI) (CuSO_4) is electrolyzed using copper electrodes, the reactions at the electrodes are:

At the cathode (negative electrode):

$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ (Copper ions gain electrons and deposit as solid copper.)

At the anode (positive electrode):

$\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$ (Copper atoms lose electrons and go into solution as Cu^{2+} ions.)

Since copper from the anode dissolves into solution, *the anode decreases in mass.* At the same time, *copper from the solution is deposited on the cathode, increasing its mass.*

2. Which of these are monoprotic

I. HCl

II. CH_3COOH

III. H_2SO_4

IV. HNO_3

A. I, II, III and IV

B. I, II, and IV

C. I, II, and III

D. II, III and IV

Answer;

B-I, II, and IV

Explanation;

A monoprotic acid is an acid that donates only one proton (H^+) per molecule in an aqueous solution.

Let's analyze the given acids:

1. HCl (Hydrochloric acid) $\rightarrow$ Monoprotic (donates one H^+)

2. CH_3COOH (Acetic acid) $\rightarrow$ Monoprotic (donates one H^+)

3. H_2SO_4 (Sulfuric acid) $\rightarrow$ Diprotic (donates two H^+ ions in two steps)

4. HNO_3 (Nitric acid) $\rightarrow$ Monoprotic (donates one H^+)

3. What is the pH of 0.001mol/dm³ concentration in a solution

- A.1
- B.4
- C.3
- D.5

Answer;

Option C

Explanation;

The pH of a solution is calculated using the formula:

$$pH = -\log[H^+]$$

For a solution with 0.001 mol/dm³ (10⁻³mol/dm³) H⁺ ion concentration,

$$pH = -\log(10^{-3})$$

$$pH=3$$

4. ion that is pollutant to drinking water even in trace amount is

A. Ca²⁺

B. Mg²⁺

C. Cu²⁺

D. Hg²⁺

Answer;

Option D

Explanation;

Mercury ions (Hg²⁺) are highly toxic, even in trace amounts, and can cause serious health issues, including neurological damage and kidney failure. Mercury contamination in drinking water is a major environmental concern.

In contrast:

Ca²⁺ (Calcium ion) and Mg²⁺ (Magnesium ion) are essential minerals and contribute to water hardness but are not toxic.

Cu²⁺ (Copper ion) can be harmful at high concentrations but is not as dangerous as mercury in trace amounts.

5. The drying effect of CaCl₂ is due to which property of the compound

A. Delinquent

B Efflorescent

C. Hygroscopic

D. Reactive

Answer;

Option A

Explanation; Calcium chloride (CaCl₂) is

deliquescent, meaning it absorbs moisture from the air and eventually dissolves in the absorbed water to form a solution. This property makes it an effective drying agent (desiccant).

Explanation of other options:

Efflorescent : A substance that loses water to the air (opposite of deliquescent).

Hygroscopic : A substance that absorbs moisture but does not dissolve in it.

Reactive : Describes a substance that readily undergoes chemical reactions, which is not the main reason for CaCl_2 's drying effect.

6. Which of the following will not be used in the laboratory in the process of evaporation

I. evaporating dish

II. Burette

III. Wire gauze

IV. Filter funnel

A. IV only

B. II only

C. II and IV only

D. I and III only

Answer;

C. II and IV

Explanation; :

The process of evaporation in the laboratory typically involves heating a solution to remove the solvent, usually using an evaporating dish and a wire gauze (to support the dish over a flame).

Now, let's analyze the given options:

1. Evaporating dish → Used (to hold the solution during evaporation)

2. Burette → Not used (it is used for titration, not evaporation)

3. Wire gauze → Used (to support the evaporating dish while heating)

4. Filter funnel → Not used (used for filtration, not evaporation)

7. Arrange the following according to single bond, double bond and triple bond

I. Chlorine gas

II. Nitrogen gas

III. Oxygen gas

A. I, II and III

B. I, II and III

C. III, II, I

D. I, III, II

Answer;

Option A

Explanation;

To arrange the gases according to single, double, and triple bonds, let's analyze their molecular structures:

1. Chlorine gas (Cl_2) → single bond ($\text{Cl}-\text{Cl}$)

2. Oxygen gas (O_2) → double bond ($\text{O}=\text{O}$)

3. Nitrogen gas (N_2) → triple bond ($\text{N} \equiv \text{N}$)

Now, arranging them in order of single →

double → triple bond:

Chlorine(1)→Oxygen(II) → Nitrogen (II)

8. Which of the the following separation techniques is used to separate mixture of common salt and iodine

I. Chromatography

II. Precipitation

III. Fractional distillation

IV. Sublimation

A.I only

B.II only

C.III only

D.IV only

Answer;

D- IV. Sublimation

Explanation:Iodine (I₂) is a sublimable substance, meaning it can change directly from a solid to a gas upon heating without passing through the liquid phase.

Common salt (NaCl) does not sublime and remains as a solid when heated.By heating the mixture, iodine sublimates and can be collected as solid iodine crystals upon cooling, while salt remains behind.

Checking the options:

1. Chromatography → Used for separating dissolved substances, not solid-solid mixtures. 
2. Precipitation → Used for separating insoluble solids from solutions, not applicable here. 
3. Fractional distillation → Used for separating liquid-liquid mixtures, not solid-solid. 
4. Sublimation → The correct method for separating iodine from salt. 

9.Which of the following is or are correct about graphite

I. Shape of graphite is hexagonal

II. Graphite is hard as diamond

III. Graphite is a good conductor

IV. Graphite is octahedral in shape

Shape

A. II and IV

B.II only

C.I and III

D.IV only

Answer;

C.I and III

Explanation;Let's analyze the statements about graphite: 

□{✓}1a10)

1. Shape of graphite is hexagonal → (Graphite has a hexagonal crystal structure.)

III. Combustion

IV. Cracking

2. Graphite is as hard as diamond → X

Incorrect (Graphite is soft because its layers slide over each other, unlike diamond, which is extremely hard.)

A. I only

3. Graphite is a good conductor → ☒ Correct (Graphite has free-moving electrons, making it a good conductor of electricity.)

B. I-IV only

C. II only

4. Graphite is octahedral in shape → X Incorrect (Graphite has a hexagonal structure, not octahedral).

D. III and IV only

Answer;

A. I only

10. The reaction occurring in Daniel cell is

A. Atomization

Explanation:

B. Neutralization - C_8H_{18} (octane)* is an *alkane*, which is a saturated hydrocarbon (no double or triple bonds).

D. Redox

C. Double decomposition

hydrocarbons (like alkenes or alkynes) where Answer; double or triple bonds are present. Since octane is saturated, it *cannot undergo addition reactions*.*

- *Addition reactions** occur in unsaturated

- *Substitution reactions** are typical for alkanes, where hydrogen atoms are replaced

Explanation:

by other atoms or groups.

D. Redox
- The *Daniel cell** is a type of electrochemical cell that involves a *redox* (reduction-oxidation)* reaction.

- In the Daniel cell, zinc metal (Zn) is oxidized at the anode, losing electrons to form Zn^{2+} ions, while copper ions (Cu^{2+}) are reduced at the cathode, gaining electrons to form copper metal (Cu).

- *Combustion** is a reaction where alkanes burn in the presence of oxygen to produce carbon dioxide and water.

- *Cracking** is a process where large hydrocarbon molecules (like octane) are broken down into smaller molecules.

II. Substitution
12. Benzene burns with more soot than ethene because

11.1. C_8H_{18} undergoes the following except I. Addition

A. Benzene contains higher carbon content than ethene
 B. Benzene contains lower carbon content than ethene
 C. Benzene has a higher degree of unsaturation than ethene
 D. Benzene has a lower degree of unsaturation than ethene

Answer;

A. Benzene contains higher carbon content than ethene.

Explanation;

Soot formation in combustion is mainly due to incomplete burning of carbon-rich fuels. The higher the carbon-to-hydrogen ratio in a compound, the more likely it is to produce soot when burned.

Benzene(C_6H_6) has a higher carbon content compared to ethene(C_2H_4)

Because of this, benzene tends to produce more carbon particles (soot) when it burns.

Ethene, having a lower carbon content, burns with a less sooty flame.

13. Which of the following can be used to determine the number of moles contained in a particle?

- A. Ideal gas equation
- B. Charles' law
- C. Graham's law
- D. Boyle's law

Answer:

A. Ideal gas equation

Explanation:

- The *ideal gas equation ($PV = nRT$)* is used to calculate the number of moles (n) of a gas when the pressure (P), volume (V), and temperature (T) are known. Here, R is the universal gas constant.
- *Charles' law (B)* relates volume and temperature at constant pressure, but it does not directly determine the number of moles.
- *Graham's law ©* describes the rate of effusion or diffusion of gases, which is unrelated to calculating moles.
- *Boyle's law (D)* relates pressure and volume at constant temperature, but it also does not determine the number of moles.

Thus, the *ideal gas equation (A)* is the correct choice for determining the number of moles.

14. Which correctly indicates Boyle's law?

- I. P is directly proportional to v with T being constant
- II. P is inversely proportional to v with T being constant
- III. $PV = K$ with T being constant

IV. V is directly proportional to T with P being constant

A.I only

B.II only

C. II and III only

D.IV only

Answer:

C. II and III only

Explanation:

- *Boyle's law* states that the pressure (P) of a gas is *inversely proportional* to its volume (V) when the temperature (T) is held constant. This can be expressed as:

* $P \propto \frac{1}{V}$ * (1 is correct), or

- * $PV = K$ * (III is correct), where K is a constant.

- *I. P is directly proportional to V with T being constant* is incorrect because Boyle's law describes an inverse relationship, not a direct one.

- *IV. V is directly proportional to T with P being constant* is incorrect because this describes *Charles' law*, not Boyle's law.

15. Fats and oils are soluble in

I. Trichloromethane

II. Benzene

III. Tetrachloromethane

IV. Water

A. IV only

B.I, II and IV only

C.I, III and II only

D.II only

Answer;

C. I, III and II only.

Explanation;

* *Fats and oils are nonpolar:* They are composed primarily of long hydrocarbon chains, making them hydrophobic (water-repelling).

* *Solubility:* "Like dissolves like." Nonpolar substances dissolve in nonpolar solvents.

* *Trichloromethane (Chloroform), Benzene, and Tetrachloromethane (Carbon tetrachloride) are nonpolar solvents:* Therefore, fats and oils will dissolve in them.

* *Water is a polar solvent:* Fats and oils do not dissolve in water.

16. Which of the following undergoes both substitution and addition reactions?

I. Pentane

II. Hexene

III. Benzene

IV. Ethene

A.I only

B.III only

C. III and IV only

D.II only

Answer:

B. III only

Explanation: *Benzene(111)* is the only compound listed that undergoes *both substitution and addition reactions*.

- Benzene undergoes *substitution reactions* (e.g., halogenation, nitration) due to its stable aromatic ring structure.

- Under certain conditions (e.g., hydrogenation), benzene can also undergo *addition reactions*, though this is less common due to the stability of the aromatic ring.

- *Pentane (I)* is an alkane and only undergoes *substitution reactions* (e.g., halogenation).

- *Hexene (II)* and *Ethene (IV)* are alkenes and primarily undergo *addition reactions* (e.g., hydrogenation, halogenation). They do not typically undergo substitution reactions.

Thus, the correct answer is *B. III only*.

17. The temperature of a gas is 27 °C. If the volume is doubled and the pressure is constant, what's the new temperature?

A. 900K

B. 600K

C. 300K

D. 150K

Answer:

B. 600K

Explanation;

*Charles's Law states that for a fixed amount of gas at constant pressure, the volume is directly proportional to the absolute temperature. This means if you increase the temperature, the volume will increase proportionally, and vice versa.

* The formula for Charles's Law is:

$V_1/T_1 = V_2/T_2$, where:

* V_1 is the initial volume

* T_1 is the initial absolute temperature

* V_2 is the final volume

* T_2 is the final absolute temperature

Steps to Solve

1. *Convert Celsius to Kelvin:*

Temperature in Kelvin(K) =
Temperature in Celsius(°C) + 273.15

$T_1 = 27^{\circ}\text{C} + 273.15 = 300.15\text{K}$. For

simplification in multiple choice questions, we can use 273 instead of 273.15, so $T_1 = 27 + 273 = 300\text{K}$.

2. *Understand the volume change:*

The volume is doubled,
so $V_2 = 2 * V_1$.

3. *Apply Charles's Law:*

$V_1/T_1 = V_2/T_2$

$V_1/300 = (2 * V_1)/T_2$

4. *Solve for T2:*

$$T_2 = (2 \cdot V_1 \cdot 300) / 1$$

$$T_2 = 2 \cdot 300$$

$$T_2 = 600\text{K}$$

18. A water contaminated with spilled oil can best be treated

With I. Kerosene. II. Ammonia III. Detergent.

IV. Dispersants

(a) I-IV

(b) I&II

(c) I,II&IV

(d) II only

(e) II&IV

Answer;

Let's consider each of the options and their effect on oil

Dispersants: These are chemicals that break down oil into smaller droplets, making it easier for natural processes to degrade the oil. This is perfect for our aim

Kerosene: Kerosene itself is a hydrocarbon. Adding kerosene would just add another type of oil to the water, worsening the contamination.
Not what we're looking for

Ammonia: Ammonia would not effectively break down or remove oil.

Detergent: Detergents can help break down oil is also correct

However, in large scale oil spills dispersants are more effective and designed for the environment but since the question doesn't specify the quantity, we go with it.

Thus, III and IV



19. What are the similarities between acid bases and salts

A. They dissociate exothermically

B. They liberate OH⁻

C. They are electrolytes

D. They liberate H⁺

Answer;

C. They are electrolytes

Explanation;

-A. They dissociate exothermically: This is not always true. While some acids, bases, and salts may dissociate exothermically, others may dissociate endothermically. It depends on the specific substance. Example, the dissociation of Ammonium Chloride NH₄Cl is endothermic.

-B. They liberate OH⁻: This is only true for bases, not for acids or salts. Bases release hydroxide ions (OH⁻) in solution, but acids and salts do not.

-C. They are electrolytes: This is true for acids, bases, and salts. All three can dissociate into ions in solution, making them electrolytes that can conduct electricity.

-D. They liberate H_3O^+ : This is only true for acids, not for bases or salts. Acids release hydronium ions (H_3O^+) in solution, but bases and salts do not

20. What is the definition of a reaction that can go backwards and forward

- A. Isothermal
- B. Isobaric
- C. Reversible
- D. Isochoric

Answer:

C. Reversible

Explanation:

- *Reversible reaction*: A reaction that can proceed in both the forward and backward directions under suitable conditions. This means the products can react to reform the reactants.

- *Isothermal*: A process that occurs at a constant temperature.

- *Isobaric*: A process that occurs at a constant pressure.

- *Isochoric*: A process that occurs at a constant volume.

21. Alkanols undergo the following reaction except
I oxidation

II saponification

III esterification

IV fermentation

a. I only

b. II and III only

c. III and IV only

d. II only

Answer:

C. III and IV only

Explanation:

Alkanols (alcohols) undergo the following reactions:

Oxidation: Primary and secondary alcohols can be oxidized to form aldehydes, ketones, or carboxylic acids.

Esterification: Alcohols react with carboxylic acids to form esters.

However, *saponification* is not a reaction that alkanols undergo. Saponification is the hydrolysis of esters in the presence of a base to form alcohols and carboxylates, and it is specific to esters, not alcohols.

Likewise, alcohols are produced from fermentation of sugars. It is not a reaction they undergo.

22. The atomic number of an element is equal to

- A. The number of protons in its nucleus
- B. The number of electrons in its nucleus
- C. The approximate mass of its nucleus
- D. A and B

Answer;

A. The number of protons in its nucleus.

Explanation:

- The *atomic number* of an element is defined as the number of *protons* in the nucleus of an atom. This is a fundamental property of an element and determines its identity.
- *Electrons are not located in the nucleus* ;they orbit the nucleus, so option B is incorrect.
- The *approximate mass of the nucleus* is determined by the sum of protons and neutrons (mass number), so option C is incorrect
- Option D is incorrect because only option A is correct.

23. If 5 drops of 2 mol dm^{-3} of HCl is added to 10 cm^3 of a solution of pH 5. The resulting solution will be

- A. Greater than 5
- B. Less than 5
- C. Equal to 5
- D. Will not change

Answer;

B. Less than 5.

Explanation:

- The initial solution has a pH of 5, which means it is slightly acidic.
- Adding 5 drops of $2\text{ mol dm}^{-3}\text{HCl}$ (a strong acid) will increase the concentration of hydrogen ions (H^+) in the solution.
- Since pH is defined as $\text{pH} = -\log[\text{H}^+]$, an increase in the concentration of H^+ ions will decrease the pH value.

Thus, the resulting solution will have a *lower pH* than the original pH of 5. The correct answer is

24. K which is a stable shell is

- A. $n=1$
- B. $n=2$
- C. $n=3$
- D. $n=4$

Answer;

A.n=1

Explanation:

-The *K shell* is the innermost electron shell of an atom, corresponding to the principal quantum number $n = 1$.

- This shell is the most stable because it is closest to the nucleus and has the lowest energy level.

25. Which one of these convert red litmus paper blue

A.

NH₄Cl

B. Na₂CO₃

C.CH₃COOH

D.H₂SO₄

Answer:

B.Na₂CO₃.

Explanation:

- *Red litmus paper turns blue in the presence of a base (alkaline solution)*.

– *Na₂CO₃(Sodium carbonate)*: This is a basic salt that forms an alkaline solution when dissolved in water, turning red litmus paper blue.

NH₄Cl(Ammonium chloride): This is an acidic salt that forms an acidic solution, so it would not turn red litmus paper blue.

*CH₃COOH(Acetic acid) *: This is a weak acid and would not turn red litmus paper blue.

– *H₂SO₄(Sulfuric acid)*: This is a strong acid and would not turn red litmus paper blue.

Thus, the correct answer is *B.Na₂CO₃

CHEMISTRY 2025

1. Which of the gases below can be dried by Phosphorus (V) oxide. select all that apply

I.CO₂

II.NH₃

IV.H₂

III.H₂S

A. I and IV only

B.I-IV

C.none of the above

D.I, III and IV

E. II, III and IV

2. Which of the following is required to break crystals

I) lattice energy

II) atomization energy

III) bond energy

3. Heat evolved or absorbed is known as

I) heat of combustion

II) heat of neutralization

III) heat of combustion

IV) heat of formation

4. Rainbow formation is due to

A. Refraction

B. Diffraction

C reflection

D interference

5. A mixture of iron fillings and sulphur can be separated by

A. Precipitation

B. Magnetisation

C.

6. Which of the following are isomers

Methoxypropane

Butan-2-ol

2-methyl butanol

Propan-2-ol

7. 100cm³ of Methane diffuses through a porous plug in t seconds, 400cm³ of oxygen diffuses through a porous plug in 100s. Find t

8. The rate of diffusion of a gas Y is 2 times that of X. Ratio of R.M.M of Y to X 9. Which of the following can undergo addition and Substitution reaction?

I. Heptane

II. Hexane

III. Pentane

IV. Benzene

10. Alkane are generally used as

a Domestic fuel

11. H_2SO_4 cannot be used to dry these

a HCl

b NH_3

c H_2 d...

12. which of these are isomers of butanol

a 2 methyl 2 propanol

b Methoxy propanol

c 1 propanol

13. Which of the following is not part of natural water

I Rain water

II water Vapour

III deionised water

IV distilled water

14. During photosynthesis oxygen is gotten form__

15. C_8H_{18} does not undergo which of the following reactions?

i. Addition

ii. Cracking

iii. Substitution

iv. Combustion

A. i and iv only

B. ii, iii and iv only

C. I only

D. ii and iv only

16. a ketone reacts with hydroxylamine to give

a. hydrozone

b. a nitroso compound

c. anoxime

d. alkanonitrile

e. all of the above

17. a charged car battery can be converted to what form of energy

a light energy and chemical energy

b. mechanical energy

c. Chemical energy and mechanical energy

d. all of the above

i. d

ii. a and b

iii. c

i. a-c

18. drops of HCl is added to 10cm³ of a solution of pH 5. The resulting solution will be

19. The unit of rate of reaction is

- A. mol dm⁻³ s⁻¹
- B. mol⁻¹ dm³ s
- C. mol dm⁻³ s⁻¹
- D. mol dm³ s⁻¹

20. Which of the following is true

I. Catalyst have an effect on equilibrium by producing a greater yield of the product

II. It has no effect on the equilibrium

III. It reduces the concentration of the reactants in the equilibrium.

- A. I and II
- B. II
- C. III
- D. I and III.

21. What is the loss and gain of heat during a chemical reaction

- A. Heat of reaction
- B. Heat of combustion
- C. standard heat of formation
- D. Entropy

22. Which gas is insoluble in water and can be collected over water?

- A) Ammonia
- B) Hydrogen chloride
- C) Oxygen
- D) Sulphur dioxide

23. Which of the following has the highest ionization energy?

- A) Sodium
- B) Neon
- C) Magnesium
- D) Aluminium

24. The colour of sodium flame test is..

- A) Red
- B) Yellow
- C) Blue-green
- D) Violet

25. Which process describes liquid turning into gas at any temperature?

- A) Boiling
- B) Evaporation
- C) Condensation
- D) Sublimation

26. Which of these is NOT an acid anhydride?

- A) SO₂
- B) CO₂
- C) N₂O₅
- D) Na₂O

27. The following is/are true about evaporation and boiling point of water

I. Evaporation occurs on the surface

II. Boiling takes place throughout the water body

III. Boiling point of water at 1atm is 100 °C and evaporation occurs at no specific temperature

28. Which will change moist red litmus paper to blue?



D. Na₂CO₃

29. Which other name is attributed to the azimuthal quantum number I. Subsidiary quantum number II. Spin quantum number III. Magnetic quantum number IV. Principal quantum number

30. Which of the following fractions is/are used as raw materials in cracking process I. Kerosene II. Lubricating oil III. Diesel oil IV. Bitumen 31. Which of the following factors affect the rate of nuclear reaction I. Catalyst II. Temperature III. Pressure IV. concentration

32. Which gas(es) will be liberated when an alkali metal reacts with cold water I. Oxygen II. Carbon(II)oxide III. Carbon(IV)oxide IV. Hydrogen 33. The product obtained from reaction of CaO and water I. Slaked lime II. Mortar III. Cement IV. Quicklime

34. Which of the following is/are alcohol

I. CH₃OCH₃ II. CH₃CHO

35. How many number of shell (a) 2 (b) 3 (c) 4 (d) none

36. When calcium carbide react with water it liberate

I. Mortar II. Slaked lime III. lime water IV. Chalk

37. When alkane reacts with halogen what type of reaction does it undergo

A. addition B. Substitution
C. Cracking D. polymerization

38. Which of the following is used in making aircraft

A. Brass B. Bronze C. Duralumin

39. Which of the following is liquid

A. Mercury B. Bromine C. Sodium C. Gallium

40. Which of the following can conduct electricity

A. Diamond B. Graphite C. P8 D. S8

41. Which of the following contains tin

I. Bronze II. Brass III. Soft solder IV. Steel

42. The properties of acid

I. It liberates H₂ when it reacts with trioxocarbonates

II. They are slippery

III. They are corrosive when concentrated

IV. They react with base to form salt and water

43. How many number of orbital can a 'd' orbital accommodate

A. 8 B. 4 C. 10 D. None

44. An element that can access three electron is in which group

A.Group 1 B.Group 2 C.Group 3 D.Group 4

45. when cao and water are mixed the product formed is

i slaked lime

ii limestone

iii cement

iii mortar

iv quick lime

46. azimuthal quantum number can also be called

- a) principal quantum number b) magnetic quantum number c) spin quantum number d) subsidiary quantum number

47. which of the following is/are correct;

- i when pressure is applied to ice, it lowers the melting point and increases the boiling point.
ii when pressure is applied to ice, it increases the melting point and decreases the boiling point.
iii when impurities are added to ice both the melting and boiling points are raised.

48. the number of orbitals in the d subshell is

a) 2 b) 3 c) 5 d) 7

49. arrange the following in order of increasing energy levels;

45,4p,4d,4f

i) 4p < 4s < 4d < 4f

ii) 4s < 4p < 4d < 4f

iii) 4f < 4d < 4p < 4s

iv) 4d < 4f < 4s < 4p

50. the number of electrons that can completely fill the d orbital is

a) 2 b) 8 c) 10 d) 4

51. when an alkali metal is dissolved in hcl, it liberates;

a) H₂ b) H₂S c) CO₂

52. the number of electrons in the 2p subshell is

a) 2 b) 6 c) 4

53. which of the following is used as a disinfectant

a) phenol

b) benzene

54. Gases liberated when an alkali reacts with cold water are

i. Oxygen

ii. Carbon (ii) oxide

iii. Carbon (iv) oxide

iv. Hydrogen

A. i and iii

B. i and iv

C. iv only

55. The minimum number of electrons in 2Px subshell is (a) 8 (b) 6 (c) 10 (d) 256. Arrange in order of increasing energy (i) 4s (ii) 4d (iii) 4p (iv) 4f

A. $1 > 11 > 11 > 1$

B. $i > iii > ii > iv$

C. $i > ii > iii > iv$

57. The number of electrons found in a Double quantum shell is (a) 2 (b) 8 (c) 32 (d) 1058. Which of the following is an Alcohol

I. CH3COOH

II. CH3COH

III. CH3CHCOOH

IV. COOHCH3

a. I only b. I, II, IV only c. I and II. d. None of the above

59. Copper is present in which of the following I. Brass II. Bronze III. Diamond iv. Duralumin

A. I and II B. I and IV C. I, II and III D. I-IV

60. Which of the following is an appropriate material for constructing airplane.

A. Brass B. Duralumin C. Copper D. iron

61. Tin is present in which of the following i. Bronze ii. Brass III solder paper. iv. corundum.

A. I and iii

B. I only

C. I and II

D. II and III

E. I, II and iv

62. which of these are isomers

I butanol

II 2 methyl 2 propanol

III Methoxypropane

IV propanol

63. Which of the following can undergo addition and substitutions reaction

A heptane

B Hexane

C Pentane

D Benzene

64. Which of the Following can be used as a solvent for Oil and fats

(I) Chloroform

(II) Tetrachloromethane

(III) Water

(IV) Ammonia

Forgotten how the Roman numerals was structured

65. Which of the following can react with Bromine water

A propanoic Acid

B propane

C Propene

66. Coffee is a

- A Elements
- B Compound
- C heterogenous Mixture
- D Homogeneous Mixture

67. CO₂ is a/an

- A Elements
- B Compound
- C homogeneous Mixture
- D heterogenous Mixture

68. Which raw material is used in cracking process

- A. Kerosene
- B. Lubricating oil
- C. Diesel

69. I - Respiration:

li - photosynthesis

lii - combustion

Iv - decomposition

V - Sequestration

Which of the following adds CO₂ to the atmosphere?

- A. I and Iv
- B. li only
- C I iii iv and v
- D I iii and iv

70. Which of the following affects the rate of nuclear reaction

- I. Temperature
 - li. Pressure
 - lii catalyst
 - Iv. Concentration
- A. I only
 - B. li iii and iv only
 - C all of the options
 - D. None from I-iv

71. The incomplete oxidation of ethanol gives

- A. Ethanal
- B. Benzene
- C. Ethyne
- D Ethanoic acid

72. Which of the following conducts electricity

- I Graphite
 - li Diamond
 - lii P₅
 - Iv S₈
- A. I and iv
 - B. li only
 - C. I only
 - D all of the above

73. Which of the following can not be gotten from the electrolysis of brine

using Mercury Cathode and
carbonanode
A NAOH

is

unyeH8
UOaNY

75. The no of eletrons in CO_3^{2-} is

B28

D 43

$$\text{CH}_3\text{C}(\text{CH}_3)_2\text{CH}_2\text{CH}_2\text{CH}_3$$

B 2 2 dimethyl butane

C 2 2 dimethyl Pentane

current enters and leaves the electrolyte

B Voltmeter

C Voltmeter

78. Which of the following are used for road surfacing

- a. Petroleum gas
- b. Bitumen
- c. Diesel

79. Arrange the following accordingly
 O_2 , Cl_2 and N_2 I. Single bond II. Double bond III. Triple bond (a) I II, III (b) II, I, II (c) I I, III (d) III, I I, I

80. The quality of petrol as a engine fuel improves with and increase in the amount of I straight chain alkanes II. straight chain alkenes III. branched chain alkanes IV. branched chain alkenes (a) I & II (b) I & III (c) IV only (d) III only

81. Oxidation number of hydrogen in CaH_2 , H_2O and H_2 respectively are (a) +10, +2 (b) -1, +1, -2 (c) +1, -1, 0 (d) -1, +1, 0 (e) -1, +1, +1

82. Reaction between wood ash and vegetable oil is I. Neutralization II. Hydrolysis III. Hydrogenation IV. Hydroxylation (a) II & IV (b) IV Only (c) III only (d) I only (e) II only

140 + 100 = 240

84. Which of these is a monoprotic acid? I.HCl II.HNO₃ III. Ethanoic acid IV.H₂SO₄ (a) I-IV (b) I only (c) I&II (d) I II&III (e) I,II&III

85. Which of these will change red litmus blue I. Ammonium chloride II.HCl III.Na₂CO₃ IV. KNO₃ (a)I-IV (b) I,III&IV (c) III only (d) III&IV.

86. Formalin is an aqueous solution of (A.)Methyl acetate B. Propanone C.Ethanol D. Methanal

87. Which of the following does not react with dilute hydrochloric acid

I. Copper. II. Iron. III. Zinc. IV. Tin.

A.I and III

B. II only

C. All except III

D. IV only

88. Helium is used in place of Hydrogen in filling balloons because the latter is

A. Corrosive

B. Inflammable

C..... can't remember the other options

89. What is the abundant noble gas in nature

A. Helium

B.Argon

C.Neon

D.Krypton

90.Which of these gases are harmful to the atmosphere

I. Carbon (IV) oxide

II. Carbon (II) oxide

III.Sulphur (IV) oxide

IV.CFC

91. . When iron (II) trioxonitrate(V) is warmed with stirring in an Aluminium pot what happens?

A. H₂ is liberated

B. Al is reduced

C. The pot decrease in size

D. The pot increases in size

92. In surgery metal pins are used for joining together broken bone .These metal pins remain uncorroded in the body. What is the Material of these pins

I.Aluminum

II.Iron

III.Copper

IV. Titanium

93. Of the following metals, which one pollutes the air of a big city?

1. Chromium

II. Copper

III. Cadmium

IV. Lead

94. Haematite is an ore of

Fe_2O_3

95. The main chemical constituent of clay is

Silicon oxide

- A. Aluminum borosilicate B.
C. Zeolites D. Aluminum
silicate

96. Which of the following compounds

is composed of Al, Si, O and H?

1. Epsom salt
II. Limestone
III. Clay
IV. Urea

97. Which one causes photochemical smog?

- A. O_3 , PAN and NO_2
B. O_3 , PAN and CO
C. HCN, NO and PAN
D. O_2 , PAN and NO_2

98. Which does not react with
dilute hydrochloric acid?

1. Copper
II. Iron
III. Zinc
IV. Tin

99. The mineral containing both
magnesium and calcium is A. Magnesite

- B. Carnallite
C. Dolomite

100. The maximum number of
isomers for an alkane with
Molecular formula C_4H_{10} is

- A. 5
B. 4
C. 2
D. 3

101. Which of the following is an halide ore

- A. Cassiterite
B. Anglesite
C. Siderite
D. Carnallite

102. The isomerism which exists
between CH_3CHCl_2 and
 $\text{CH}_2\text{ClCH}_2\text{Cl}$ is

- A. chain isomerism
B. Functional group Isomerism
C. Positional isomerism

103. Which one
of the following
statements
are correct?

1.
solids.

II. Solutions have more Entropy than solids.
 107. A sample of a hydrocarbon was

III. Gases and their liquids have equal combusted producing 1.054g of CO₂ and Entropy.
 IV. Gases have less entropy than their solids 0.2168 g of H₂O. What is the empirical formula For the hydrocarbon?
 A. CH

V. Entropy of a substance increases as its B.
 CH₂temperature increases.

C. C₂H₂
 D. C₂H₄

104. How many unpaired electrons can be
 Cu²⁺

108. Which of the following will not dissolve
 the oil spilled on Water except found in

1. perfumes
1. kerosene
- II. 5detergents
- III. 1ammonia
- IV. 1ammonia

IV. 2

- A. I only
 B. I&III only
 C. I&IV
 D. II&IV
 E. IV only

105. The pH of a solution is 10 and that of another is 12. When
 equal Volumes of these two are mixed, the pH of the resulting solution is

- A. 0
 B. 12

C. 11

electricity 109. Which of the
 following conducts

D. 10.30
 106. Which of these is used in ripening of
 fruits

1. Ethene
- II. Ethyne
- IV. Ethanol

- A. graphite
 B. Diamond
 C. P₄
 D. S₈

Ethanoic
number is/are?110. Another name for
subsidiary quantum

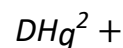
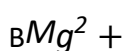
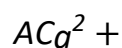
- 1, Angular
Magnetic
11. 111. Principal
IV. Primary
V. Azimuthal
A. II only
B. II & I
V
C. V only
D. I & V
E. none of the above

111. Which of the following ions is/are harmful to aquatic habitats even when present in traces.

- I) Fe^{2+}
II) Hg^{+}
III) Cu^{2+}
IV) Ag^{+}

A) Hg
B) Cu & Fe only
C) Ag
D) Cu

112. ion that is pollutant to drinking water even in trace amount is



- bases and salts
A The dissociate exothermically
B They liberate OH^{-}
C They liberate H_3O^{+}

D They are electrolytes

114. Which of the following undergoes both substitution and addition reactions? I. Pentane II. Hexene III. Benzene IV. Ethene

A I only

B III only

C III and IV only

D II only

115. A vessel was cold when touched, which of the following explains that

A Endothermic

B Exothermic

C Unstable

D Stable

116. Catalyst helps do what in a reaction?

- i) Create an alternative pathway ii) increase surface area iii) decrease the activation barrier iv) increase concentration

A.) I and III

B ii and iv

C iv only

D iii and iv

113. What are the similarities between amines

doue Suelcke
BUNOllay
e4yo4alu ZTT

A disorderliness

B heat content

C? D?

118. In which of the following is the enthalpy of the product greater than the that of the reactants

i) Lattice energy ii) Heat of Neutralization

iii) Heat of combustion iv) Hydrolysis

119. Reactions that can go back and forth are called....reversible reactions

120. Which of these explains the addition of an external constraint to affect the equilibrium of a Reaction

A) Le Chatelier

B) Dalton's law

C) Graham's law

121. Which of the following is linear

I. H₂O

II. CO₂

III NH₃

IV. Cl₂

A. I and II

B. IV only

C. III and IV

D. II and IV

E. II only

122. Which of the following is responsible to the sour taste in palm wine.

I. Methanol

II. Ethanoic Acid

III. Methanoic acid

IV Ethanol

A. I and III

B. I and II

C. II and IV

D. II only

E. II and IV

123. How many unpaired electron has fluorine

A. 3

B. 1

C. 0

D. 2

124. Which of the following is responsible for coloured ion formed by transition metal.

I. Partially filled s-orbital

II. Partially filled p-orbital

III Partially filled d-orbital

IV/. No. of valence electron

A. II and III

B.II and IV

125. What is the shape of H₂O

leau 7
/neou 8
7ueg 3

D. Linear

126. And element has electronic configuration 1s² 2s² 2p⁶ 3s² 3p⁵, which of the following is/are true of the element

I The valence electron is 5

II it is an halogen

III I've forgotten IV It belongs to period 3 of the periodic table

A. I and II

B II only

C IV only

D. II and IV

127. Alkanol has higher boiling point than the corresponding alkane due to:

I. Presence of intramolecular hydrogen bond II presence of intermolecular hydrogen bond III Non-aromatic properties of Alkane

A. II and III

B. I and III

Clonly

D II only

128. A compound with high melting and boiling point as a result of which of the following?

A. Covalent bond

B. Dative bond

C. Electrovalent bond

D. Dipole-dipole

E. Hydrogen bond

129. The energy that must be attained before chemical reaction can occur is:

A. Ionization energy

B. Activation energy

C. Heat energy

D. I can't remember!

130. The shape of carbon (IV) oxide is:

A. Bent

B. Linear

C Tetrahedral

D. Triangonal

131. Alkanol reacts with alkanolic acid in the presence of H₂SO₄ to form an ester.

B. Hydrolysis

- C. Saponification
- D. Esterification
132. Ethene reacts with cold water to form
- A. Acidic solution
- B basic solution
- C. Alkanol
- D. No reaction occur
133. Which of the following is/are true
- I Oxygen and Ozone are allotrope
- II Oxygen and Ozone are isotope
- III Oxygen and Ozone have same atomic number
- IV. I can't remember and the options combinations too.
134. Why is soap referred to as salt?
1. It does not contain hydroxide ion
- II. It has water of crystallization
- III. It is prepared by Reacting fatty acid with a strong base

9. IV
- 10.A
- 11.B
- 12.A
13. III, IV
14. WATER
- 15.C
- 16.C
- 17.D
18. ACIDIC
- 19.C
- 20.B
- 21.A
- 22.C
- 23.B
- 24.B
- 25.B
- 26.D
- 27.ALL
- 28.D
- 29.I ONLY
- 30.III
31. NONE
32. IV ONLY
- 33.I ONLY
34. III ONLY
- 35.B
- 36.II ONLY
- 37.B
- 38.C
39. A&B
- 40.B
- 41.1,I1
- 42.III,IV
- 43.C
- 44..
- 45.I ONLY
- 46.D
- 47.I ONLY
- 48.C
- 49.II ONLY
- 50.C
- 51.A
- 52.A

ANSWERS

1. D
- 2.1
3. Heat of reaction
4. A
5. B
6. 1,II
7. 17.7S
- 8.1:4

53.A	97.A	
54.C	98.1 ONLY	
55.D	99.C	
56.B	100.	D
57.B	101.	D
58.D	102.	C
59.A	103.	1,11,V
60.B	104.	II
61.A,11,111	105.	C
62.A,11,111	107. 106. B	IONLY
63.D	108.	B
64.1,11	109.	A
65.C	110.	D
66.D	111.	A
67.B	112.	D
68.C	113.	D
69.C	114.	C
70.D	115.	A
71.A	116.	A
72.C	117.	A
73.C	118.	IV
74.D	119.	Reversible reaction
75.A	120.	A
76.C	121.	D
77.A	122.	D
78.B	123.	B
79.B	124.	C
80.D	125.	B
81.D	126.	D
82.E	127.	D
83.D	128.	C
84.D	129.	B
85.C	130.	B
86.D	131.	D
87. Only copper is correct	132.	C
88.B	133.	I& III ONLY
89.B	134.	III ONLY
90.I,II,III,IV		
91.C		
92.IV		
93.IV		
94.A		
95.D		
96.III		



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PAST QUESTIONS &ANSWERS

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OLAGUNJU AHMAD	MBBS
MUHEEDEN FATHIA	PHARMACY
AYODELE HAMMED	PHARMACY
OKOYE DANIEL	MBBS
OZUEM JOSEPH	MBBS
OGUNTINMEHIN YINKA	MBBS
OSHO TINI DENTISTRY	LAW
FERANMI PETER	LAW
ADESINA PLEASANT	PHARMACY
OMIGIE FRANCES	LAW
ADEAGBO MERCY	LAW
OMOLEKAN DEBORAH	LAW
ABISOLA MARIAM	THEATER ART
NWAORGWU CHIAMAKA	LAW
OYEDIRAN TOHEEBAT	PHARMACY
NNUBIA ESTHER	NURSING
OGUNBIYI PRAISE	LAW
KAMORUDEEN ABUBAKR	CIVIL ENGINEERING
ADEAGBO GOODNESS	ELECT ENGINEERING
SHARON ONYENE	LAW
AKANJI JEHU	ELECT ENGINEERING

And So Many More...